

Machine Design for Auger Dosing of Powders by utilizing DEM Simulations and Multivariate-Regression-Based Ex- perimental Analysis

Maschinendesign für die Schneckendosierung von Pulvern durch Einsatz von DEM-Simulationen und multivariater regressionsbasierter experimenteller Analyse

Von der Fakultät für Maschinenwesen der Rheinisch-Westfälischen Technischen Hochschule
Aachen zur Erlangung des akademischen Grades eines Doktors der Ingenieurwissenschaften
genehmigte Dissertation

vorgelegt von

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Tag der mündlichen Prüfung: 08. Februar 2021

Report No. IAM-08

Institute of General Mechanics / Institut für Allgemeine Mechanik (IAM)

RWTH Aachen University, Germany, 2021

Editor:

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PREFACE

The work described in this doctoral dissertation was carried out as a scientific assistant between the years 2016 and 2019 at Robert Bosch Packaging Technology GmbH (now as Syntegon Technology GmbH) in Waiblingen and Institute of General Mechanics IAM of the RWTH Aachen University. In this period of my occupational career, I gained profound knowledge in the design and manufacturing process, the experimental measurements and regression analyses, and the particle discrete element simulations. However, during the years many people have crossed my path and contributed in many ways to the realization of this work. I would like to sincerely thank all of them.

In particular, I would like to express my sincere gratitude to my supervisor, Professor Bernd Markert, for giving me the opportunity to prepare my doctoral study under his guidance and allowing me to be part of the IAM institute. His encouragement, flexibility, genuine caring, and concern enabled me to finish my dissertation. I am also sincerely grateful to Dr. Yousef Heider for his constant assistance, precious advices and endless efforts that helped me improve my critical thinking, research methodology, and think beyond the horizons. For this and much more, I am forever thankful.

I address a special thanks to Dr. Thomas Brinz, my tutor, for offering me the opportunity to write my thesis at Bosch. I am deeply appreciative for his knowledge and support, where he has always encouraged me and offered me all the scientific freedom to make this work possible. Furthermore, I am also very grateful to my colleagues from different departments at Bosch, who always shared their experience and knowledge with me and contributed with technical/laboratory support. Special acknowledgments to Stephanie Otterbach, Nizar Salloum, Bernhard Wagner, Tobias Burk, Olaf Eistert, Peter Strobel, Shailendrasingh Gahirwal, Osman Usta and Jamal Badawieh for their valuable assistance in realizing my academic and experimental work. Many thanks to my amazing friends in Aachen and Stuttgart for their various encouragements to continue with my thesis and the great time we spent together that relieved my stress.

Additionally, I dedicate this work to my family especially my dearly beloved mother and father who played above all a vital role in finishing this dissertation by inspiring me and teaching me tolerance, endurance, and self-control. Thank you all!

Ulm, February 18, 2021

Bilal El Kassem

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Summary

The aim of the underlying work is to develop a statistical methodology that identifies the correlations between critical material attributes, critical process parameters and critical quality attributes of a volumetric vertical micro-auger dosing process and to end up with building a loss in weight (LiW) feeder. In this, shear cell analysis and other various empirical tests are used to characterize 30 material attributes of ten different powders ranging between free-flowing and cohesive ones that are widely used in the drug industry. The analysis is done using principal component analysis, which reveals the (dis)similarities between the powders and identifies overarching attributes. Thereafter, regression analysis is applied to find possible correlations between the studied factors. Moreover, as a numerical approach, Discrete Element Method (DEM) simulations are performed to model the granular material flow in an auger dosing process. Two main studies are carried out to calibrate and validate the studied powders with the experimental reference values. The successful application of the multivariate regression analysis results in predicting the bulk behavior of the powder by clustering different predicted behaviors of various responses together which allows to dramatically reduce the admissible parameter combinations. Consequently, an optimized set of calibrated DEM parameters is chosen, where the simulation results accurately match with the experimental data. The essential goal of this contribution is to present the alternative vertical LiW feeder dosing concept. In a systematic design process, based on functional design specifications, the semi-automated vertical LiW feeder for dosing a wide range of powders, especially cohesive ones, is developed. Experimental sensitivity analysis and powder dosing tests to study the functionality of the machine and to investigate the weight variability of a weighing platform is successfully accomplished. As a result, the proof of concept prototype is validated.

Zusammenfassung

Das Ziel der vorliegenden Arbeit besteht darin, aus einer statistischen Methodik zur Korrelation zwischen kritischen Materialeigenschaften, kritischen Prozessparametern und kritischen Qualitätseigenschaften die Auslegung und Optimierung eines volumetrischen vertikalen Mikroschneckendosierungsprozesses zu entwickeln und am Ende eine Differentialdosierwaage (Loss in Weight Feeder) zu bauen. Dabei werden Scherzellenanalyse und verschiedene, empirische Tests verwendet, um 30 Materialeigenschaften, von zehn verschiedenen Pulvern, zu charakterisieren, die zwischen frei fließenden und kohäsiven Pulvern liegen. Diese Pulver sind in der pharmazeutischen Industrie weit verbreitet. Mithilfe der sogenannten Hauptkomponentenanalyse sollen Ähnlichkeiten, Unterschiede und übergreifende Attribute der unterschiedlichen Pulver zueinander, bestimmt werden. Des Weiteren wird eine Regressionsanalyse durchgeführt, um mögliche Korrelationen zwischen den untersuchten Faktoren herauszufinden. Darüber hinaus werden, als numerischer Ansatz, Simulationen anhand der Diskrete-Elemente-Methode (DEM) eingeführt, um den körnigen Materialfluss in einem Schneckendosierungsprozess zu modellieren. Zudem werden zwei Hauptstudien durchgeführt, um experimentelle Referenzwerte der zu untersuchenden Pulver, kalibrieren und validieren zu können. Die erfolgreiche Anwendung der multivariaten Regressionsanalyse führt zur Vorhersage des Schüttgutverhaltens des Pulvers durch Zusammenfügen verschiedener vorhergesagter Verhaltensweisen verschiedener Reaktionen, wodurch die zulässigen Parameterkombinationen drastisch reduziert werden können. Folglich wird ein optimierter Satz von Parametern der kalibrierten Diskrete-Elemente-Methode gewählt, bei dem die Simulationsergebnisse genau mit den experimentellen Daten übereinstimmen. Das Ziel dieser Arbeit ist die Darstellung des vertikalen Verlusts im Dosiersystem bei Gewichtszuführungen. Der Entwurfs- und Konstruktionsprozess, sowie die Installation des Dosiersystems, machen ein Teil der Hauptaufgaben dieser Arbeit aus. Ein systematischer Entwurfsprozessansatz zum Entwerfen, Erstellen und experimentellen Validieren eines Proof-of-Concept-Prototyps wird definiert. Eine Sensitivitätsanalyse und experimentelle Validierung zur Untersuchung der Funktionalität der Maschine und zur Untersuchung der Gewichtsvariabilität einer Wägezelle wird erfolgreich durchgeführt. Dadurch wird der Proof-of-Concept-Prototyp validiert.

Motivation

Die Forschungsbereiche „Schüttguthandhabung“ und „Partikeltechnologie“ sind aufgrund der Berücksichtigung unterschiedlicher Disziplinen, hochinteressante Themen. Die Komplexität des Problems und die Überschneidung verschiedener Forschungsbereiche, wie Materialwissenschaften, Maschinenbau, Dosierungs- und Messtechnik, Produktherstellung, Automatisierung und Verfahrenstechnik, Verpackungstechnik sowie Gesundheitswesen sind notwendig, um die Phänomene der Fließfähigkeit von Schüttgütern zu verstehen oder einen richtigen Einblick diesbezüglich zu gewinnen. Weitere Bereiche

umfassen Lagerung, Fließen, Verarbeitung und Transport von Schüttgütern, z.B. Pulvern. Ein wichtiger Bereich ist hierbei die Implementierung einer multivariaten statistischen Datenanalyse, die für das Verständnis des Prozesses und den Aufbau eines robusten Systems eine wichtige Rolle spielt. Darüber hinaus trägt die Erstellung eines numerischen Modells zur Simulation eines realen Dosierungsprozesses bei der Analyse und Optimierung des Produktionsprozesses. Somit können die Kosten und der Aufwand reduziert werden. Ein weiteres wichtiges Feld ist die Erstellung und Validierung eines Proof-of-Concept Prototyps für Dosiermaschinen durch Verwendung eines systematischen Entwurfsansatzes. Zusammenfassend ist dieses Forschungsfeld aufgrund der Berücksichtigung verschiedener Disziplinen und herausfordernder Themen von großem Interesse, wodurch interdisziplinäres Wissen in verschiedenen Bereichen angeeignet werden kann.

In fast allen Industrien werden Pulver und Schüttgüter gehandhabt, gefördert, transportiert und gelagert, von der Pulverbeschichtung bis zu Lebensmitteln, von nanoskaligen Pulvern bis zu Produkten wie Zement, Kohle und Erz, von trockenen Materialien wie Flugasche bis zu feuchten Schüttgütern wie Filterkuchen und Ton [1]. Daher ist bei der Entwicklung fester Darreichungsformen die Charakterisierung von Pulvern, hinsichtlich ihrer physikalischen Fließeigenschaften, von großer Bedeutung, wobei das Verständnis des Fließverhaltens darauf abzielt, das Design von Pulverhandhabungsgeräten und damit die Qualität eines pharmazeutischen Produkts zu optimieren [2,3]. Es ist wiederum von großer Bedeutung, die Qualität des Endprodukts bereits in den frühesten Phasen der Produktentwicklung zu berücksichtigen [4–6]. Pulver sind komplex und ihr Verhalten wird nur teilweise verstanden, weil sie in der Regel nicht mit einem einzigen Parameter beschrieben werden können. Ihr Verhalten ist einer Vielzahl von Faktoren ausgesetzt, wie z.B. dem Wechsel des Zustands von einem fluidisierten zu einem gepackten Pulverbett, und dies hängt von der Arbeitsweise der Anlage und den verwendeten Prozessparametern ab [7]. Schneckenförderer werden häufig eingesetzt, um den Materialfluss von einem Trichter in einen bestimmten Herstellungsprozess, z.B. einen Mischprozess, zu steuern. Die Leistung eines Schneckenförderers wird durch die Materialeigenschaften sowie durch die Geometrie der Schnecke, des Rührwerks und des Trichters beeinflusst [8]. Tatsächlich zeigen eine Reihe von Dosierproblemen, beispielsweise niedrige und schwankende Fördermengen, hohe Schneckendrehmomente und Pulveranhäufungen an den Innenwänden eines Dosierbehälters, eine schlecht abgestimmte Kombination von Pulver-Prozessfaktoren. Clayton [9] beschrieb einige Faktoren, die die Spezifikation einer Schnecke oder eines Schneckenförderers beeinflussen, darunter hauptsächlich Installationsbeschränkungen, Prozessanforderungen und Materialeigenschaften. Daher ist es wichtig, die Konstruktion eines bestimmten Systems an das zu handhabende Produkt anzupassen. Es ist entscheidend, das Zufuhrvolumen für ein bestimmtes Pulver vorherzusagen, ohne die besten Kombinationen zwischen den verschiedenen Materialattributen und Prozessparametern zu definieren. Da es nicht möglich ist, die Materialcharakterisierung zu ändern, können einige Entwurfsvariablen manipuliert werden, um Spezifikationskriterien zu erfüllen, einschließlich der Geometrie einer Schnecke, d.h. des Innen- und Außendurchmessers, des Innenwinkels und der Steigung, der Geometrie des Rührers und der Geometrie des Trichters. Die Partikelsimulation ist ein beliebter

Ansatz zur Modellierung und Vorhersage des Verhaltens von Schüttgütern in verschiedenen Systemen. Diese Systeme können verschiedene Anwendungen umfassen, wie z.B. die Lagerung, den Transport oder die Verarbeitung von körnigen Materialien, z.B. Sand, Gieß, Pigmente, pharmazeutische Pulver, kristallisierte Produkte wie Ammoniumsulfat, Düngemittel, Polyethylenpulver, Ernteprodukte oder andere Schüttgüter in der Lebensmittel-, Chemie-, Bergbau- oder Pharmaindustrie. In dieser Studie wird die Diskrete-Elemente-Methode [10] zur Modellierung der Granulatströmung auf der Ebene der Partikel eingesetzt.

Zielsetzung und Vorgehensweise

In der vorliegenden Arbeit wird die Korrelation zwischen den Pulverattributen und Prozessparametern untersucht, ein Simulationsmodell und die Entwicklung, die Konstruktion und die Umsetzung einer Schneckendosieranlage für pharmazeutische Abfülllösungen betrachtet. Durch die Flexibilität und Modularität der Dosiermaschine kann ein breites Spektrum unterschiedlicher Pulver mit hoher Genauigkeit dosiert werden. Dazu wird ein systematischer Entwurfsprozessansatz zum Entwerfen, Konstruktion und experimentellen Validierung eines Proof-of-Concept-Prototyps verfolgt. Die Funktionalität der Maschine und ihre Fähigkeit, Pulver genau zu dosieren, werden durch Sensitivitätsanalyse und Experimente untersucht. Eine Reihe von Pulvercharakterisierungsverfahren wird verwendet, um 30 Materialeigenschaften von zehn verschiedenen Pulvern zu charakterisieren. Durch die entwickelte multivariate Datenanalyse können die möglichen Korrelationen und Wechselwirkungen zwischen den untersuchten Faktoren und den Massenantworten angelegt werden. Die Analyse basiert dabei auf der Versuchsplanung, der Hauptkomponentenanalyse und der Regressionsanalyse. Die Partikelsimulation, nach der Diskreten Elementen Methode (DEM), basiert auf dem Hertz-Mindlin-Kontaktmodell. Das entwickelte und validierte numerische Modell reduziert, mit Hilfe der multivariaten Regressionsanalyse, reduziert die Simulationszeit und ermöglicht ein besseres Wissen über die Einflussfaktoren auf die Dosierungsleistung. Es wird in dieser Arbeit ein halbautomatischer Kalibrierungsansatz entwickelt, um die optimierten Parameterkombinationen herauszufinden, die zu ähnlichen Ergebnissen wie in den Experimenten führen.

Gliederung der Arbeit

In *Kapitel 1* wird die Arbeit eingeleitet und der Schneckendosierungsprozess vorgestellt. Des Weiteren wird ein umfangreicher Überblick über den Stand der Forschung hinsichtlich der Korrelationen zwischen Pulverattributen und Prozessparametern sowie Kalibrierungsansätzen für die Partikelsimulation dargestellt und erläutert.

Danach werden in *Kapitel 2* die theoretischen Grundlagen der Pulvercharakterisierung der Materialeigenschaften vorgestellt. Dabei werden verschiedene Pulvercharakterisierungsverfahren verwendet, um

30 Materialeigenschaften von zehn verschiedenen Pulvern zu charakterisieren, die zwischen frei fließenden und kohäsiven Pulvern liegen.

In *Kapitel 3* wird eine statistische Methode beschrieben, die die Korrelationen zwischen kritischen Materialattributen, kritischen Prozessparametern und kritischen Qualitätsattributen eines volumetrischen vertikalen Mikroschneckendosierungsprozesses identifiziert. Das Design von Experimenten, die Hauptkomponentenanalyse und die multivariate Regressionsanalyse werden implementiert.

Die numerische Behandlung des Partikelflusses wird in *Kapitel 4* vorgestellt. In diesem Zusammenhang werden die maßgeblichen Gleichungen und die implementierten Partikelwechselwirkungsmodelle vorgestellt, gefolgt von zwei Fallstudien, die einen entwickelten Kalibrierungs- und Validierungsansatz abdecken.

Schließlich wird in *Kapitel 5* ein systematischer Entwurfsprozessansatz zum Entwerfen, Erstellen und experimentellen Validieren eines Proof-of-Concept eines vertikalen LiW-Feeder-Prototyps definiert.

Eine abschließende Zusammenfassung und Diskussion wird im *letzten Kapitel* erstellt, einschließlich einiger Aussagen zur Denkbarekeit weiterer Entwicklungen und Zukunftspotenzial.

Nomenclature

The nomenclature throughout this thesis follows the conventions and symbols that are commonly used in the equations of motion between particles.

Conventions

Index and suffix conventions

i, j	indices as sub- or superscripts range from 1 to N; between particle i and j
coh	of cohesion
$grav$	gravity
max	maximum
min	minimum
n	normal
pp	particle-particle
pw	particle-wall
r	rolling
t	tangential
ww	wall-wall

Symbols

Symbol	Unit	Description
$\dot{m}$	kg/s	Mass flow rate/feed rate
ρ	kg/m ³	Density
ρ_{true}	kg/m ³	Particle density
ν	-	Poisson's ratio
μ_s	-	Coefficient of sliding friction
μ_{ss}	-	Shear slider
μ_r	-	Coefficient of rolling friction
β	-	Damping ratio
Δt	s	Time step
δ	-	Contact overlap
ω	1/s	Rotational velocity

v	m/s	Relative velocity
γ	-	Viscoelastic damping constant
τ	Pa	Shear stress
σ	Pa	Normal stress
ε	%	Porosity
A	μm^2	Projected area of a particle
a	m^2	Particle contact area
AE	mJ	Aeration Energy
AIF	°	Angle of Internal Friction
AoR	°	Angle of Repose
AR	-	Aeration Ratio
ASR	-	Aspect Ratio
BD	kg/m^3	Bulk Density
BFE	mJ	Basic Flowability Energy
nBFE	mJ/g	Normalized Basic Flowability Energy
C	kPa	Cohesion
C	J/m^3	Cohesion energy density
CI	-	Carr's Index
CPS	%	Compressibility
CT	min	Computation Time
d	m	Diameter of particle
d10	μm	10 % cumulative undersize of the volumetric particle size distribution
d50	μm	50 % cumulative undersize of the volumetric particle size distribution
d90	μm	90 % cumulative undersize of the volumetric particle size distribution
dist	m	Distance between particle i and j
D[4,3]	m	Volume Moment Mean
DT	min	Dosing Time
e	-	Coefficient of restitution
Fe	μm	Feret diameter
F	N	Force
ff_c	-	Flow function coefficient
I	-	Irregularity factor
I	kg m^2	Moment of Inertia
G	Pa	Shear modulus

G^*	Pa	Equivalent shear modulus
g	m/s ²	Acceleration due to gravity
k	-	Elastic constant
MPS	kPa	Major Principal Stress
MC	%	Moisture Content
m	kg	Mass of particle
m^*	kg	Effective mass of particle
NAS	s/mm	Normalized Aeration Sensitivity
P	μm	Perimeter of projection
PoDM	%	Percentage of Dosed Mass
PR	cm ²	Permeability
R	m	Radius of particle
R^2	-	Coefficient of determination
Ro	-	Roundness
RSD	%	Relative Standard Deviation
S_f	-	Scaling factor
S	-	Contact stiffness
SE	mJ/g	Specific Energy
SI	-	Stability Index
StD	g/s	Standard Deviation
TD	kg/m ³	Tapped Density
TKE	J	Translational Kinetic Energy
T_R	s	Critical time step
UYS	Pa	Unconfined Yield Strength
V	mm ³	Volume
WFA	°	Wall Friction Angle
Y	MPa	Young's Modulus
Y^*	Pa	Equivalent Young's Modulus
$\hat{z}$	-	Unit vector in z-direction

List of Abbreviations

3D	Three-dimensional
API	Active Pharmaceutical Ingredient
CAD	Computer Aided Design
CG	Coarse-Graining
CMA	Critical Material Attribute
CPP	Critical Process Parameter
CQA	Critical Quality Attribute
DEM	Discrete Element Method
DoE	Design of Experiment
EPSD	Elastic-Plastic Spring-Dashpot
FDS	Functional Design Specification
FT4	Freeman Technology Powder Rheometer version 4
GiW	Gain in Weight
LAMMPS	Open-access software package: Large-scale Atomic and Molecular Massively Parallel Simulator
LIGGGHTS	Open-access software package: LAMMPS Improved for General Granular and Granular Heat Transfer Simulations
LiW	Loss in Weight
MCC	Microcrystalline Cellulose
MVDA	Multivariate Data Analysis
MVRA	Multivariate Regression Analysis
PCA	Principal Component Analysis
PC	Principal Component
PLC	Programmable Logic Controller
PLS	Partial Least Square
PTFE	Polytetrafluoroethylene
QbD	Quality by Design
SJKR	Simplified Johnson-Kendall-Roberts

1 Introduction

The research fields of bulk solid handling and particulate technology are highly interesting topics in diverse disciplines of engineering and technology. Understanding the complexity of the problem and the overlapping of different research fields, such as material science, mechanical design engineering, dosing and measurement technologies, product manufacturing, automation, process engineering, packaging technology and health care are necessary to gain a proper insight of the bulk flowability phenomena. Additional areas include storage, flow, processing, and transportation of bulk solids, e.g., powders. An important approach in powder analysis is the implementation of a multivariate statistical data analysis that helps to understand the process and build a robust system. Moreover, building a numerical model to simulate a real dosing process aids in analyzing and optimizing the production process and thus saves a lot of efforts and costs. Another essential field is building and validating a proof of concept dosing machine prototype using a systematic design approach. To sum up, this research field is of great interest due to the consideration of diverse disciplines and challenging topics, where the researcher is able to gain interdisciplinary knowledge in various domains.

1.1 Motivation

In almost all industries, powders and bulk solids are being handled, conveyed, transported and stored [1]. Therefore, in solid dosage forms development, the characterization of powders regarding their physical and flow properties play a significant role, where understanding the flow behavior aims in optimizing the design of powder handling equipment and thus the quality of a pharmaceutical product [2,3]. It is of great importance to consider the final product quality in the earliest stages of product development [4–6]. Powders are complex and their behavior is poorly understood, where they cannot be described using a single parameter. Their behavior is subjected to a wide range of factors, such as changing the state from a fluidized to a packed powder bed, and that is depending on the unit operation and the process parameters used [7]. Auger feeders are widely utilized to control the flow of material from a hopper into a certain manufacturing process, e.g., mixing process. The performance of an auger feeder is influenced by the material attributes as well as the auger, stirrer and hopper geometries [8]. In fact, a number of dosing problems, for instance, low and fluctuated feed rates, high auger torques and powder accumulation on the inner walls of a dosing container, reveal a poorly matched powder-process factors combination. Clayton [9] mentioned some factors that influence an auger or screw-feeder specification, which mainly include installation constraints, process requirements, and material attributes. Therefore, it is essential to adapt the design of any given system to the product being handled. It is critical to predict

the feed volume for any given powder without understanding the best combinations between the different material attributes and process parameters. Since it is not possible to alter the material characterization (attributes), i.e., particle sizes, density, etc., some design variables can be varied to meet the auger feeder specification criteria. These design variables include the geometry of the auger, i.e., inner and outer diameters, internal angle, and the pitch, the geometry of the stirrer, the geometry of the hopper, and the rotation speed. Particle simulation is a popular approach to model and predict the behavior of bulk materials involved in different systems. These systems may include several applications such as storing, transporting, or processing granular materials, for instance, sand, semolina, pigments, pharmaceutical powders, crystallized products such as ammonium sulfate, fertilizers, polyethylene powder, harvest products, or other bulk materials in food, chemical, mining, or pharmaceutical industries. In this study, the discrete element method (DEM) [10] is utilized to model granular flow at the particle level.

1.2 State of the Art

One of the big challenges in the pharmaceutical industry is to optimize the powder solid handling process performance. This significantly relies on the flowability of the granular material in different process stages including hopper discharge, powder feeding, blending, mixing, and dies filling. Flowability, which describes the behavior of powder, is dependent on a range of fundamental powder attributes/properties. The flow properties of powders can be affected by various essential particle properties, including mean particle size and distribution, particle shape, surface roughness and moisture content, etc. Powder flow also depends on the interparticle interactions, where, for instance, flowability decreases with increasing cohesiveness of the powder. Literature review relevant to the different topics studied in the thesis is discussed in the following.

Powder Dosing Systems

In the pharmaceutical industry, a broad variety of solid powder dosing systems is used. These systems employ different technologies and principles such as vibration, compression, vacuum, gravitational and pneumatic system. The appropriate selection of one of these systems is dictated by, e.g., the process application, type of material powder(s) and the amount of dosage required. Vibration-assisted filling method uses a mechanical vibrator to de-agglomerate the powder and allow it to flow to fill the hard capsules. A dosing wheel or drum filler is a micro-filling system to meter powder or granular. The principle lays on using vacuum pressure to fill the cavities and then after the wheel rotates 180° the powder is discharged under gravity. Tamping filling machine or dosing disc system uses tamping pins to fill capsules under different applied pressure loads. It has a cylindrical powder bowl that contains a removal dosing disk with six dosing holes. Moreover, a dosator nozzle machine comprises two parts, the body, and the pin. The dosator moves into the powder bed and collects the desired volume of powder, where it then transfers the compacted dosage to the packing receptacle.

Screw conveyors or auger feeders are utilized extensively in process engineering as mechanisms to transport bulk materials (e.g., granular materials). As the screw rotates, bulk materials are dispensed from the container and conveyed along the screw casing. Typically, the auger is operated in conjunction with a rotating stirrer/agitator. There are two main dosing principles that can express and control the feed rate of the bulk material: gravimetric and volumetric dosing. The gravimetric dosing is based on weighing the amount of material to be dosed, whereas the volumetric system doses the material according to the space it occupies. Although the mechanical system is quite simple, understanding and controlling the material behavior towards an optimal process might be challenging [11–13]. In fact, several problems might be experienced if the machine is not designed properly in connection with the nature of the conveyed material and mechanical behavior. These problems can be defined, for instance, as surging and unsteady flow rates, inhomogeneity of the product, inaccurate metering and dosing, product degradation, high start-up torques, excessive power consumption, high equipment wear, variable residence time, and segregation [13,14].

Correlations between Powder and Process Attributes

In the literature, a lot of studies attempted to correlate raw material attributes with process performance [15–18]. ChamCarthy et al. [15] indicated that starch grades with higher surface energy result in stronger compacts. Fu et al. [16] studied the effect of particle shape and size on flow properties of lactose on the flow characteristics of lactose under a wide range of stress conditions. Herting and Kleinebudde [17] showed an increase in the tensile strength of tablets when the particle size of the active pharmaceutical ingredient (API) or excipient decreased for a roller compaction/dry granulation process. Steckel et al. [18] investigated the effect of milling and sieving on the functionality of dry powder inhalation products, where milling of lactose affects the carrier surface and thus the aerosolization properties of drug blends produced.

Other studies have also conducted the approach of linking the material attributes with the process behavior via multivariate models at different unit operations of continuous manufacturing line (e.g., feeders, blender), granulation and tableting processes [9,19,28–30,20–27]. Fonteyne et al. [21] studied the influence of raw material attributes of different binders and microcrystalline cellulose (MCC) grades on critical quality attributes (CQA) of continuously produced granules and tablets. The results revealed that the variation in the water binding capacity of MCC affected the granule size distribution. Willecke et al. [22] used principal component analysis (PCA) of wet granulation fillers and binders to reveal similarities and differences in excipient characteristics and linked their properties with twin-screw granulation. Garcia-Munoz [23] developed two novel methods to analyze the combined effect of multiple raw-materials and processing conditions on the final attributes of the product and predicted the process behavior. Van Snick et al. [29,30] correlated different raw material and blend attributes with the feeding, blending, die-filling and compression step of a continuous direct compression system for sustained-release tablets and

low-dose drug products. In this, the density at the screw flight determined the amount of material transported per screw revolution, whereas the sensitivity of the feed factor to the hopper fill level was regulated by the compressibility of the powder bed height. Moreover, they showed that the consistency of the screw filling was less steady for cohesive powders, whereas at the continuous mixing stage, the residence time in the mixer was directly correlated with the bulk density. Van Snick et al. [26] proposed a holistic material characterization approach and developed an extensive raw material property database of APIs and excipients. In this, a PCA model was developed, which allowed to rationalize the number of critical characterization techniques in routine characterization and to identify surrogates for use during early drug product development stages. Bostijn et al. [25] correlated volumetric and gravimetric feeding behavior of different pharmaceutical powders on a low rate feeder with their material attributes through partial least squares (PLS) regression. They used the raw material property database and the PCA models developed by Van Snick et al. [26]. Different volumetric and gravimetric feeding responses were determined, where the volumetric responses were better predicted than the gravimetric ones. Stauffer et al. [31] investigated the impact of the batch-to-batch variability of a cohesive API on the feeding capacity during twin-screw blend feeding. PLS regression models showed a clear correlation between blends properties (i.e., density, mass charge, cohesiveness) and the feeding capacity. In addition, Escotet-Espinoza et al. [27] used PCA and hierarchical clustering to evaluate a material property database and to find similarities between materials. They evaluated the similarities with the performance of materials on powder feeding, refilling, and continuous blending equipment. Moreover, Wang et al. [28] used a similar analysis approach as in [27] to establish a material library and evaluate its potential application to the development of a continuous manufacturing process for solid dosage forms. The analysis approach highlighted the importance of selecting a reduced process-related testing conditions and measurements for material characterization and found correlations between material attributes and the process performance of a material loss in weight feeding.

Particle DEM Modeling

For modeling a granular material, several numerical approaches on different scales and with different degrees of complexity and computational costs can be found in the literature. The microscopic DEM (Cundall and Strack, [10]) accounts for the most popular schemes and provides an effective approach for the understanding of granular material's mechanical behavior and flowability on the microscale. In particular, DEM is a particle-based method used to model movement and interaction of bulk material [32]. Each particle is treated individually and its motion is described by Newton's equation of motion. In the underlying study, a soft-sphere approach is used, where colliding particles are allowed to slightly overlap resulting in a repulsive force [33]. DEM has been proven to be an important and efficient tool in several fields such as powder metallurgy, chemical engineering, pharmaceuticals, and agriculture. To mention some, DEM simulations have been applied in granular-related processes such as die

filling [34], bulk compression [35], powder mixing [36], particles packing [37,38], agglomerates fragmentation [39], flow in screw conveyor [13,40,41], and outflow from a hopper. Apart from DEM, other approaches, such as continuum porous media mechanics, can be applied to compute granular material mechanical responses on larger (macroscopic) scales and capture important behaviors such as wave propagation or fracture [42–44]. However, these approaches assume the continuity of the matter and cannot explicitly capture the interactions between the particles. Therefore, the focus in this work will be merely on DEM approach.

DEM Calibration Approaches

Notwithstanding the wide range of applicability of DEM, choosing a set of micromechanical parameters sufficient for representing the macroscopic behavior of granular material always lacks a calibration process. Therefore, many research works can be found in the literature and are dedicated to this aim. The conventional calibration approach is the “trial and error”, where in many cases; one factor is changed at a time to reach a certain relation between the inputs and outputs of the system. One of the most challenging aspects in applying efficiently the DEM numerical approach is choosing the appropriate contact models and then finding the right values of the input particle parameters involved in any used model. As particles introduced in granular materials are considerably small in size, measuring the properties of every single particle is considered very costly, challenging, and time-consuming. Yet, some trials using atomic force microscopy have been given to measuring the individual particles’ yield stress [45], Young’s modulus (Y) [46], and interfacial energy [47]. The state-of-the-art approaches for calibration of DEM parameters in most cases are based on the trial-and-error, where the track of the calibration process is lost in many cases by the human calibrators, especially when many factors are involved [48]. It is often the case that the DEM model is calibrated against bulk measurements and the input parameters are adjusted until the outputs of the model match with the experimental observations [49]. Therefore, it is of great interest for researchers to optimize the process of calibration. In the literature, different tools for DEM parameters calibration, some of which are (semi-)automatized, can be found. Al-Hashemi et al. [50] presented an extensive review related to the angle of repose (AoR), methods of measurements, appropriate applications and the influencing factors of different parameters. Coetzee [51] discussed different DEM calibration approaches covered in the literature over the last 25 years to assist future researchers to improve on the existing ones. Roessler et al. [48] developed a standard calibration procedure for the DEM parameters of cohesionless bulk materials using a modified drawdown test to solve the problem of ambiguity. Liu et al. [52] examined the effect of particle aspect ratio (shape) on granular flow in a cylindrical hopper, where ellipsoidal particles decreased the mixed region adjacent to the side-wall, increased the stagnant zone at the bottom corner of the hopper, decreased the pressure on the wall, and affected the discharge rate. They were able then to develop a modified description of the classic Beverloo equation. Zhou et al. [53] studied the influence of particle–particle and particle–wall coefficients of friction, i.e., static and sliding on the AoR of a sandpile test. They constructed a power-law

relationship between measured AoR and the input parameters and showed that all friction coefficients followed a positive dependency, whereas particle size followed a significant negative dependency.

Approaches that are more efficient are being recently following systematic data analysis such as Design of Experiments (DoE). Yan et al. [54] utilized a multilevel statistical analysis approach on hopper flow in DEM to study the influence of coefficients of friction and restitution, as well as Young's modulus. Using PCA, the empirical power-law approach was found to agree with a semi empirical approach, where it compared AoR to contact model parameter sensitivity matrices. The results in all cases showed that the coefficient of static friction was the critical parameter, which allows the gap to be bridged between bulk and interparticle interactions. In addition, it is shown that reducing the shear modulus (i.e., from 10^{11} to 10^7 Pa) proved to have an insignificant effect on the powder flowability using free-flowing material which was also proven by Lommen et al. [55]. Boukouvala et al. [56] used a reduced-order approach to assess the importance of DEM parameters where PCA was coupled with surrogate models mapping which identified key areas of sensitivity. Major principal components enabled a reduced and computationally more efficient model to be engaged for future simulations. El Kassem et al. [3] calibrated the AoR of a pharmaceutical powder by varying coefficients of rolling friction and restitution using DoE, while determining the coefficient of static friction using the FT4 powder rheometer. In an advanced study, El Kassem et al. [57] developed a systematic numerical approach, using DoE and multivariate regression analysis (MVRA), to efficiently study the influence of different DEM input parameters on various bulk responses. The statistical prediction model of the AoR and porosity, of free-flowing and cohesive powders, yielded a good quantitative agreement between DEM simulation and experimental data results. Souihi et al. [58] used DoE to provide a more efficient approach in investigating roller compaction. Orthogonal PLS regression was used to analyze the results of a reduced central composite face-centered design in DoE. Wilkinson et al. [59] added that such approach using regression analysis has not been used so far for DEM simulations, but is highly recommended for developing more universal models for particle flow in a more efficient way. He proved that the Y has a negative impact on flow energy under specific conditions using a cohesive material. In this work, the effect of Y with the combination of varying the particle size in the same design model is studied which was not covered by [54] and [59]. To add up, Johnstone [60] used DoE to perform different calibration and validation methods based on experimental measurements. Benvenuti et al. [61] used artificial neural network to identify DEM simulation parameters. However, Rackl et al. [62] used a methodical calibration approach based on Latin hypercube sampling and Kriging, where they showed that different parameter combinations might lead to similar results. Wei et al. [63] studied the effect of three particle shapes on the AoR and bottom porosity distribution of a heap formed by natural piling using DEM. They found that the static friction has the major impact on the AoR, followed by the porosity and then the particle shape. Moreover, Li et al. [64] investigated the influence of three factors, namely moisture content, rice velocity and depth of the rice layer on the porosity of the flowing rice layer based on mass conservation and the poroelasticity theory. These results showed that the latter three factors have a significant influence on

the porosity. Cheng et al. [65] developed a DEM calibration approach of granular soils based on the sequential quasi-Monte Carlo filter. They calibrated the micromechanical parameters of the contact laws against the stress–strain behavior of Toyoura sand in drained triaxial compression conditions at different confining pressures. Do et al. [66] developed two automated DEM calibration approaches using genetic and direct optimization algorithms.

Moreover, in addition to the calibration process, one of the existing challenges in DEM is the high computational cost needed to perform real industrial-scale simulations because of the extremely large number of particles being studied simultaneously at every time step. One of the most efficient approaches utilized to overcome this “obstacle” is to scale up the particle sizes and thus reduce the total number of particles in the system according to coarse-graining method [67–71]. The most challenging issue in following such a method lies in adapting the material properties in such a way that compensates for the enlargement of the particles. In this study, the coarse-graining scheme proposed by Bierwisch et al. [70] was applied. The basis of this scheme is the preservation of the kinetic and gravitational potential energy densities as well as the volume fraction of particles as the original system with unscaled particles.

Screw Feeding and DEM

DEM is applied to simulate particle systems in different industrial applications [72] as e.g., in connection with screw feeding process. In particular, DEM is used to model the behavior of particles flowing in screw feeders in several studies leading to a number of valuable findings [12,13,73]. Different responses obtained from DEM modeling of screw conveyor process showed a good agreement with analytical and experimental reference results [74]. These responses include mass flow rate, required screw power, and particle transfer velocity [13,75,76]. Pezo et al. [77] studied the influences of screw length and different screw design types on the performance of screw conveyor-mixer during material transport. In this, DEM modeling and experimental validation were performed, where it was shown that the modified screw conveyor geometry leads to better mixing quality.

Loss in Weight Feeding

Feeding of raw materials is generally the first unit operation in a continuous manufacturing line and it is essential towards the success of the complete manufacturing process [78]. In the literature, some studies showed that continuous manufacturing can significantly decrease production costs while ensuring product quality improvement [79]. Impaired product quality can be caused by inaccurate and inconsistent feeding of raw materials since the composition of the final product is determined by the feed rate of the individual raw materials [80]. Flow problems result in downtime, lost production hours, and more importantly low-quality drugs that may pose health risks to end-users, i.e., patients. High accurate feeding solutions for the growing demands of continuous [81,82] as well as batch pharmaceutical processing are required. Powder dosing can be performed in two different setups; gain in weight (GiW) and loss in

weight (LiW) dosing. In GiW application, the dosing is driven by the equipment receiving the powder, which must be placed on a load cell, whereas in LiW application, the dosing is controlled by the equipment that is dosing the powder [83]. LiW feeders can be operated in continuous dosing to deliver a constant mass flow for a continuous downstream process. The principle work of the LiW feeding is that the entire feeder, hopper, and material, are continuously weighted. As the feeder discharges material, the rate of decline in system weight is computed and feeder speed is adjusted to maintain the desired (set) rate. Therefore, when the material is fed through the feeder, the weight of the system decreases. In LiW feeders, a feedback control loop from the signal of the load cell to the speed of the motor is maintained. The rate of discharge of the product is continuously calculated over a very short interval of time to check if the throughput is lower or greater than expected. Accordingly, the signal gives the order to increase or decrease the dosing speed, respectively. A LiW feeder can operate in two modes, i.e., gravimetric or volumetric, where the gravimetric mode is characterized by the displacement of a constant mass per unit time, whereas the volumetric mode is characterized by the displacement of a constant material volume per unit time [25].

Machine Design – Proof of Concept

In general, designing a new machine can be carried out in an iteration procedure until specified desired needs are fulfilled or certain problems are solved [84]. The transition from an idea to conceptual hardware design is a challenging process, which entails numerous uncertainties. LiW feeders are commonly oriented in a horizontal way. In this work, a LiW feeder prototype is designed, constructed, and its performance is finally analyzed. In particular, special attention has been paid to solve main problems faced in the horizontal LiW feeders to achieve better powder dosing accuracy results. The developed vertical LiW feeder can be operated in both batch and continuous dosing processes. The flexible and modular design concept is essential to quickly deal with different materials as the operation and performance of dosing equipment can significantly be influenced by the flow characteristics and the bulk density of the newly considered material. Therefore, it is crucial to have a flexible powder container design, where the auger, the stirrer and the dosing tube are easily interchangeable and have easy access to all parts for cleaning and maintenance. For instance, inappropriate stirrer design can cause bridging of the powder inside a hopper or powder container and thus cause abnormally long discharge times.

One of the main challenges in getting high dosing accuracy results is selecting a robust, accurate, and reliable load cell. [85] presented several factors that affect the weighing accuracy and suggested recommendations to avoid their drawbacks. These factors include the functionality and the specifications of the load cell itself, load factors if applied as specified by the manufacturer, i.e., improperly applied loads or conditions, such as a twisting load, wind loading, shock loading, vibration, large temperature changes, high moisture, pressure differentials, stability of the mounting equipment, and electric noises i.e., radio frequency interference and electromechanical interferences. To eliminate the negative impacts of the factors on the weighing accuracy, it is crucial to use filters to remove any left or sudden uncontrolled

disturbances. Many studies investigated the filtration of noise in the load cell signal. [86] examined analytically and experimentally in time and frequency domains the noise in load cell signal in an automatic weighing system based on a belt conveyor to enhance signal quality. Using a noise reduction algorithm, they achieved better reading accuracy. Coperion K-Tron, as one of the leading companies that manufacture LiW feeders, employs digital filtration in its load cells to minimize the effects of shock and vibration transmitted to the feeder from the surrounding environment [87,88]. The digital filtering is highly effective in suppressing the effects of vibration throughout a full range of frequencies encountered in a typical plant environment. Moreover, the digital filtering permits accurate detection of even small vibrations and fluctuations in weight, where, on the other hand, conventional filtering techniques fail to discriminate between forces induced by vibration and actual weight changes [87]. In this study, it was essential to use digital filtering to get more accurate dosing results.

Research Questions

After reviewing the state of the art of auger dosing in literature, the core challenges of this thesis are concerned with three main research questions:

- i) ***RQ1: how to efficiently identify and build a systematic analysis approach that figures out critical material attributes and critical process parameters that have significant impact on the auger dosing performance?*** This research question mainly focuses on establishing a methodical technique that can effectively analyze a data set of experimental results and find possible relationships between the studied factors. The challenge is to find out an optimized approach by running a limited number of experiments that are sufficient to reach the goals of the underlying study.
- ii) ***RQ2: how to successfully develop a methodical experimental validation approach to calibrate a simulation model?*** This question relies first on identifying an appropriate numerical approach to simulate an auger dosing process. The main challenge is to build a systematic calibration approach that can validate the simulation outcomes with the experimental data results rather than following a trial-and-error method. Besides, it is important to identify which contact models and input parameters are the most important and adequate in this study.

- iii) ***RQ3: what are the particular design specifications and functionalities of an applicable dosing machine that can accurately dose certain ranges of pharmaceutical powders and how to experimentally validate a proof of concept prototype?*** This research question covers a wide range of sub-questions regarding identifying and constructing an industrial machine that fulfills certain requirements and features. Three related sub-questions, hence, i) what design process steps to follow? ii) how to build a flexible modular machine? and iii) how to prove the functionality of this prototype machine?

1.3 Goal & Overview

The goal of this work is to develop a new auger dosing machine to fulfill the research requirements and the validation of the numerical model. The flexible and modular design of the dosing machine can lead to dose various powders with high dosing accuracy. The developed multivariate data analysis can figure out the potential correlations and interactions between the studied factors and the bulk responses. Furthermore, the developed and validated numerical model based on the multivariate regression analysis reduces the simulation time and obtain a better knowledge of the affecting input factors on the dosing performance.

To begin with, in *Chapter 2*, the theoretical fundamentals of powder characterization of the material attributes are reviewed. In *Chapter 3*, a statistical methodology that identifies the correlations between critical material attributes, critical process parameters and critical quality attributes of a volumetric vertical micro-auger dosing process is developed. The numerical treatment of the particle flow is shown in *Chapter 4*. In this regard, the governing equations and the implemented particle interaction models are presented, followed by two main case studies covering a developed calibration and validation approach. Finally, a systematic design process approach for designing, building, and experimentally validating a proof of concept of a vertical LiW feeder prototype is defined in *Chapter 5*. A final conclusion and discussion are given in the last Chapter, including some statements on the conceivability of further developments and future potential.

2 Powder Characterization

In order to accurately describe the dosage of powders, it is necessary to analyze their bulk characteristics. Critical material attributes and critical process parameters need to be compatible to achieve high productivity and better-quality results during any powder dosing method, especially using auger feeders. Due to the variety of factors affecting the behavior of powders, it is very challenging to predict the process performance. Therefore, as a first step in correlating powder material attributes and process parameters in the underlying study, ten powders ranging between free-flowing and cohesive ones were experimentally characterized at Bosch's labs using a number of empirical methods such as, the shear cell in the FT4 powder rheometer from Freeman Technology Ltd (UK) [89].

2.1 Materials

Ten different powders spanning a range between free-flowing and cohesive ones were extensively characterized in the underlying study, where eight of them are pharmaceutical and the remaining two are natural products. All powders were sourced from commercial suppliers. In this, Cellulose Arbocel P 290, crystalline powder Emcompress (calcium hydrogen phosphate) beside both MCC Vivapur 12 and Vivapur 101 were purchased from JRS Pharma GmbH (Germany). Crystalline lactose monohydrate InhaLac 251 and SpheroLac 100 were purchased from Meggle (Germany). Lactose Glucidex IT 19 (Maltodextrin) and Maize Starch extra white powder were purchased from Roquette (France). Ascorbic acid (Elco P-100 K) was purchased from Muehlenchemie GmbH (Germany), whereas Magnesium Stearate (NutriMag ST-v) was purchased from Calmags GmbH (Germany).

2.2 Material Flow Attribute Measurements

Samples of each powder were subjected to comprehensive testing with a high degree of repeatability (Relative Standard Deviation, $RSD < 5\%$) in each case. All the standard powder flow tests were performed in triplicate for all samples. Table 2.1 summarizes the 30 material attributes and their characterization techniques that were considered to characterize the ten different powders used in the experiments.

Categories	Characterization technique	Material attribute descriptor	Abbreviation	Unit
Particle size characterization	Laser diffraction	10 %, 50 % and 90 % cumulative undersize of the volumetric particle size distribution	d10, d50, d90	μm

Bulk properties	Camsizer XT	The span of the volumetric particle size distribution	Span	-
		Aspect ratio	ASR	-
		Roundness	Ro	-
		Irregularity factor	I	-
	Jolting volumeter	Bulk density	BD	g/cm ³
		Tapped density	TD	g/cm ³
		Porosity	ϵ	%
	Helium pycnometer	True density	ρ_{true}	g/cm ³
	FT4 powder rheometer	Compressibility at 15 kPa	CPS	%
		Permeability at 15 kPa	PR	cm ²
Flowability indicators	Halogen moisture analyzer	Moisture content	MC	%
	Jolting volumeter	Carr's index	CI	%
	Static funnel flow	Angle of repose	AoR	°
Shear properties	FT4 powder rheometer	Cohesion at 6 kPa	C	kPa
		Unconfined yield strength at 6 kPa	UYS	kPa
		Major principal stress at 6 kPa	MPS	kPa
		Flowing function coefficient at 6 kPa	ff _c	-
		Angle of internal friction at 6 kPa	AIF	°
		Wall friction angle at 6 kPa	WFA	°
Dynamic properties	FT4 powder rheometer	Basic flowability energy	BFE	mJ
		Normalized basic flowability energy	nBFE	mJ/g
		Stability index	SI	-
		Flow rate index	FRI	-
		Specific energy	SE	mJ/g
		Aeration energy	AE	mJ
		Aeration ratio	AR	-
		Normalized aeration sensitivity	NAS	s/mm

Table 2.1: Summary of the applied quantitative characterization techniques, the corresponding material attributes descriptor and the used abbreviations.

2.2.1 Particle Size and Shape Characterization

HELOS/BR in combination with RODOS/M and VIBRI from SympaTec GmbH (Germany) was used to measure particle size distribution (PSD). The frequency of particles in a defined quantity is determined by a laser diffraction measurement. It finds out the particle size distribution at x10, x50 and x90, which are the cumulative undersize of volumetric PSD. Based on this, the span of volumetric PSD is then computed. In general, as the particle sizes are very small, the powder is very cohesive and it resists flow under gravity [90].

Camsizer XT from Retsch Technology GmbH (Germany) was used to measure the shape of particles via dynamic image analysis. The particles are described in two-dimensional images in terms of the minimum and maximum Feret diameters (Fe_{min} , Fe_{max}), i.e., Fe_{min} is the length of the minor axis and Fe_{max} is the length of the major axis. The ratio of Fe_{min} to Fe_{max} , called in the following the aspect ratio (ASR), has a value between 0 and 1. It reflects the elongation of a particle and deviation from a sphere, where small values correspond to elongated particles and higher values correspond to spherical particles:

$$ASR = \frac{Fe_{min}}{Fe_{max}}. \quad (2.1)$$

According to Li et al. [91], a shape factor called circularity or roundness (Ro) is an additional factor that indicates to what extent a particle has a spherical shape. It is based on the projected area (A) of the particle and the overall perimeter of the projection (P) according to the following equation:

$$Ro = \frac{4 \times \pi \times A}{P^2}. \quad (2.2)$$

According to Podczek et al. [92], irregularity factor (I) is another descriptor, which is the ratio of the projected perimeter P of the particle to the length of the major axis Fe_{max} . It illustrates that lower values of I indicate elongated shapes, whereas higher values indicate irregularities.

2.2.2 Density and Porosity

The bulk density (BD) and tapped density (TD) were analyzed via the jolting volumeter (Jel STAV II) from J. Engelsmann AG (Germany). The bulk density represents an average density of the powder sample, where it depends on both the density of the particles and their spatial arrangement. The volume of a bulk solid depends on the size and the shape of the particles. A certain mass of powder was filled into a graduated cylinder (of 100 ml capacity) and the volume and mass were then recorded. On the other hand, the tapped density was attained after mechanically compacting the powder sample by up to 1250 taps.

The true density (ρ_{true}) was estimated using the Ultrapyc 1200e, Automatic Density Analyzer from Quantachrome Instruments (USA). It is a pycnometer that measures the true density of powder samples using an inert gas (helium) to measure its volume by applying Archimedes' principle of displacement

and the technique of gas expansion (Boyle's law). Consequently, the powder bed porosity (ε) was calculated as described in the following equation:

$$\varepsilon = \left(1 - \frac{BD}{\rho_{\text{true}}}\right) \times 100. \quad (2.3)$$

2.2.3 Compressibility

Compressibility (CPS) is a bulk property measurement that measures the volume change of powder samples after consolidation stress is being applied [93]. Cohesive powders consisting of smaller particles are more compressible than that consisting of granular particles. Compressibility was measured via the FT4 powder rheometer, where the powder was first conditioned by a helical blade that moves gently downwards and upwards, with a tip speed of 60 mm/s, in order to create a homogeneous and reproducible packing state. Then, a vented piston is used to compress the sample powder leading the trapped air to escape under increasing normal stress. During the test, the pressure starts at 1 kPa and increases step wisely until 15 kPa, and simultaneously, it calculates the ratio between the density and bulk density for each compaction step that takes place.

2.2.4 Air Permeability

Air permeability (PR) is a measurement of the ability of air to flow through the porous medium. It is measured by air pressure drop under 15 kPa stress across a powder bed [16,94]. It is measured using the FT4 powder rheometer, where a vented piston compresses the powder under increasing normal stress from 1 kPa to 15 kPa, whilst at the same time, a constant flow rate of air is passed [94]. For powders, PR is influenced by many physical properties such as particle size distribution, cohesivity, particle stiffness, shape, surface texture, and bulk density. Permeability is highly influenced by several external factors such as consolidation stress, where they alter the porosity and particle contact surface areas, making it more difficult for the air to pass through the bulk [95]. In general, cohesive powders consisting of very small particle sizes are less permeable, whereas powders of a granular nature are typically more permeable [95]. More details on this and other methodologies used in the FT4 rheometer can be found in the literature [93,96].

2.2.5 Moisture Content

The moisture content (MC) interprets important characteristics such as cohesion and agglomeration tendency. Depending on the percent of water and material characteristics, MC might enhance or obstruct the flowability of the powder. The MC is determined by the halogen drying method using the Halogen Moisture Analyzer type HR73 from Mettler Toledo (Germany). A sample amount of 5 g is dried at 105 °C by a halogen lamp, and thus the moisture is extracted from the powder. Due to this drying process, the bulk loses some weight, and hence the moisture loss is calculated from the difference between

the initial and final weights. This moisture loss corresponds to the moisture content of the powder in the initial state.

2.2.6 Powder Flow Properties

2.2.6.1 Carr's Index

Based on the BD and TD, the powder flowability of a bulk solid could be characterized by Carr's index (CI) as a compressibility index [1,97], where larger values of CI indicate poor flowability [98]. It is a measurement of the relative difference between tapped volume and untapped volume of the powder [2] and can be expressed as

$$CI = \left(\frac{TD-BD}{TD} \right) \times 100 \% . \quad (2.4)$$

In this, powders with $CI < 15 \%$ have good flowability, whereas a poor flowability corresponds to $CI > 25 \%$.

2.2.6.2 Angle of Repose

The angle of repose (AoR) is defined as the slope of a poured conical pile of loose uncompacted bulk solid material [1]. It is a characteristic related to the cohesiveness of the powder [2] and used as a common method to characterize powders according to their flowability [99–103]. Inter-particulate forces are high in cohesive powders, where powders are characterized by low flowability and higher AoR values. In this, an AoR less than 30° indicates good flowability, while AoR greater than 40° indicates bad flowability [104].

2.2.7 Shear Properties

Shear flow properties include the parameters flow function and unconfined yield strength, which are related to the cohesivity of a powder stored under consolidation and are widely used to design powder hoppers [1,105,106]. Shear cell and wall friction analysis were performed using the FT4 powder rheometer to characterize the static flow properties of powder materials in a consolidated state (at 6 kPa) [96] and to determine how difficult it is for a powder to begin to flow. Powders are always subjected to some levels of consolidation stress during their treatment, which causes changes in density and mechanical inter-particulate forces. For flow to occur, it is necessary to overcome the yielding point of the powder. Measuring shear properties will provide important information about powder flow, i.e., whether it will happen through the process or whether bridging, blockages or stoppages are likely to occur. Therefore, the shear test, as a static test, measures the powder behavior when moving from no-flow to flow [96].

2.2.7.1 Cohesion and Flow Function Coefficient

The testing procedure consists of four steps, i.e., conditioning, consolidation, pre-shearing and shearing [96,107]. In consolidation, a vented piston moves downwards applying initial consolidation stress to the powder in the split vessel of inner diameter 25 mm and total capacity of 20 ml. Then, a shear head moves downwards inserting its blades into the powder and inducing normal stress. The shear head continues to move downwards until the required normal stress is established. Then, slow rotation of the shear head begins, inducing shear stress. As the powder bed resists the rotation of the shear head, the shear stress increases until the powder bed fails, at which maximum stress is observed. This maximum shear stress represents the Yield Point [96].

A commonly used approach is the Mohr-Coulomb model that fits the Mohr stress circles (Figure 2.1) to the yield locus, which in turn identifies the major principal stress (MPS) or consolidation stress (σ_1) and unconfined yield strength (UYS) or compressive strength (σ_c) [1,106]. The flowability of powder is classified by the flow function coefficient ff_c , developed by Jenike, which is defined as the ratio between the MPS and the UYS [96]. In addition, the cohesion (C) value is identified by the shear cell test, where it is the shear stress at which the normal stress is equal to zero for the best-fit of the shear and applied normal stress plot [108]. Generally, a cohesive powder will have higher values for cohesion and UYS and consequently a low ff_c . The following equation describes the ff_c [1] as

$$ff_c = \frac{MPS}{UYS}. \quad (2.5)$$

Powders with ff_c values below 4 denote poor flow and values above 10 denote good flow [1,106].

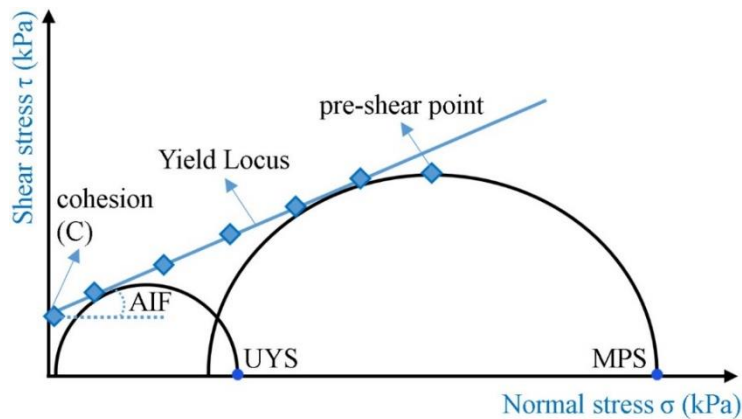


Figure 2.1: The schematic of yield locus and pre-shear point. Mohr circle analysis was used to derive cohesion (C), unconfined yield strength (UYS) and major principal stress (MPS).

2.2.7.2 Static Friction

The angle of internal friction (AIF) between particles is computed based on the linearized yield locus of the powder. It is the angle between the axis of normal stress (abscissa) and the tangent to the yield locus

[96]. It measures the shear stress (τ) for various normal stresses (σ) to describe the magnitude of the shear stress that a powder can sustain [96].

2.2.7.3 Wall Friction Angle

The wall friction test provides a measurement of the sliding resistance between the powder and the surface of the process equipment [1,109]. This test is important for understanding the discharge behavior from a container. The measurement principle of a wall friction test is very similar to the shear cell test, where in this case, the powder is sheared against a material resembling process equipment wall. The used process equipment wall is a rounded stainless steel 316L disc with surface roughness (Ra) 0.05 μm . The data from the test is represented as a plot of shear stress against normal stress, allowing the determination of the wall friction angle (WFA). It is the arctan of the ratio of the wall shear stress to the wall-normal stress [96]. The greater the wall friction angle, the higher the resistance between the powder and the wall.

2.2.8 Stability and Variable Flow Rate

As a necessity to measure powder flowability under conditions that closely represent the process environment, dynamic test methodologies using the FT4 powder rheometer were introduced [99,110]. Measuring the torque and force acting on a helical blade as it rotates through a powder sample along a defined path helps in determining the dynamic powder properties [96]. As the blade moves on the downward and upward traverse, the energy consumed to induce the powder to flow is measured. After a conditioning step, a testing cycle is repeated seven times with measured flow energies E_1 - E_7 . Many related flow behaviors in different process steps, such as auger dosing, can be correlated to such comprehensible measurement principle.

2.2.8.1 Basic Flowability Energy

Basic flowability energy (BFE) is a measure of the flow properties of a powder when the powder is in a loosely packed state (following conditioning). It is measured on a downward movement of the blade, which forces the powder against the base of the test vessel [96]. It reflects how a powder will flow under specific consolidation conditions [110,111]. It is the energy needed to move the blade through a conditioned powder sample from the top to the bottom of the vessel [108]. The BFE does not account for the density of the material; therefore, the normalized basic flowability energy (nBFE) is commonly used [7],

$$\text{BFE (mJ)} = E_7, \quad (2.6)$$

$$\text{nBFE } \left(\frac{\text{mJ}}{\text{g}} \right) = \frac{\text{BFE (mJ)}}{\text{Sample mass (g)}}, \quad (2.7)$$

where the sample mass is the amount of powder in the sampling measuring vessel.

2.2.8.2 Specific Energy

In contrast to BFE, specific energy (SE) is the energy needed to displace conditioned powder as the blade travels upwards to produce shear [96]. This parameter reflects more the behavior in a low stress, unconfined environment. $SE < 5$ corresponds to low cohesion powder, between 5 and 10 moderate cohesion, and $SE > 10$ represents cohesion powder.

2.2.8.3 Stability Index

The stability test is designed to find out if a powder is going to change after being made to flow over time. Therefore, any changes in the flow energy during the test describe how unstable the powder is and the direction of change (increase or decrease in energy) indicates the possible causes. The stability index is calculated by dividing the total energy of test number 7 to the total energy of test number 1 performed by the same blade tip speed (100 mm/s):

$$SI = \frac{E_7}{E_1}. \quad (2.8)$$

A SI value of 1 indicates that the material is stable and non-friable. If SI is larger than 1, it indicates that the material requires more energy to flow over time, possibly due to de-aeration, agglomeration, segregation, moisture uptake or electrostatic charges. If SI is smaller than 1, then it is an indication of attrition or de-agglomeration [96].

2.2.8.4 Flow Rate Index

An important parameter when describing the flow properties of a powder is its sensitivity to changing the flow rate. Generally, cohesive powders are more sensitive to changes in flow rate than non-cohesive or granular materials, mainly because of the high air content in the cohesive materials. In this test, four repetitions were performed with different shear rates (different blade tip speeds ranging from 10 to 100 mm/s) after measuring the BFE and SI. For this, the flow rate index (FRI) was employed, which is defined as the ratio between the required flow energy from Energy Test 11 with tip speed 10 mm/s and Energy Test 8 with tip speed 100 mm/s:

$$FRI = \frac{E_{11}}{E_8}. \quad (2.9)$$

where E_{11} is the basic flow energy at a blade tip speed of 10 mm/s and E_8 is the basic flow energy at 100 mm/s. FRI values are typically higher ($FRI > 3$) with cohesive materials because they are more sensitive to changes in flow rate [93].

2.2.9 Aeration

2.2.9.1 Aeration Energy

In powders, interaction of particles is influenced by the void ratio (here air content), which consequently influences their flow properties. The measured resistance to flow, aeration energy (AE), quantifies the strength of the cohesive forces [112,113]. High AE represents a cohesive powder, while low AE represents a free-flowing powder as the powder fluidizes. The aeration test was done using the FT4 powder rheometer, where initially the flow energy (AE 0) was measured without aeration. In this, air in-flow is applied at the base of the powder column with an increasing air velocity (AV) until reaching 10 mm/s. This allows quantifying the changes in the flow properties by measuring the reduction in the flow energy (AE AV). The transitioning from a densely-packed powder to a fluidized powder bed caused the reduction in flow energy [96]. This is quantified by normalizing the aerated flow energies with the initial flow energy (AN AV). Consequently, the maximum normalized aeration sensitivity (NAS) is calculated as the difference in normalized flow energy divided by the difference in air velocity.

$$AN\ AV = \frac{AE\ AV}{AE\ 0}. \quad (2.10)$$

2.2.9.2 Aeration Ratio

In the aeration test, the aeration ratio is calculated as the ratio between energy at air velocity 0 and the energy at air velocity n, i.e.,

$$AR = \frac{\text{Energy (Air Velocity 0)}}{\text{Energy (Air Velocity n)}}. \quad (2.11)$$

If AR is equal to 1, it indicates that a material is not sensitive to aeration, and thus it may represent a very cohesive powder. If AR is between 2 and 20, it indicates that a material is to some extent sensitive to aeration. If AR is much greater than 20, it indicates that the material is very sensitive to aeration and probably become fluidized due to the air flow. In general, the larger the AR and the lower the AE, the less cohesive is the powder [96].

2.3 Material Attributes Results

Particle size and shape characterization of the ten powder materials are listed in Table 2.2. Emcompress, Glucidex IT 19, SpheroLac 100 and Vivapur 12 have relatively big particle sizes compared to the other powders, whereas, for instance, Magnesium Stearate and Maize Starch extra white have very small particle sizes. Arbocel P 290, Vivapur 12 and Vivapur 101 have fiber shaped particles, while the other powders have particles close to the spherical shapes with some irregularities, such as Magnesium Stearate and Maize Starch extra white.

Material	d10 (μm)	d50 (μm)	d90 (μm)	Span	ASR	Ro	I
Arbocel P 290	26.07	62.66	125.27	1.58	0.597	0.722	0.319
Ascorbic acid	9.48	31.95	77.91	2.14	0.704	0.852	0.428
Emcompress	63.72	197.4	302.16	1.64	0.756	0.851	0.197
Glucidex IT 19	14.39	167.43	341.14	1.95	0.697	0.849	0.219
InhaLac 251	9.61	43.88	86.65	1.76	0.726	0.878	0.384
Magnesium Stearate	0.69	2.46	12.08	4.68	0.797	0.845	0.855
Maize Starch extra white	7.86	14.26	21.05	0.93	0.866	0.86	0.762
SpheroLac 100	53.78	123.8	197.57	1.16	0.746	0.891	0.266
Vivapur 12	38.96	161.74	323.81	1.76	0.681	0.800	0.194
Vivapur 101	14.3	46.01	109.13	2.06	0.622	0.787	0.332

Table 2.2: Particle size and shape characterization properties. The abbreviations of the material attributes are described in the material flow attribute measurements section.

Bulk material attributes of the ten powders are listed in Table 2.3. The ten powders have different density values, and thus porosities and Carr's indices. The wide ranges of CI and AoR values express different flowabilities that characterize the powders between free-flowing and cohesive ones, where, for instance, Emcompress and SpheroLac 100 represent free-flowing powders by having low values of CI and AoR. In addition, the tested materials showed dissimilar values of compressibility and permeability values, where cohesive powders are more compressible and less permeable like, Ascorbic acid, Magnesium Stearate, and Maize Starch extra white.

Material	BD (g/cm^3)	TD (g/cm^3)	ρ_{true} (g/cm^3)	ε (%)	CPS (%)	PR (cm^2)	MC (%)	CI (%)	AoR ($^\circ$)
Arbocel P 290	0.319	0.486	1.543	79.33	22.4	0.55	5.05	34.33	42.4
Ascorbic acid	0.523	0.767	1.694	69.12	43.55	10.4	0.83	31.89	52
Emcompress	0.925	1.09	2.307	59.9	2.91	0.16	1.7	15.27	30.7
Glucidex IT 19	0.374	0.481	1.379	72.88	5.65	0.12	3.3	22.33	35.2
InhaLac 251	0.626	0.89	1.539	59.34	21.76	3.69	0.08	29.69	40.2
Magnesium Stearate	0.265	0.378	1.098	75.87	39.18	94.25	4.56	30	47
Maize Starch extra white	0.543	0.714	1.494	63.64	25.31	8.09	10.55	24	50
SpheroLac 100	0.714	0.85	1.536	53.52	6.3	0.61	0.08	16	31.8
Vivapur 12	0.365	0.472	1.552	76.48	10.76	0.29	2.68	22.67	34.7
Vivapur 101	0.361	0.479	1.553	76.75	17.53	1.21	2.75	24.67	44.4

Table 2.3: Bulk and flow powder properties. The abbreviations of the material attributes are described in the material flow attribute measurements section.

Figure 2.2 compares the results from the shear cell and wall friction angle tests at an initial consolidation stress of 6 kPa. The high values of the ff_c of Emcompress, SpheroLac 100 and Vivapur 12 indicate the good flowability of these powders.

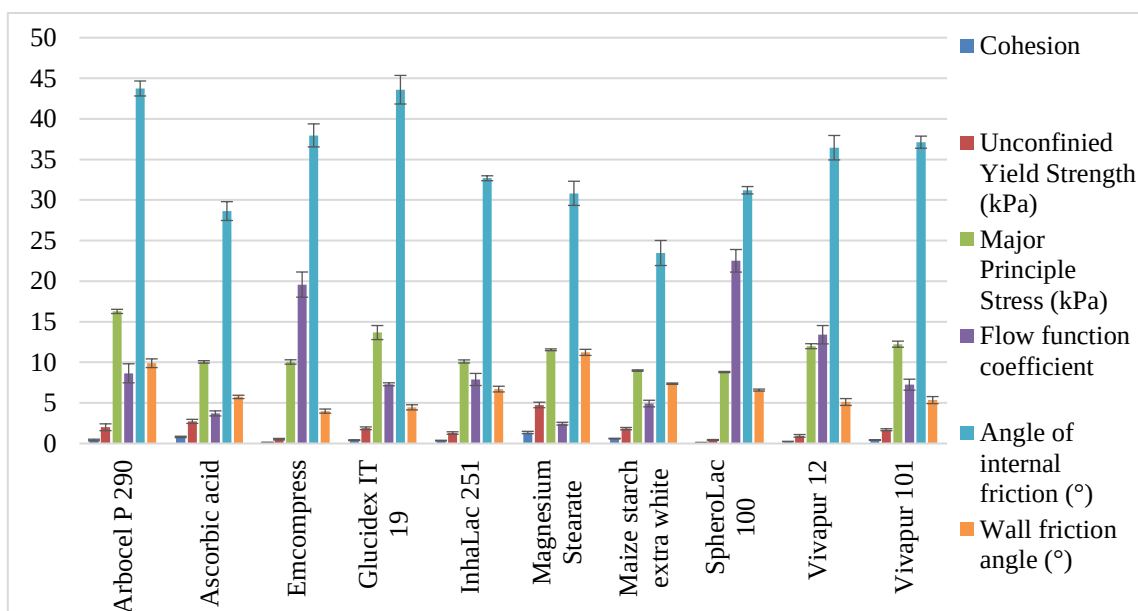


Figure 2.2: Material characterization results from shear cell and wall friction tests at an initial consolidation stress of 6 kPa.

Figure 2.3 compares the results from the stability and variable flow rate and aeration tests of the ten materials. The results indicate big differences between the materials, and thus their flowabilities.

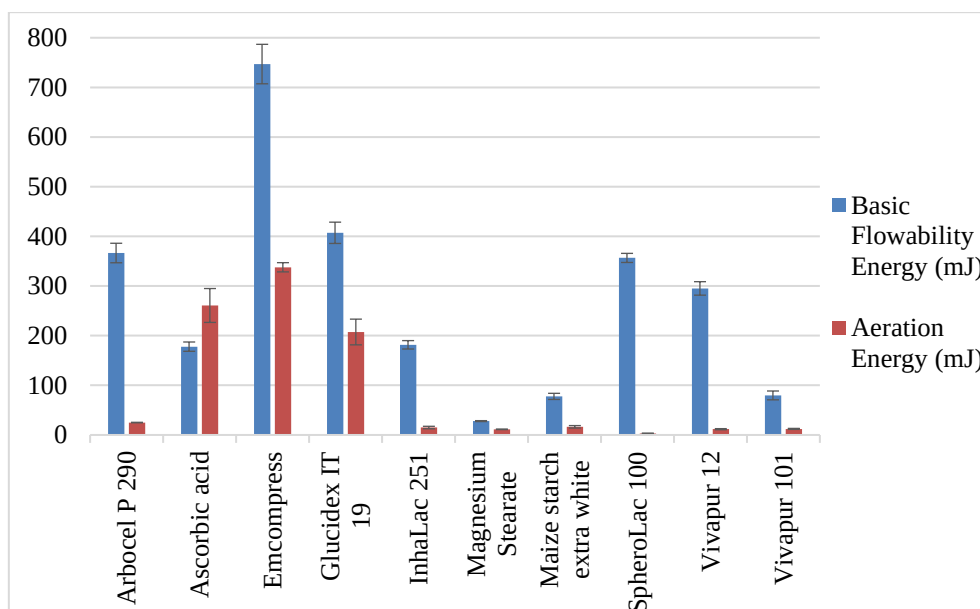


Figure 2.3: Material characterization results from stability and variable flow rate and aeration tests, part 1.

Figure 2.4 shows the second part of the results from the stability and variable flow rate and aeration tests of the ten materials. The values of the nBFE and the AR vary largely between the ten studied materials.

The smaller values of AR indicate that the powder is more cohesive, but this is not a necessity to reflect this material attribute by itself without considering other attributes. For instance, Emcompress has an AR value of 1.2 and ff_c was 19.58, which seems contradicting. This simply means that one material attribute has more impact and weight than the other does, where the auger dosing tests revealed how good the flowability of Emcompress was. So, it is always recommended to consider a bigger amount of material attributes together in an analysis to feasibly understand more the flowability behavior of a material.

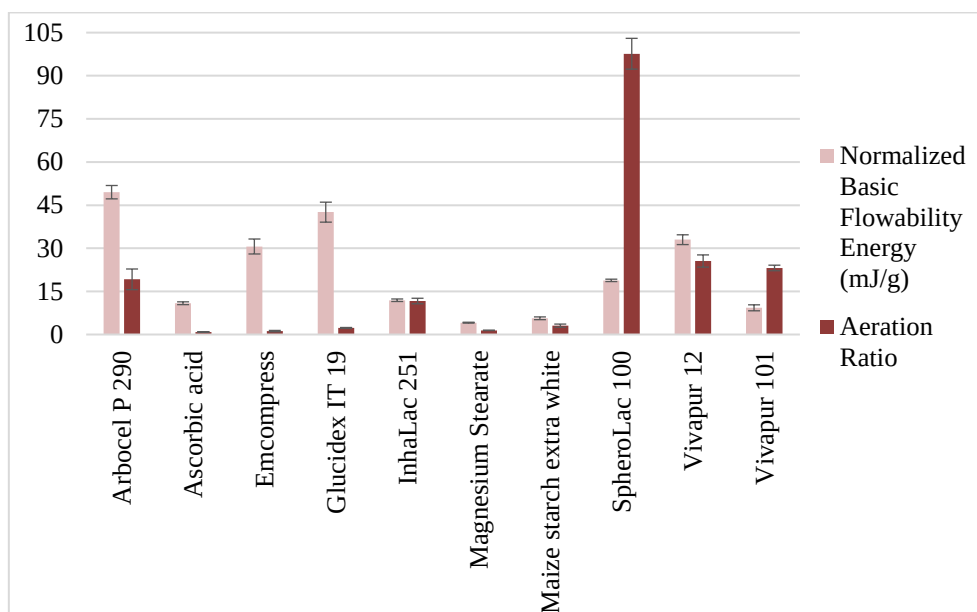


Figure 2.4: Material characterization results from stability and variable flow rate and aeration tests, part 2.

Besides the two previous graphs, Figure 2.5 shows the third part of the results from the stability and variable flow rate and aeration tests. The values of the SI, FRI, SE and NAS vary differently between the ten powders, where some powders represent similar values of material attributes.

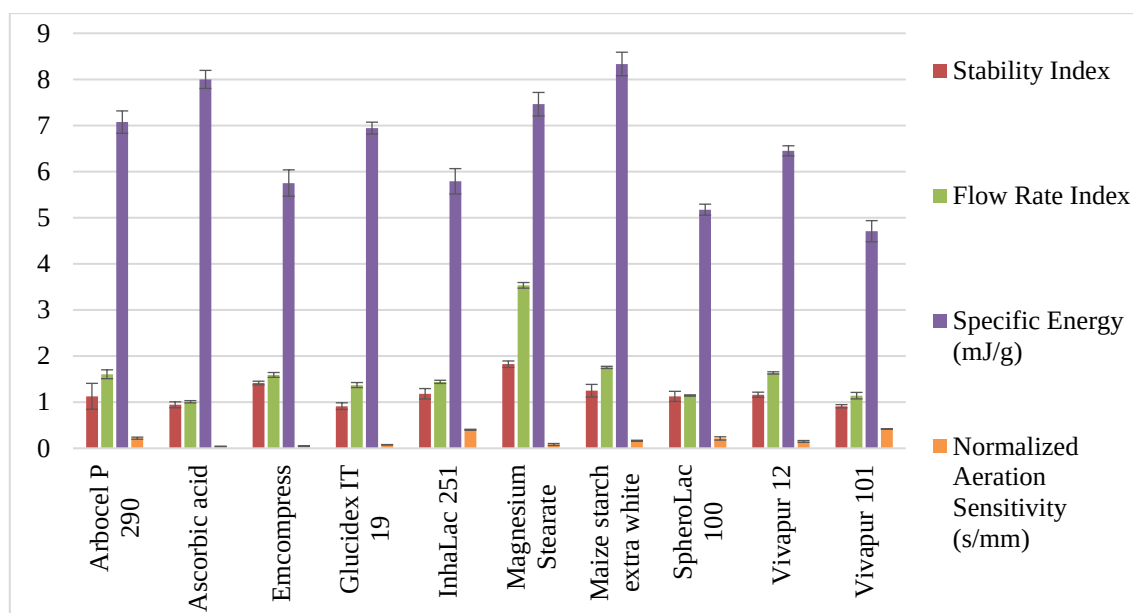


Figure 2.5: Material characterization results from stability and variable flow rate and aeration tests, part 3.

As it is evident, the ten powders are differently characterized. Some powders have in common similar material attribute values, but this does not mean that they will behave similarly during a given dosing process, which will be thoroughly discussed in the coming chapter.

3 Correlations between the Attributes

A systematic approach that captures the variability in the powder material attributes and in the process parameters in context with auger dosing behavior is developed. This is an essential step to improve the quality of drug products and accelerate their process development. For the aim of understanding the flow behavior of the studied powders in the scope of volumetric feeding experiments based on a vertical micro-auger dosing system, 30 material attributes for each powder were determined. For the dosing experiments of the ten diverse powders, three distinct auger geometries running at three different constant rotational speeds were considered as process parameters in the DoE. The powders were dosed continuously into a catching container, placed on a laboratory weighing balance until there was no more powder discharge from the container. Four CQA related to auger feeder performance are studied. The four CQA are the feed rate, the relative standard deviation that quantifies the consistency of the feed rate, percentage of dosed mass and the dosing time for each test combination. Therefore, PCA, as a data dimensionality reduction method, is used to reveal the correlations between the material attributes, process parameters, and the performance of the feeder. Regression analysis is then followed to figure out the critical factors that impact a given auger dosing application [114]. Finding and understanding the correlations allows building an optimized process for the aim of achieving better dosing quality. Figure 3.1 summarizes the workflow of the analysis approach followed in this study.

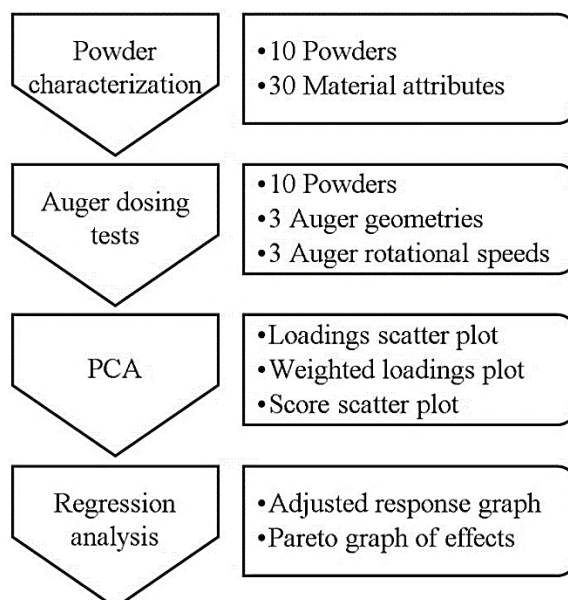


Figure 3.1: The workflow of the proposed correlation analysis approach.

3.1 Mass Flow Rate Experimental Setup

An auger powder container is used in running the mass flow rate tests of the ten studied powders. The container consists of a cover, a funnel, an opaque cylindrical glass tube, a shaft connected to a rotational motor, a stirrer, and an auger. The inner diameter of the powder container is 64 mm with a total height of 224 mm. Two air vents were added to compensate the negative pressure (vacuum) inside the container during the dosing process. In the case of a closed container, a negative pressure during powder discharge evolves, especially when dosing very fine powders where the permeability of air is low. The absence of air exchange causes a vacuum, which negatively affects the powder flowability. Three augers were used in conducting the experiments, where they have an overall length of 58 mm. Table 3.1 summarizes the geometry dimensions of the three tested augers.

Auger	Outer diameter (mm)	Inner diameter (mm)	Pitch (mm)	Auger filling volume (mm ³ /rev)	Geometry form
A	5	3.1	3	25.52	Straight
B	5.9	2.2	3.5	58.59	Straight
C	5.8	3	4.5	67.33	Conical

Table 3.1: Augers used in auger dosing experiments.

Figure 3.2 represents the geometry of the three augers with their auger filling volume per one revolution. Moreover, as an illustration example, Figure 3.2 shows the inner and outer diameters and the pitch of auger type A besides the geometries of the other two augers.

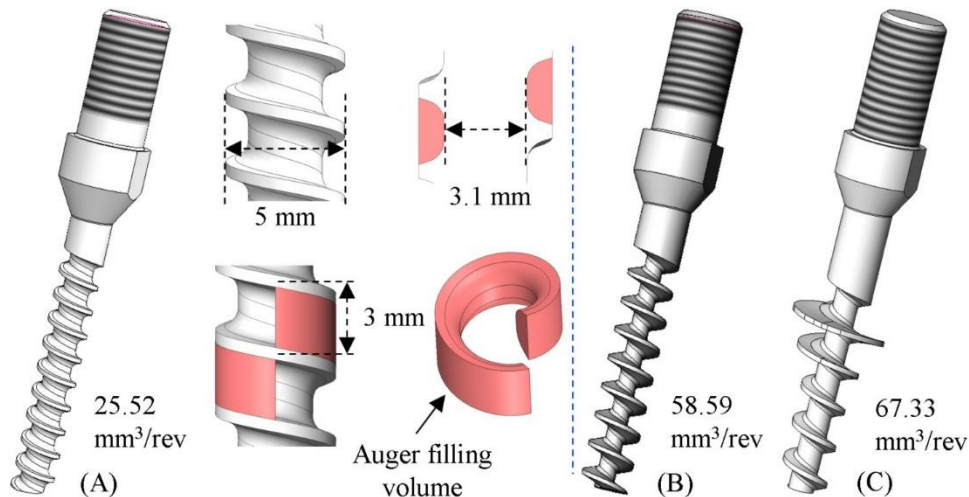


Figure 3.2: CAD models of the three augers used in the tests (A) 25.52 mm³/rev, (B) 58.59 mm³/rev, and (C) 67.33 mm³/rev.

The stirrer geometry is modeled in Figure 3.3. It has a total width of 48 mm and a total height of 37 mm. The two wings are twisted with an angle of 25°, where the inclination angle of the long wing is 45° and

that of the short wing is 57° . The advantage of using such design is the mechanism that conveys the powder downwards towards the center of the auger, and thus helping in maximizing the mass output.

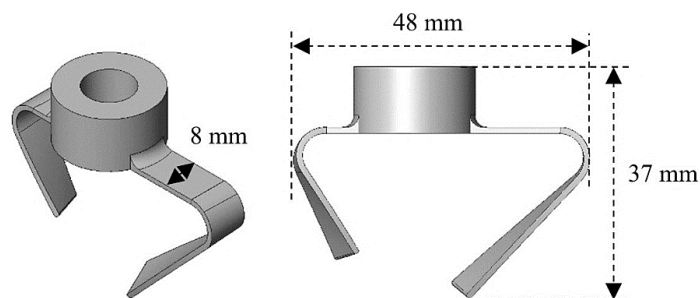


Figure 3.3: CAD model of the stirrer used in all the auger dosing tests.

As an example, Figure 3.4 represents the assembly of auger type A and the stirrer inside the powder container. It displays how the auger and the stirrer are coupled on one common shaft in the powder container, and thus they both run with the same rotational speed. The shaft is connected directly to the drive axis of an overhead stirrer rotating motor RZR 2102 control from Heidolph Instruments (Germany).

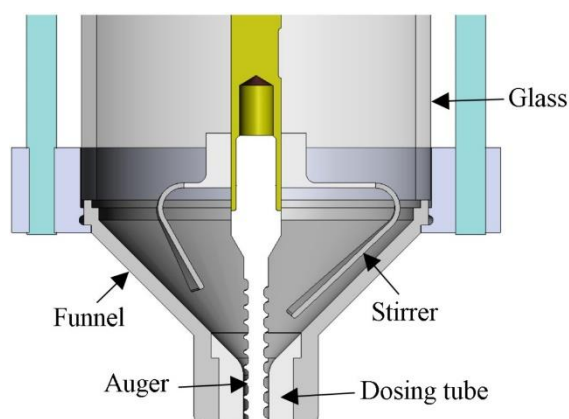


Figure 3.4: CAD cross-sectional view showing the assembly of the auger and the stirrer inside the powder container.

A laboratory weighing balance XPE 204, used as a catch scale, was positioned under the outlet of the powder container to record the powder mass flow every 0.5 s, where the dosed powder was collected in a transparent acrylic catching container. The XPE 204 was allocated on a Regupol sound 17 isolation pad to avoid transmission of any external vibration into it. The XPE 204 has readability down to 0.1 mg and a capacity up to 220 g. HTerm software was used on a computer to record the values of weight (in g) over time. Moreover, Static Line LC ionization device was installed to neutralize electrostatic charges. The entire experimental setup is shown in Figure 3.5.

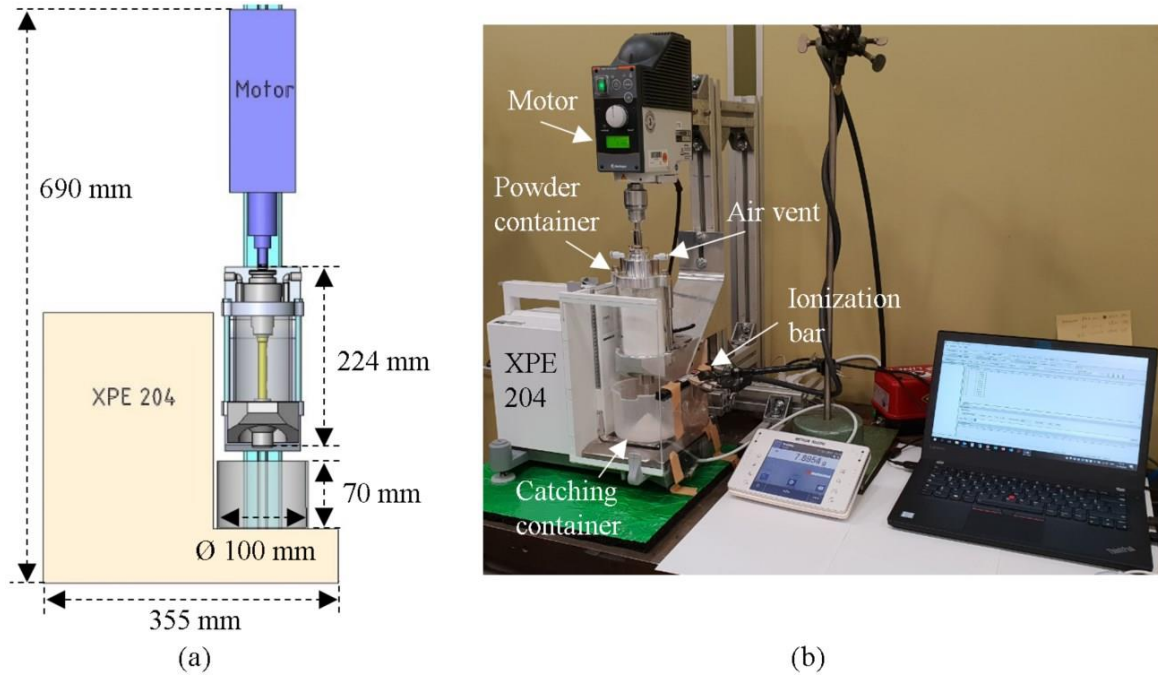


Figure 3.5: Auger dosing experimental setup in (a) CAD front view and (b) laboratory equipment setting.

Mass flow tests were executed with the target of emptying the powder container. The powder materials were stored in dry and unconsolidated conditions in aluminum foils. They were conditioned by gently shaking the foil in different directions and then loosely filling the material inside the powder dosing container. The tests were conducted at temperature around 22 °C and at relative humidity of around 38 %. The quantity of dosed mass and the dosing time (DT) were recorded for every single test. The mass flow rate or the volumetric feed rate ($\dot{m}$) is calculated using the difference in mass (Δm) obtained from the weighing balance divided by the difference in time between two consecutive weighing balance measurements (Δt):

$$\text{Feed rate } \dot{m} \left(\frac{\text{mg}}{\text{s}} \right) = \frac{\Delta m \text{ (g)}}{\Delta t \text{ (s)}} \times 1000, \quad (3.1)$$

with 1000 the conversion factor to convert g to mg. Then, the relative standard deviation of the feed rate (RSD) is calculated from the standard deviation of the feed rate (StD) and the mean value of the feed rate ($\bar{\dot{m}}$):

$$\text{StD (mg/s)} = \sqrt{\frac{\sum_{i=1}^n (\dot{m}_i - \bar{\dot{m}})^2}{n}}, \quad (3.2)$$

$$\text{RSD (\%)} = \frac{\text{StD}}{\bar{\dot{m}}} \times 100, \quad (3.3)$$

where n is the number of time points of an experiment. The RSD quantifies the consistency of the feed rate.

As the considered ten tested powders have different bulk densities, the net dosed mass out of the powder container was normalized for the maximum filled powder inside the container for each powder. It is then expressed as the percentage of dosed mass (PoDM):

$$\text{PoDM (\%)} = \frac{\text{Dosed Mass (g)}}{\rho_{\text{bulk}} \left(\frac{\text{g}}{\text{mm}^3} \right) \times V_{\text{container}} (\text{mm}^3)} \times 100, \quad (3.4)$$

where ρ_{powder} is the density of the filled powder, and $V_{\text{container}}$ is the volume of the container filled with powder. This response factor serves in building better comparisons between the feed rates profiles (graphs) of the ten studied powders.

The start-up time for the data collection and evaluation started just after few seconds of turning on the motor to ensure that the auger filling volume was full with powder. The dosing runs were stopped when no more mass flow took place. In some cases, due to a very low amount of discharge at the end of the test, high fluctuations of mass flow occurred and thus truncation of these data took place. These data are considered outliers (experimental errors) and can cause serious problems in the statistical analyses.

3.2 Design of Experiments (DoE)

Design of experiments (DoE) is a systematic methodology of planning experiments employed to figure out the cause-effect relationship between factors affecting a process and the response of that process [115]. It can help in identifying the best combinations of parameters that yield to the desired results and thus optimizing the output with minimum expended time and resources [60]. In this study, the DoE of the auger dosing experiments contains, besides the powders, two process parameters, namely the auger geometry and the auger rotational speed. Therefore, the ten powders were tested using the three previously mentioned augers running at three different rotational speeds (400, 600 and 800 rpm). Quadratic D-optimal design model was used to obtain the largest amount of information from the smallest number of experiments using Cornerstone 6.1.1.1 software from camLine GmbH (Germany). In that, a completely replicated factorial design can be saved. The mathematics behind D-optimal design is closely related to the Least Squares method [116] to find the concrete model equation on the basis of the design and the response data. The values in Cornerstone are, by default, normalized, i.e., the original variables are scaled to have a mean value of 0 and a variance of 1. This is performed since the investigated attributes have different measurement units.

3.3 Multivariate Data Analysis (MVDA)

Multivariate data analysis (MVDA) is a statistical technique that is used to identify the critical material attributes by extracting relevant information from the measured data [117]. It is implemented to identify what factors influence the dosing accuracy and build possible correlations between different critical material attributes (CMA), critical process parameters (CPP) and the CQAs of the dosed powders. Since

the process of identifying the relationships between all the factors interfering in the dosing system is complex and not easily understood, the process is simply considered a black box. In many cases, various measured output responses occur without fully understanding the influences and interactions that took place between the studied factors. Therefore, DoE and MVDA are implemented to solve this issue. Figure 3.6 represents a schematic illustration of the black box idea.

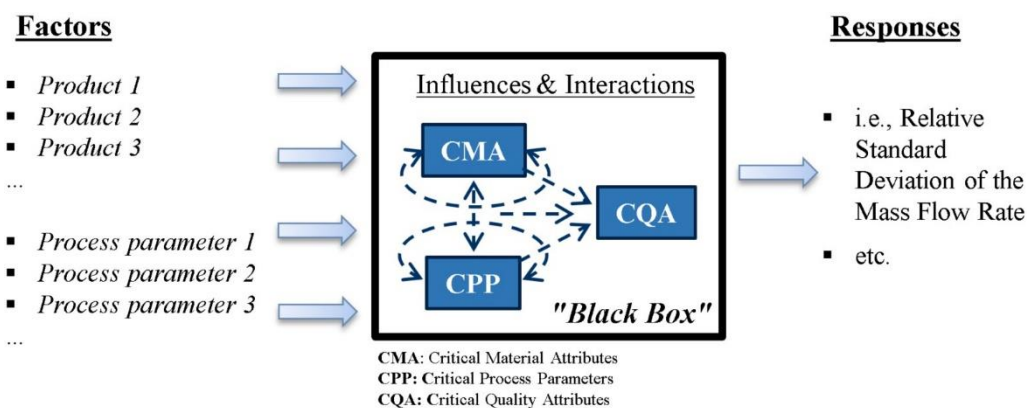


Figure 3.6: Schematic illustration of the black box process model.

PCA and regression analysis are two consecutive analysis approaches conducted in this study. PCA, a multivariate statistical analysis method developed by Pearson [118], is used in this study to select fewer significant variables for the auger dosing process. These selected variables explain most of the variance in the original data and that is by using orthogonal linear transformation. This transformation transforms the original data to a new coordinate system, where the axes are orthogonal to each other. PCA minimizes the studied factors into a set of several new independent variables (uncorrelated principal components (PCs)), where they are ordered by the amount of variation each explains in the original variables. Each PC is a linear combination of the original variables, where the value of the PC for a particular attribute is a weighted sum of the values of the original variables for that attribute. PC1 explains the greatest variability in the original variables, whereas PC2 is orthogonal to PC1 and accounts for the second-largest remaining variability. The cumulative sum of the PCs that captured at least 85 % of the overall variance in the data set is selected in the model, which is sufficient for qualitative analysis. This value limit is accepted and used in the literature [25,26]. The loadings scatter plot is utilized to illustrate the contribution of each material attribute to the PCs. The more the material attribute is located at the extremes of the axis of the graph, the more it contributes to the corresponding PC. PCA was then followed by regression analysis. The goodness of fit of the model was described by the adjusted coefficient of determination (R^2). This coefficient describes the proportion of the variability of the data explained by the model, which is adjusted for the number of factors in the model. The confidence level used in the DoE analysis was $\pm 95\%$. An adjusted response graph was demonstrated to show the effect of each PC and other process factors on the responses, where the response values are adjusted to average out the effects of the other factors. Furthermore, a Pareto graph of effects analysis was employed, where it expresses the relative size of the effect of each factor studied in the model. It represents the orthogonally

scaled effects of each investigated factor on the studied responses. Principally, orthogonal scaling of a factor α_i leads to a scaled value β_i in the range $[-1, 1]$. This is defined as $\beta_i = (\alpha_i - M_S)/R_S$ with $M_S = (\alpha_{\max} + \alpha_{\min})/2$ and $R_S = (\alpha_{\max} - \alpha_{\min})/2$ [119]. This makes the design scale invariant.

3.4 Mass Flow Rate Results

Based on the DoE, 72 tests were executed that allowed the examination of wide combinations of the three main varied factors used to study the dosing behavior of the ten different powders. Each powder was volumetrically fed with three different auger geometries at three different auger speeds. The DoE matrix and the results of the four studied responses (CQA) are summarized in Table 3.2 and Table 3.3.

Run	Auger Filling Volume (mm ³ /rev)	Dosing Speed (rpm)	Powder	m (mg/s)	RSD (%)	PoDM (%)	DT (min)
1	25.52	400	Arbocel P 290	34.5	4.54	86.95	45.9
2	25.52	800	Arbocel P 290	66.4	7.03	87.05	24.31
3	25.52	800	Arbocel P 290	69.5	8.02	86.87	23.14
4	58.59	400	Arbocel P 290	95.6	6.36	79.24	15.39
5	58.59	800	Arbocel P 290	211.4	1.72	79.61	6.9
6	67.33	400	Arbocel P 290	125	3.11	86.52	12.68
7	67.33	800	Arbocel P 290	266.1	4.54	86.64	6.08
8	25.52	400	Ascorbic acid	9.6	38.16	5.66	16.64
9	58.59	800	Ascorbic acid	537.9	2.95	59.08	3.3
10	67.33	400	Ascorbic acid	292	1.72	87.56	9.06
11	67.33	800	Ascorbic acid	554.7	2.46	91	4.97
12	67.33	800	Ascorbic acid	558.1	3.21	90.22	4.93
13	25.52	400	Emcompress	92.8	1.44	87.77	31.56
14	25.52	400	Emcompress	94.4	2.87	87.84	31.06
15	25.52	800	Emcompress	205	1.72	87.87	14.36
16	58.59	400	Emcompress	308.9	1.61	87.76	9.5
17	58.59	800	Emcompress	618.2	1.42	86.01	4.66
18	67.33	400	Emcompress	390.1	0.89	87.88	7.5
19	67.33	600	Emcompress	559.2	3.32	87.62	5.23
20	67.33	800	Emcompress	783.6	0.87	87.65	3.74
21	25.52	800	Glucidex IT 19	106.2	11.82	5.05	1.07
22	58.59	400	Glucidex IT 19	151	8.33	94.06	13.47
23	58.59	800	Glucidex IT 19	296.1	9.30	94.16	6.89
24	67.33	400	Glucidex IT 19	162.8	11.86	93.89	12.32
25	67.33	800	Glucidex IT 19	337	5.04	93.87	6.02

26	25.52	400	InhaLac 251	79.3	1.39	80.69	36.81
27	25.52	600	InhaLac 250	113.7	2.40	80.83	25.72
28	25.52	800	InhaLac 251	148.9	1.89	80.68	19.52
29	58.59	400	InhaLac 251	162.2	13.90	79.18	18.1
30	58.59	600	InhaLac 251	231.7	11.80	79.61	12.68
31	58.59	800	InhaLac 251	372.1	10.37	79.56	7.81
32	67.33	400	InhaLac 251	274.1	1.09	80.73	10.59
33	67.33	800	InhaLac 251	543.4	1.21	80.93	5.36
34	67.33	800	InhaLac 251	557.9	0.62	80.69	5.22

Table 3.2: Executed test runs of the first five powders based on the DoE and the related responses results of the feed rate ($\dot{m}$), the relative standard deviation of feed rate variability (RSD), the percentage of dosed mass (PoDM), and the dosing time (DT).

The DoE matrix and the results of the four studied responses (CQA) of the second five powders are summarized in Table 3.3.

Run	Auger Fill- ing Volume (mm ³ /rev)	Dosing Speed (rpm)	Powder	$\dot{m}$ (mg/s)	RSD (%)	PoDM (%)	DT (min)
35	25.52	400	Magnesium Stearate	32.7	24.03	76.97	37.1
36	25.52	600	Magnesium Stearate	45.6	25.27	70.78	24.62
37	25.52	800	Magnesium Stearate	62.8	24.08	69.81	17.47
38	58.59	400	Magnesium Stearate	10.4	46.95	22.83	33.2
39	58.59	600	Magnesium Stearate	5.8	48.00	13.13	35.85
40	58.59	800	Magnesium Stearate	9.2	53.44	19.51	34.14
41	67.33	400	Magnesium Stearate	120.2	18.40	89.64	11.64
42	67.33	400	Magnesium Stearate	109.5	32.54	87.53	12.9
43	67.33	800	Magnesium Stearate	236.6	32.24	88.77	6.1
44	25.52	400	Maize Starch extra white	43.2	9.90	74.73	52.9
45	25.52	800	Maize Starch extra white	134.1	4.38	80.16	18.81
46	58.59	600	Maize Starch extra white	59.4	16.18	23.1	12.39
47	58.59	800	Maize Starch extra white	117.8	29.04	46.61	12.47
48	67.33	400	Maize Starch extra white	285.2	0.98	82.74	9.1
49	67.33	400	Maize Starch extra white	267.8	0.51	82.84	9.66
50	67.33	800	Maize Starch extra white	532.9	1.67	82.06	4.81
51	25.52	400	SpheroLac 100	71.2	3.07	88.32	41.19
52	25.52	600	SpheroLac 100	111.1	3.65	88.41	26.39
53	25.52	800	SpheroLac 100	134.2	2.93	88.37	22

54	58.59	400	SpheroLac 100	227.5	0.52	88.02	12.89
55	58.59	800	SpheroLac 100	445.5	3.48	88.1	6.58
56	67.33	400	SpheroLac 100	265.7	3.60	88.37	11.02
57	67.33	400	SpheroLac 101	273.7	4.93	88.41	10.68
58	67.33	800	SpheroLac 100	490.6	4.63	88.36	5.97
59	25.52	400	Vivapur 12	38.2	2.32	87.93	48.31
60	25.52	800	Vivapur 12	79.3	2.82	87.94	23.27
61	58.59	400	Vivapur 12	107.1	3.08	87.72	17.28
62	58.59	800	Vivapur 12	210.1	3.32	87.64	8.78
63	67.33	400	Vivapur 12	152.2	2.65	87.09	12.06
64	67.33	800	Vivapur 12	283.6	1.13	86.92	6.44
65	25.52	400	Vivapur 101	40.5	3.80	88.75	45.81
66	25.52	800	Vivapur 101	81.8	2.48	88.73	22.61
67	25.52	800	Vivapur 101	78.4	3.67	88.87	23.81
68	58.59	400	Vivapur 101	96.4	5.43	84.23	18.33
69	58.59	600	Vivapur 101	138.2	15.87	83.28	12.56
70	58.59	800	Vivapur 101	178	8.43	82.23	9.55
71	67.33	400	Vivapur 101	149.7	1.69	87.56	12.13
72	67.33	800	Vivapur 101	273.3	2.52	87.5	6.7

Table 3.3: The DoE of the executed test runs and the related responses results of the feed rate ($\dot{m}$), the relative standard deviation of feed rate variability (RSD), the percentage of dosed mass (PoDM) and the dosing time (DT) of the second five powders.

The results of the auger dosing tests in Table 3.2 and Table 3.3 demonstrated the dissimilarities in the behavior of the powders that were tested under the same process conditions. Different powders tested under the same process parameter conditions showed dissimilar results, which signifies the high impact of the material attributes. Moreover, the same powder undergoing different process parameters revealed different flow behaviors and response values. Therefore, process parameters have significant influences as well.

The RSD values of the feed rate of the ten powders were varying between 0.51 % and 53.44 %. The mean RSD covering the entire dosing tests is $8.93 \% \pm 12.02$ (standard deviation). The interpretation of the results in Table 3.2 and Table 3.3 indicated that the powders could be clustered in three categories, namely good flowing powders, bad flowing powders and powders that undergo dissimilar flowability behavior (good and bad results) under different process parameters. The good flowing powders in the first category were Emcompress, SpheroLac 100 and Vivapur 12, where they have low RSD ($< 5\%$) and high PoDM ($> 86\%$) in all test cases. The second category includes powders that under some process parameters had good flowing behavior and under other conditions, their flowability quality was

poor. The powders of the second category were Arbocel P 290, Glucidex IT 19, InhaLac 251, Maize Starch extra white, Ascorbic acid, and Vivapur 101. The third category contains the powders with low flowability, which mainly included only Magnesium Stearate, which had high RSD values in all its test cases.

The maximum PoDM reached value was around 94 %, where some residuals of powders were left inside the powder container. The PoDM was varying between 5.05 % and 94.16 % with a mean value of $78.68 \% \pm 20.8$. One of the reasons that caused the powder container not to run empty is the low powder bed height at the end of the dosing process. Consequently, the low powder bed height along with the high speed of the stirrer caused a turbulence flow of the remaining powder inside the whole container because of low mass concentration. This is due to the volumetric dosing process (constant rotational speed) that was performed which could be considered as a disadvantage in some cases. Therefore, a controlled motor speed and an independent rotational speed of an auger and a stirrer would be a better solution approach for future work to regulate the feed rate, and thus improve the dosing quality.

Although the PoDM in Glucidex IT 19 was very high (around 94 %) in almost all the run cases, but when using the smallest auger, it had the lowest PoDM (5.05 %). This can be interpreted due to the sticky characteristics of Glucidex IT 19 as a maltodextrin powder. Given that the auger of filling volume $25.52 \text{ mm}^3/\text{rev}$ (type A) has small outer diameter and big inner diameter and that the particle size distribution of Glucidex IT 19 is very high, the chemical structure of the molecules was modified due to heat and pressure coming from the rotation of the auger. Thus, the powder was compressed inside the dosing tube within the auger geometry, which led to the formation of hard sticky spiral forms (not anymore in its initial powder state). Consequently, the dosing process was stopped after dosing around 5 % in 1 min.

In general, the small auger (type A) was not applicable in dosing cohesive powders, whereas auger of filling volume $67.33 \text{ mm}^3/\text{rev}$ (type C) with its conical form derived better dosing results compared to the other two augers. The conical geometry served in forwarding the powder downwards into the dosing tube, and thus aided in consistently having a high powder mass concentration around the auger.

In the following, illustrational examples of different mass flow graphs of 20 tests, under two varied process parameters, are represented. These examples compare the ten powders undergoing two different process conditions. In the first example, Figure 3.7 and Figure 3.8 demonstrate the auger dosing tests of the ten powders using an auger of filling volume $58.59 \text{ mm}^3/\text{rev}$ (type B) rotating at 800 rpm. The left column of the graph represents the mass M (g) and the right column represents $\dot{m}$ (g/s), where both are plotted over time (s). The test runs represented in Figure 3.7 are tests number 5, 9, 17, 23 and 31, where their value results are summarized in Table 3.2. The results under these two constant process parameters showed that Emcompress (run 17) was the fastest powder to be dosed with almost constant $\dot{m}$ followed by SpheroLac 100, Arbocel P 290 and Vivapur 12. On the other side, the feed rates of the other powders were highly fluctuating, where no big amount of powder was dosed, i.e., Ascorbic acid, Magnesium Stearate and Maize Starch extra white. This is actually due to their high cohesivity characteristics.

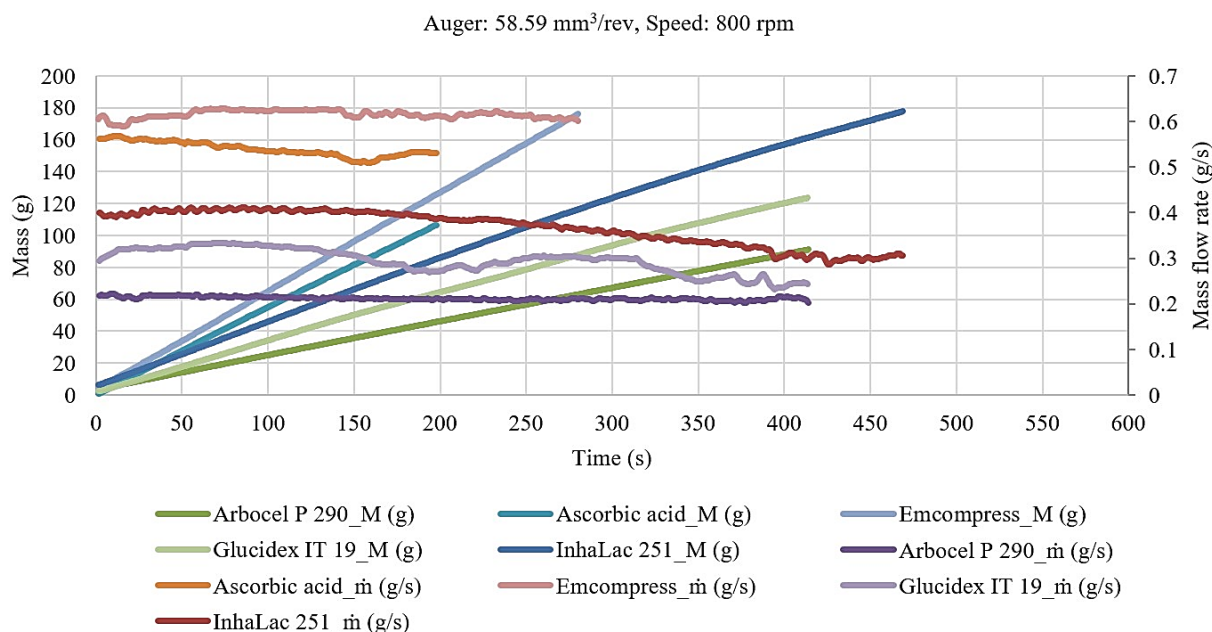


Figure 3.7: Mass flow rate of the first five powders using auger 58.59 mm³/rev at speed 800 rpm.

The test runs represented in Figure 3.8 are tests number 40, 47, 55, 62 and 70, where their value results are summarized in Table 3.3.

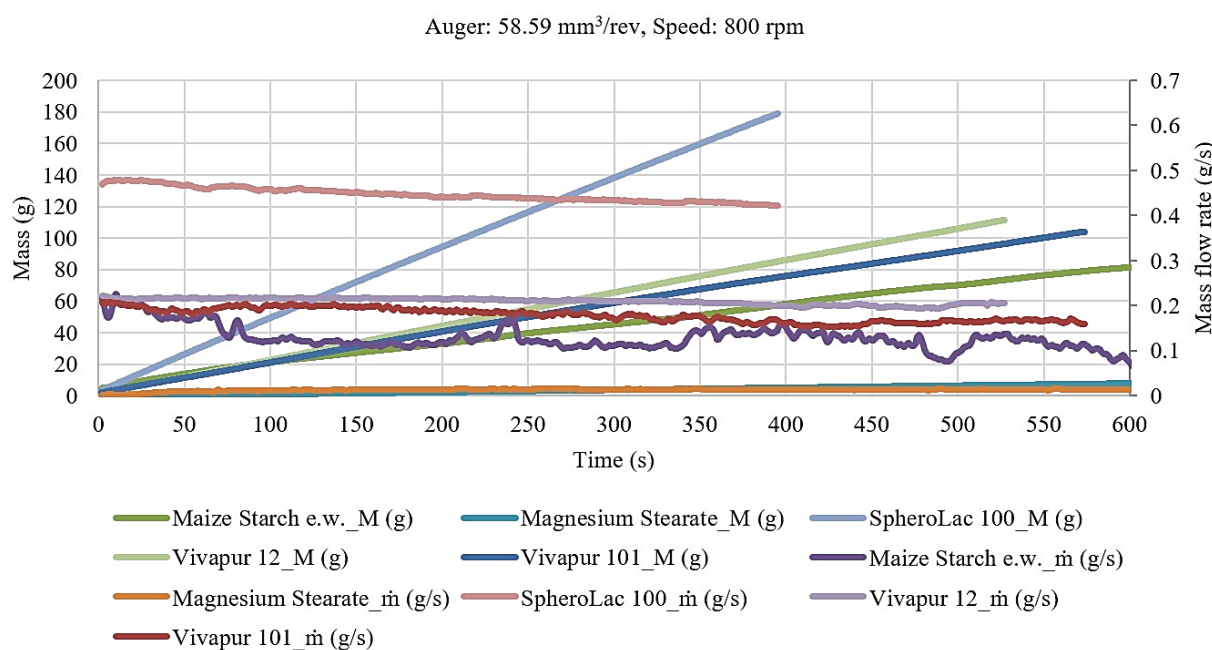


Figure 3.8: Mass flow rate of the second five powders using auger 58.59 mm³/rev at speed 800 rpm.

In the second part of the mentioned example, the mass flow results of the ten powders using an auger of filling volume 67.33 mm³/rev (type C) rotating at 400 rpm are illustrated in Figure 3.9 and Figure 3.10. The test runs represented in Figure 3.9 are the tests number 6, 10, 18, 24 and 32, where their results are summarized in Table 3.2. It is evident that when using the auger with the conical form, the results are

much better regarding the RSD. The $\dot{m}$ is almost constant except with Magnesium Stearate and Glucidex IT 19. Note that the same powders tested with the same conical auger form but at a rotational speed 800 rpm showed almost similar results with error deviations of $\pm 3\%$.

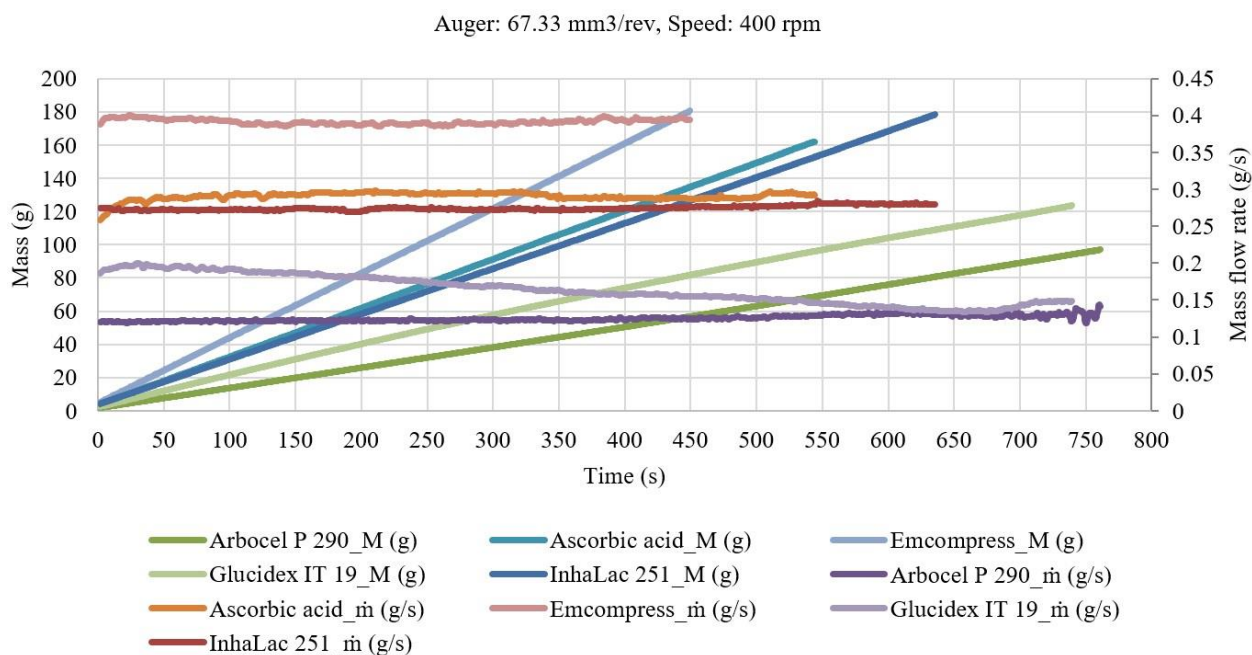


Figure 3.9: Mass flow rate of the first five powders using auger 67.33 mm³/rev at speed 400 rpm.

The test runs represented in Figure 3.10 are the tests number 42, 49, 56, 63 and 71, where their results are summarized in Table 3.3.

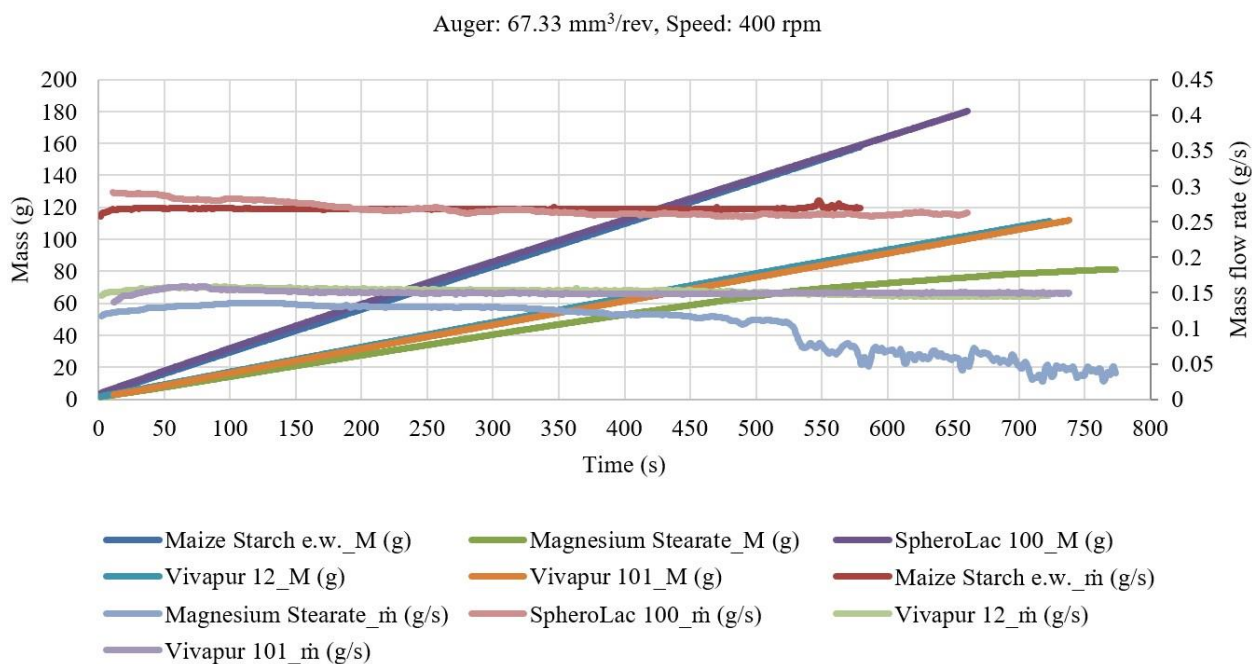


Figure 3.10: Mass flow rate of the second five powders using auger 67.33 mm³/rev at speed 400 rpm.

Overall, it was noticeable in the results that the $\dot{m}$ was high at the beginning of the dosing process, where the powder bed height inside the powder container was high and the powder was somehow compressed

under the influence of the high powder mass (consolidation stress) inside and around the dosing tube [25]. During the dosing process, the powder mass started decreasing gradually over time, where the powder became less compressed leading to a reduction in the density near the dosing tube and consequently this led to the decrease of the m [25]. Statistical analysis of the auger dosing results will be presented in the following sections.

3.5 Multivariate Analysis Results

3.5.1 Principal Component Analysis

3.5.1.1 Initial model: Loadings scatter plot

After characterizing the powders and performing auger dosing tests, a multivariate statistical analysis using PCA and regression investigations were employed to identify the critical material attributes and the critical process parameters influencing the auger dosing process. An initial PCA model using only the 30 material attributes was firstly built to figure out the (dis)similarities between the ten powders. The material attributes inserted into the initial PCA model were reduced to a few significant orthogonally PCs. The first four PCs were considered, where they captured the majority of 85.69 % (Table 3.4) of the overall variance of the original data set, which is sufficient for qualitative analysis.

PC	% Variance	Cumulative %
1	45.65	45.65
2	19.02	64.67
3	12.85	77.52
4	8.16	85.69

Table 3.4: Initial model: Percentage of the considered variance of the first four principal components covering the 30 material attributes.

For the interpretation of the PCs, the loadings scatter plots (Figure 3.11 and Figure 3.12) were presented. The loadings scatter plot was used to understand how the material attributes are related to each other and indicate the importance (weight) of the material attributes towards the PCs. The points on a loadings scatter plot are the coefficients of the original factors (material attributes) that define two PCs. Attributes with high positive or negative loadings strongly influence the PCs, whereas attributes located close to the origin of the loadings graph are not well explained by the model and are therefore not relevant for the respective PCs in the plot. In the scatter plots, an imaginary line through the origin of the PCs is drawn, where the variables can be projected on this line. The variables are positively correlated when the projected variable is located close to the other selected variable, while they are considered anti-correlated when being projected at the other side of the line. The closer the projection to the origin of the plot is, the weaker the correlation. A logarithmical (log) transformation of AE was applied because

it was non-normally distributed. This transformation helps in improving the data analysis efficiency [120]. Figure 3.11 represents the loadings scatter plot of PC1 vs. PC2.

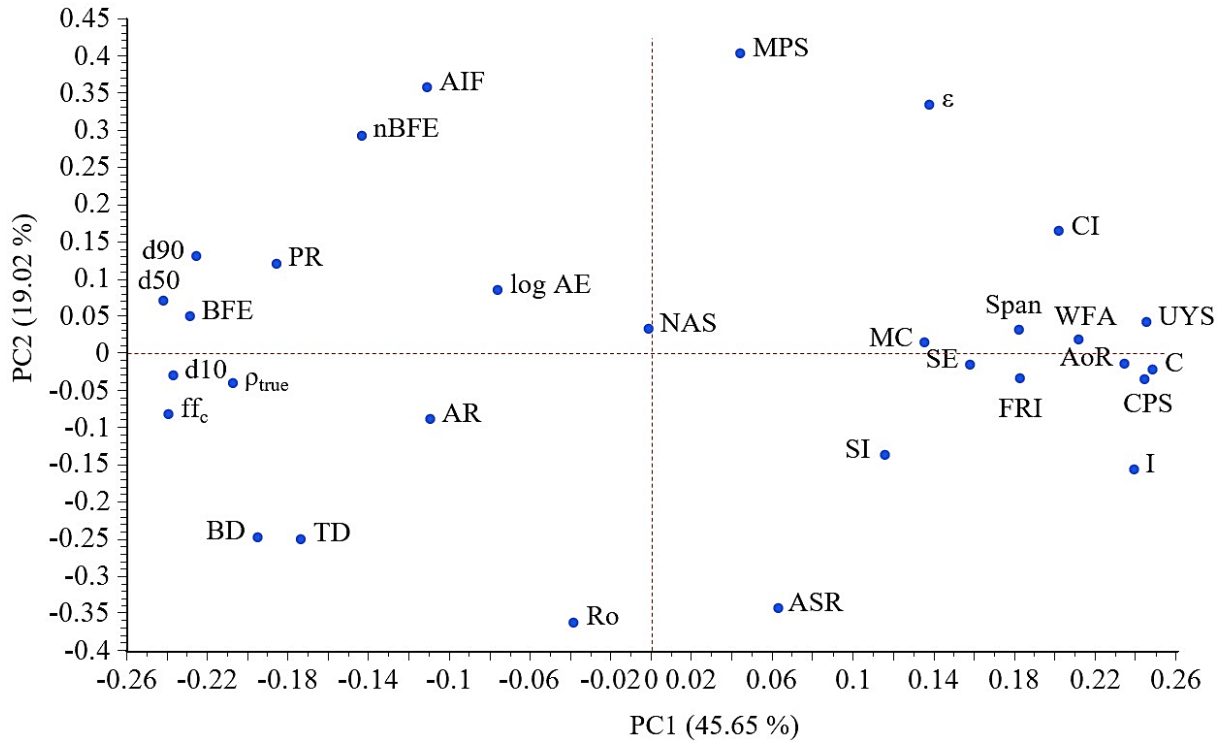


Figure 3.11: Initial model: Loadings scatter plot of PC1 vs. PC2. The abbreviations of the material attributes are described in the materials and methods section.

In the following, the high significant loading values of each PC are summarized.

PC1 loadings: *Flow-related material attributes.* PC1 is the most important PC that contains a lot of significant attributes especially the ones related to particle sizes, densities, flow energy, flowability, cohesion, and compressibility. The PC1 vs. PC2 loadings scatter plot (Figure 3.11) shows how the particle sizes, densities, flowability, flow energy and permeability are clustered at the left, which indicates that they contribute highly negatively to PC1. On the opposite side, cohesion-related attributes (C and SE), compressibility and some flowability indicators (i.e., CI, AoR and UYS) contribute positively to PC1. UYS presents the stress required to initialize the flow of a consolidated mass of material when it is in an unconfined state. Consequently, PC1 merely described flow-related material attributes.

PC2 loadings: *Shape and friction-related attributes.* The particle shape attributes ASR and Ro have strongly negative PC2 loadings values (lower part of the plot, see Figure 3.11) in contrast to friction-related attributes MPS and AIF with strongly positive loading values for PC2 (upper part of the plot, see Figure 3.11).

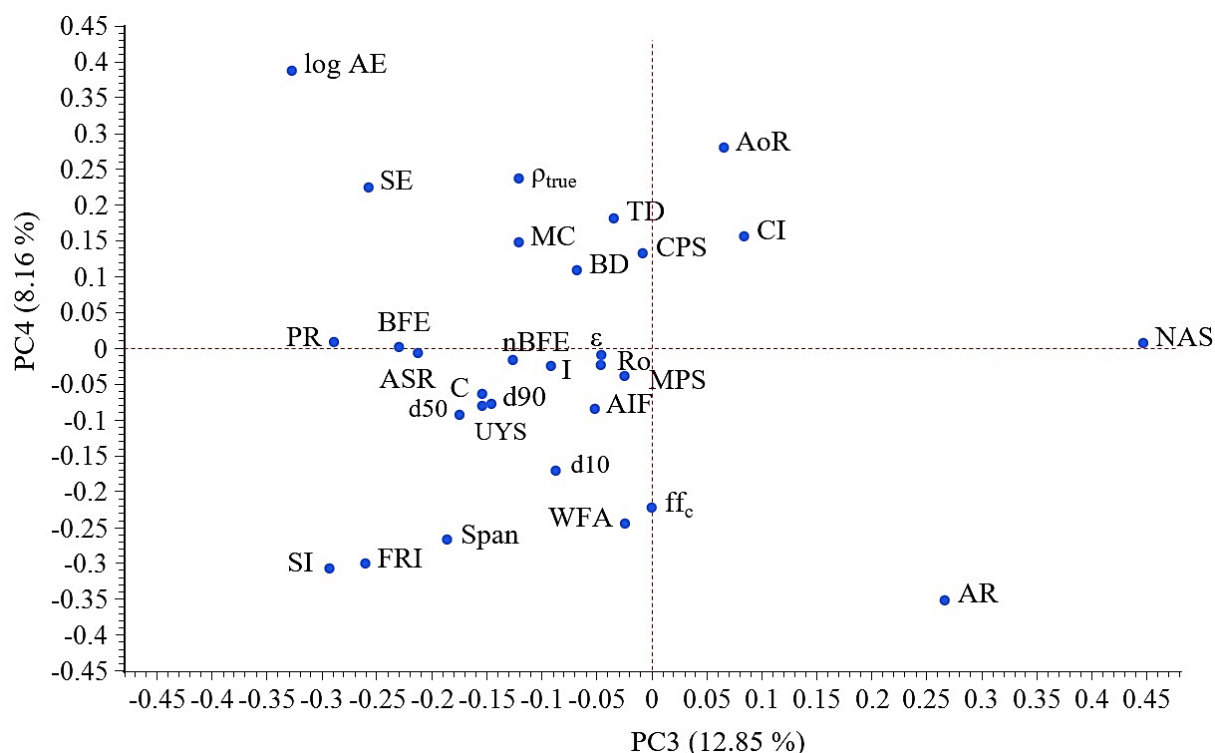


Figure 3.12: Initial model: Loadings scatter plot of PC3 vs. PC4. The abbreviations of the material attributes are described in the materials and methods section.

Similarly, the high significant loading values of PC3 and PC4 are summarized in the following.

PC3 loadings: *Fluidization-related attributes.* PC3 mainly explained the variability in air fluidization-related attributes (NAS, AR, log AE and PR), which are distributed on the left and right sides of the plot (Figure 3.12).

PC4 loadings: *Aeration and stability related attributes.* PC4 covered attributes related to aeration, the stability of flow (SI) and sensitivity to changing flow rate (FRI).

3.5.1.2 Initial model: Weighted loadings plot

The weighted loadings plot is used to identify the contribution of each material attribute on each PC. It is a bar plot of the loadings (weights) of each material attribute in each PC multiplied by the variance explained in each material attribute by the PCA model [7]. Figure 3.13 represents the weighted loadings plot of PC1 and PC2.

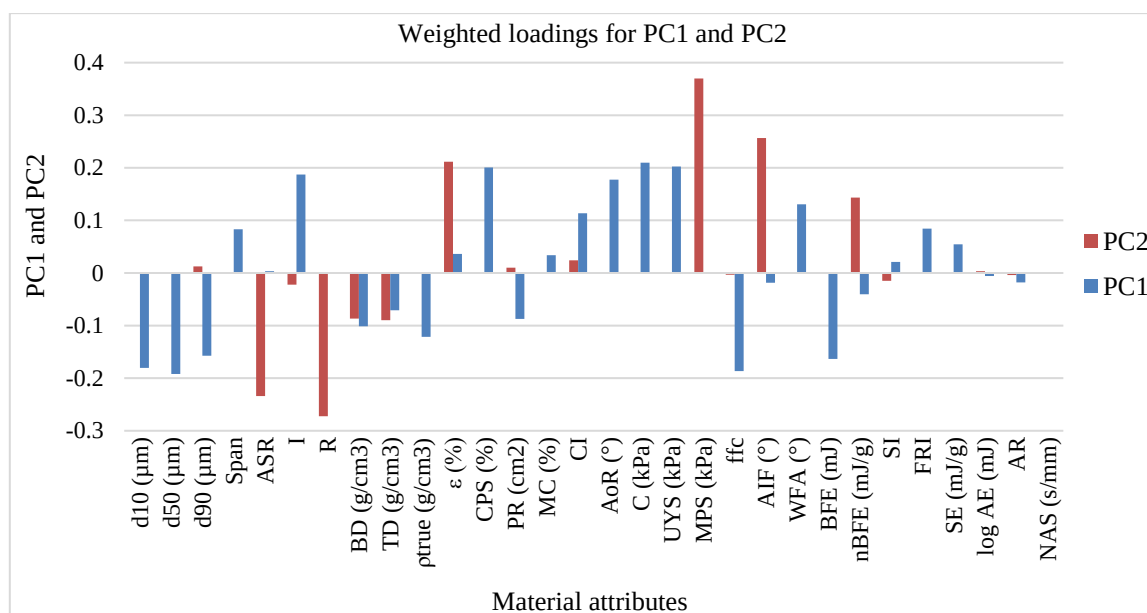


Figure 3.13: Initial model: Weighted loadings plots of PC1 and PC2.

Similarly, Figure 3.14 illustrates the weighted loadings plot of PC3 and PC4.

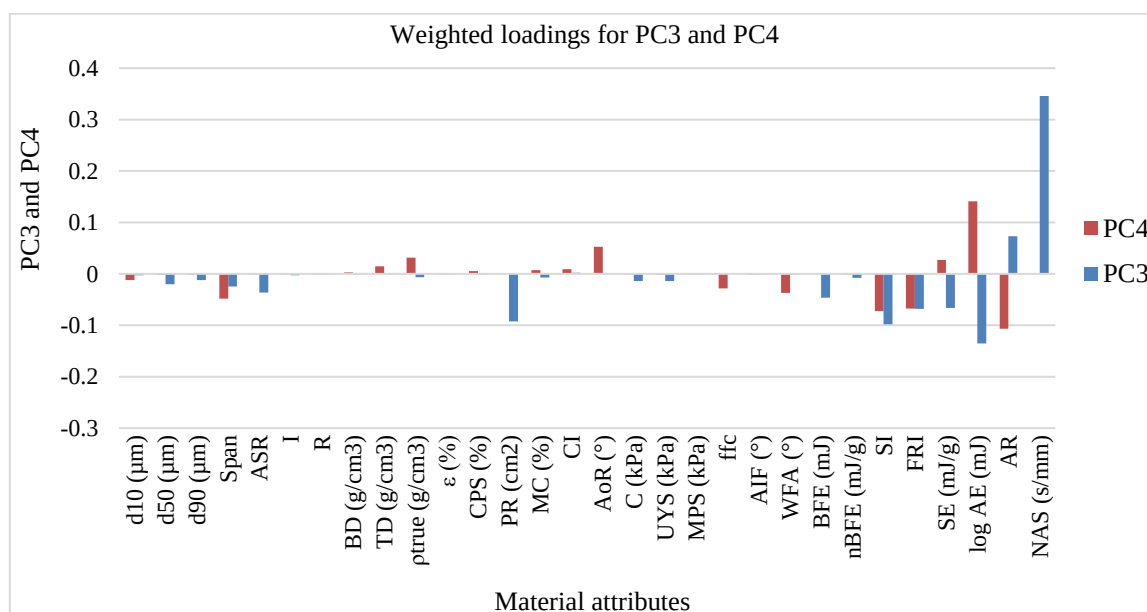


Figure 3.14: Initial model: Weighted loadings plots of PC3 and PC4.

Based on these plots, it is clearly shown that PC1 encloses at least 21 significant material attributes out of 30. Moreover, the remaining material attributes were important to the other three PCs loadings. Therefore, no PCA reduction model was required since all material attributes proved to be significant to the four PCs.

3.5.1.3 Initial model: Score scatter plot

The score scatter plot is used to compare any two variables (in this case powders) in the analysis and check to what extent the variables are correlated to each other. Powders with a specific location on the

score scatter plot possess high values for the material attributes with a similar location on the corresponding loadings scatter plot (positive correlation), and low values for material attributes at the opposite side of the origin (negative correlation) [26]. The powders that exhibit similar attributes regarding the plotted overarching material attributes (i.e., loadings) were clustered. On the contrary, powders that are located on opposite sides of the score scatter plot will have distinctly different material attributes (i.e., anti-correlation). The interpretation of the score scatter plot PC1 vs. PC2 in Figure 3.15 indicates the clustering of three groups of powders that have some similarities between their materials attributes. Free-flowing powders, namely Emcompress and SpheroLac 100, categorize group 1 (left-hand side of the plot). Group 2 consists of moderately free-flowing powders Vivapur 12, Glucidex IT 19, InhaLac 251, Vivapur 101, Arbocel P 290 and Ascorbic acid (around the origin of the plot). Moreover, group 3 contains cohesive powders, i.e., Maize Starch extra white and Magnesium Stearate (right-hand side of the plot). These results highly confirm with the initial analysis of the auger dosing test results, where almost the same three clusters were determined with only having two powders in one way or another exchanged between the categories, which are, namely Vivapur 12 and Maize Starch extra white. For elaboration, Maize Starch extra white can also be considered in group 2 since it is located relatively far away from Magnesium Stearate. Vivapur 12 is relatively close to SpheroLac 100, where it can also be clustered in group 1 as in the results of the auger dosing tests. Since PC1 describes flow-related material attributes, it is obvious to see the clustering of Emcompress and SpheroLac 100 to the left, where they have high particle sizes, densities, and ff_c values and have low cohesion and compressibility values on the other side. On the contrary, Maize Starch extra white and Magnesium Stearate have opposite values to that of group 1. Moreover, Vivapur 12 and Vivapur 101 are both MCC and have undergone the same production method (spray dried), where they principally differ in their particle size distribution. The PCA analysis proved that according to PC1, they are relatively dissimilar and according to PC2 they are clustered together representing the same shape attributes (fiber shape) and particle friction-related attributes. In addition, SpheroLac 100 and Maize Starch extra white have both almost the same negative PC2 values, where they have relatively similar particle shape and MPS values.

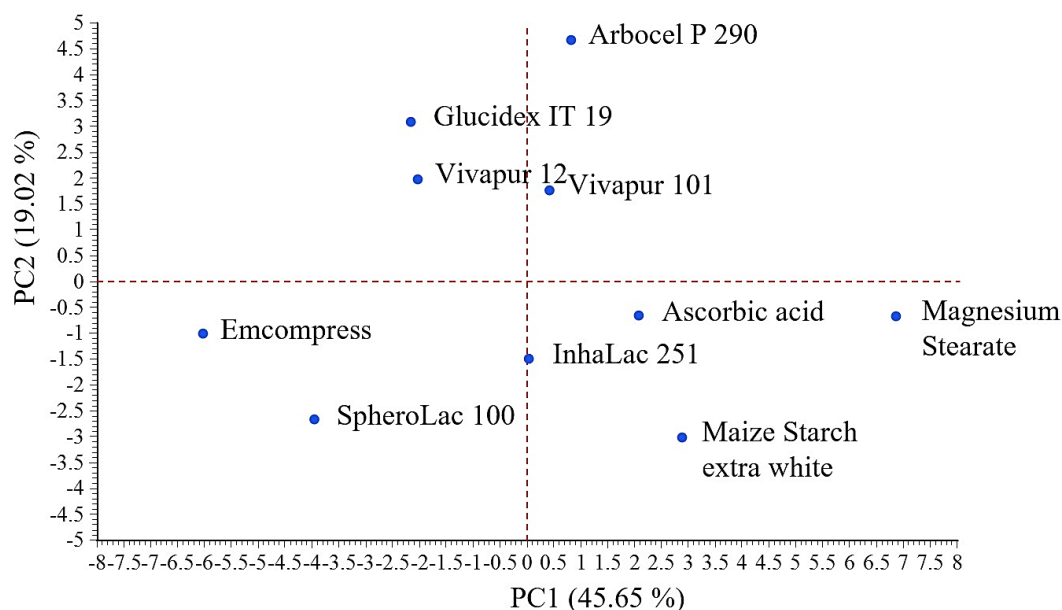


Figure 3.15: Initial model: Score scatter plot of PC1 vs. PC2.

The PC3 vs. PC4 loadings scatter plot (Figure 3.12) showed that PC3 and PC4 were related to fluidization, aeration and stability attributes, respectively. In Figure 3.16, the score scatter plot of PC3 vs. PC4 shows that Emcompress and Vivapur 101 are anti-correlated according to PC3. Since the AE has a negative contribution to PC3, Emcompress (high value of AE) has also a negative contribution to PC3, whereas Vivapur 101 (low value of AE) has a positive contribution (anti-correlated). In addition, since NAS has a positive contribution to PC3, Vivapur 101 (high NAS value) has a high positive PC3 value, whereas Vivapur 12 (low NAS value) is insignificant to PC3. SpheroLac 100 and Magnesium Stearate are anti-correlated according to PC3, which is due to their high and low aeration values, respectively. Since AR contributes positively to PC3, SpheroLac 100 (very high AR) also contributes positively to PC3, while Magnesium Stearate (very low AR) contributes negatively to it. Ascorbic acid and Magnesium Stearate are anti-correlated with PC4 since they have both low and high SI and FRI values, respectively. Therefore, since SI and FRI have negative contributions to PC4, Magnesium Stearate (with high SI and FRI values) has a negative contribution to PC4 as well.

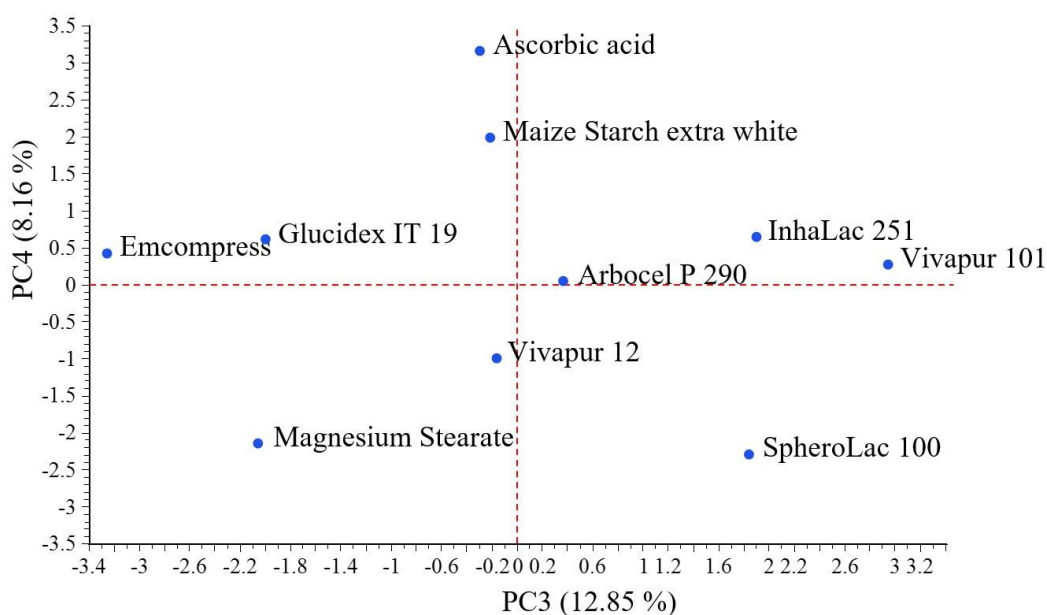


Figure 3.16: Initial model: Score scatter plot of PC3 vs. PC4.

3.5.1.4 Advanced model: Loadings scatter plot

An advanced PCA model was built, which contained, in addition to the 30 critical material attributes, the four process parameters. As listed in Table 3.1, the four process parameters include the outer and inner diameters and the pitch of an auger geometry, and the auger rotational speed. To be mentioned, due to some limitations, the auger helical form as an additional factor was not counted in the DoE and the MVDA. After including all the attributes together, the goal was here to determine the new loadings that affect the auger dosing performance. The first six PCs were selected, where they covered 85.8 % of the total explained variance of the newly expanded data set as represented in Table 3.5.

PC	% Variance	Cumulative %
1	40.29	40.29
2	16.83	57.11
3	11.37	68.48
4	7.35	75.84
5	5.57	81.41
6	4.39	85.80

Table 3.5: Advanced model: Percentage of variance and its cumulative sum of the first six principal components that cover all material attributes and process parameters.

The advanced loadings scatter plot of PC1 vs. PC2 is displayed in Figure 3.17, where both PCs still represent the same loadings related attributes as described in the initial model. The four process parameters were centered around the origin, which indicates their insignificant influence on both PC1 and PC2.

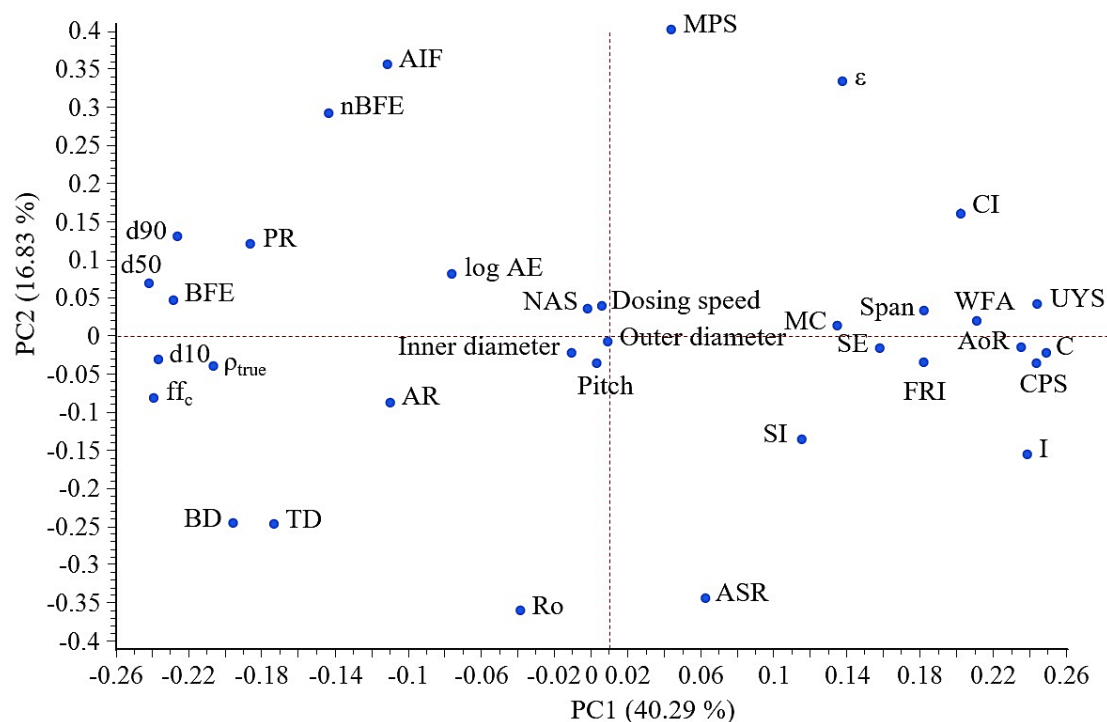


Figure 3.17: Advanced loadings scatter plot of PC1 vs. PC2. The abbreviations of the material attributes are described in the materials and methods section.

In Figure 3.18, the advanced loadings scatter plot of PC3 and PC4 represented similar loadings related attributes as in the initial model. The remarkable here is that the outer diameter and the pitch have some impact for PC4, followed by the auger rotational speed and a negligible impact of the inner diameter.

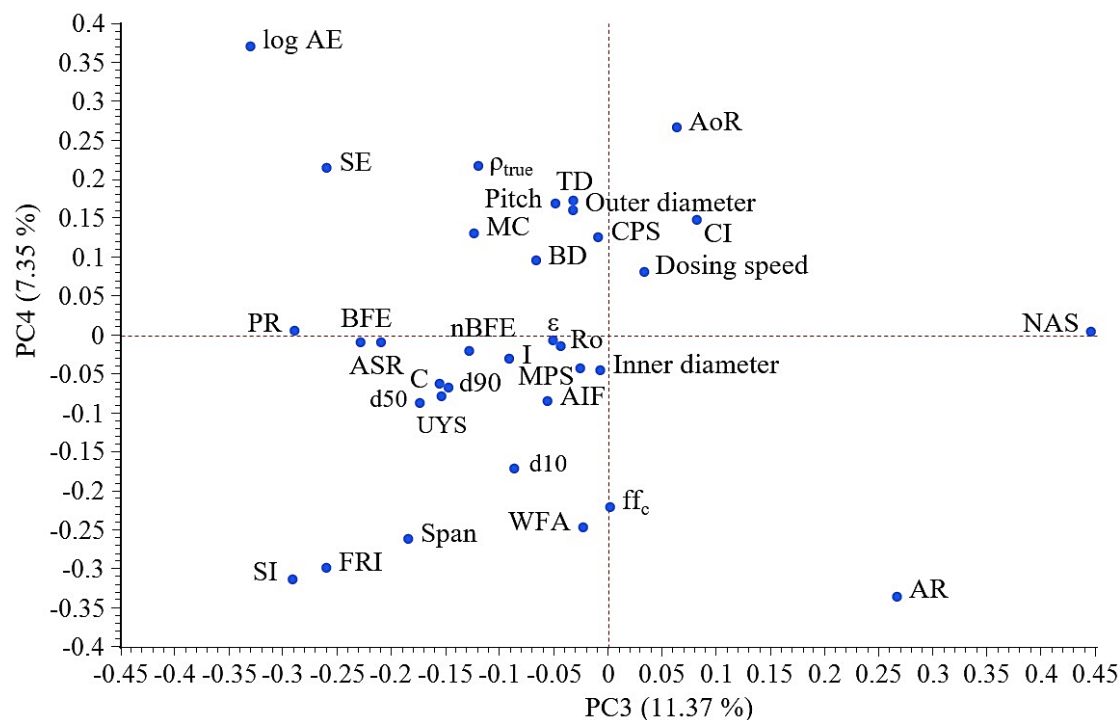


Figure 3.18: Advanced loadings scatter plot of PC3 vs. PC4. The abbreviations of the material attributes are described in the materials and methods section.

In the following, the high significant loading values of the two new additional PCs, i.e., PC5 and PC6 are summarized.

PC5 loadings: *Auger geometry related attributes.* The auger geometries, namely the outer diameter and the pitch have strongly positive loading values for PC5 as depicted in Figure 3.19, while the inner diameter has a negative loading value for it.

PC6 loadings: *Moisture content and span of particle sizes attributes.* The moisture content has a strong positive loading value for PC6 as represented in the middle top of the plot in Figure 3.19. On the other side, the span of the particle sizes has a negative contribution value for PC6.

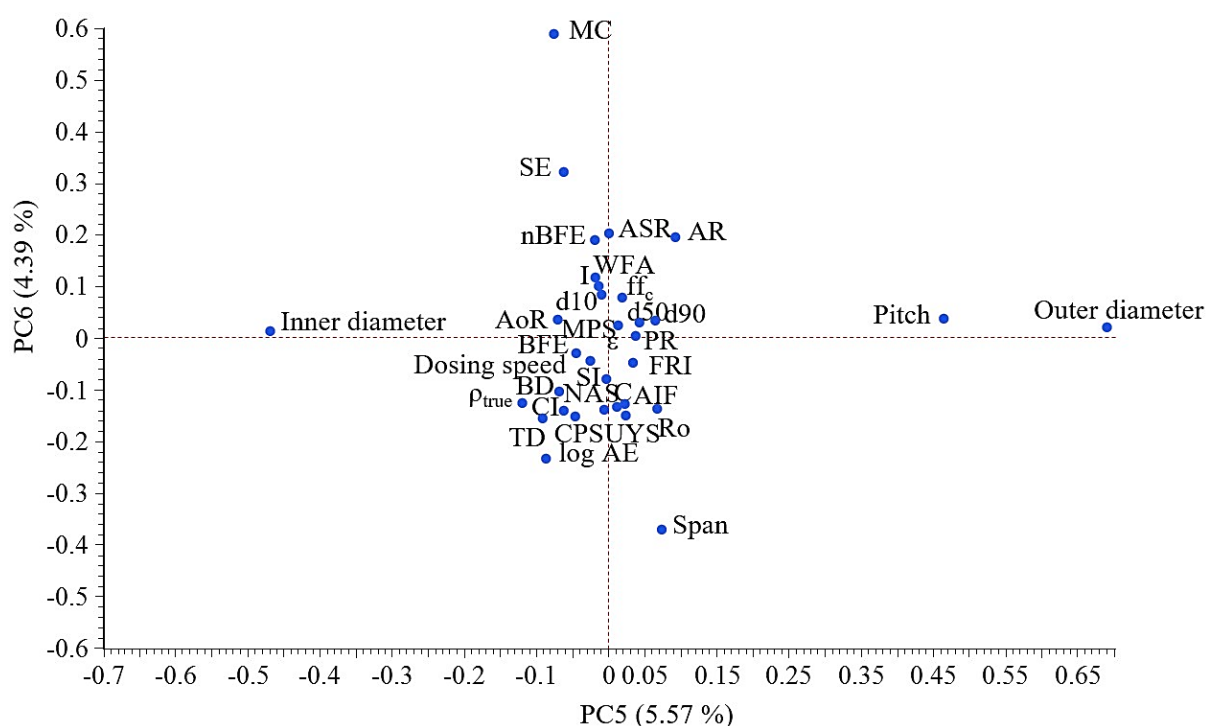


Figure 3.19: Advanced loadings scatter plot of PC5 vs. PC6. The abbreviations of the material attributes are described in the materials and methods section.

3.5.2 Multivariate Quadratic Regression Analysis

Based on the PCA results, the 30 material attributes and the four process parameters were all considered in the multivariate quadratic regression analysis. Therefore, the impact of the six PCs was studied against the four responses (CQA) as mentioned in Table 3.2 and Table 3.3. A quadratic regression model was built, where Table 3.6 shows the high-adjusted R^2 values of the four responses and thus indicating their good fit to the data points of the model.

	$\dot{m}$ (mg/s)	RSD (%)	PoDM (%)	DT (min)
Adjusted R^2	0.938	0.831	0.82	0.882

Table 3.6: Adjusted R^2 of the four studied responses in the quadratic regression model.

Using the Box-Cox plot (Figure B.1), it was determined if a transformation of the responses results was necessary. The objective was to improve the goodness of fit of the model, and thus increase the adjusted R^2 . The transformations took place for the PoDM (squared), the DT (square root), and the feed rate (square root). The RSD remained untransformed since the transformation did not lead to better results. After that, the residuals were normally distributed, and the regression model was optimized using only significant terms in the model equations. The automatic integrated algorithm in Cornerstone software allowed to determine the significant factors in the model and removed the insignificant ones outside after setting up the significance threshold values.

After applying the transformation, the adjusted response graph was generated as represented in Figure 3.20. An adjusted response graph is used to illustrate the effect of each PC on the studied responses, where it shows, for instance, the type and direction of the trend line of the effects. Each of the presented sections (row-wise) corresponds to a studied response, whereas the column-wise sections correspond to the six PC factors. In each of the latter, all the data points obtained from the experiments were presented besides the effect drawn from these points. For a data point, the effects of the other factors included in the model were averaged. Therefore, the data points were adjusted in such a way that the effect of the corresponding factor on the response is only presented [57].

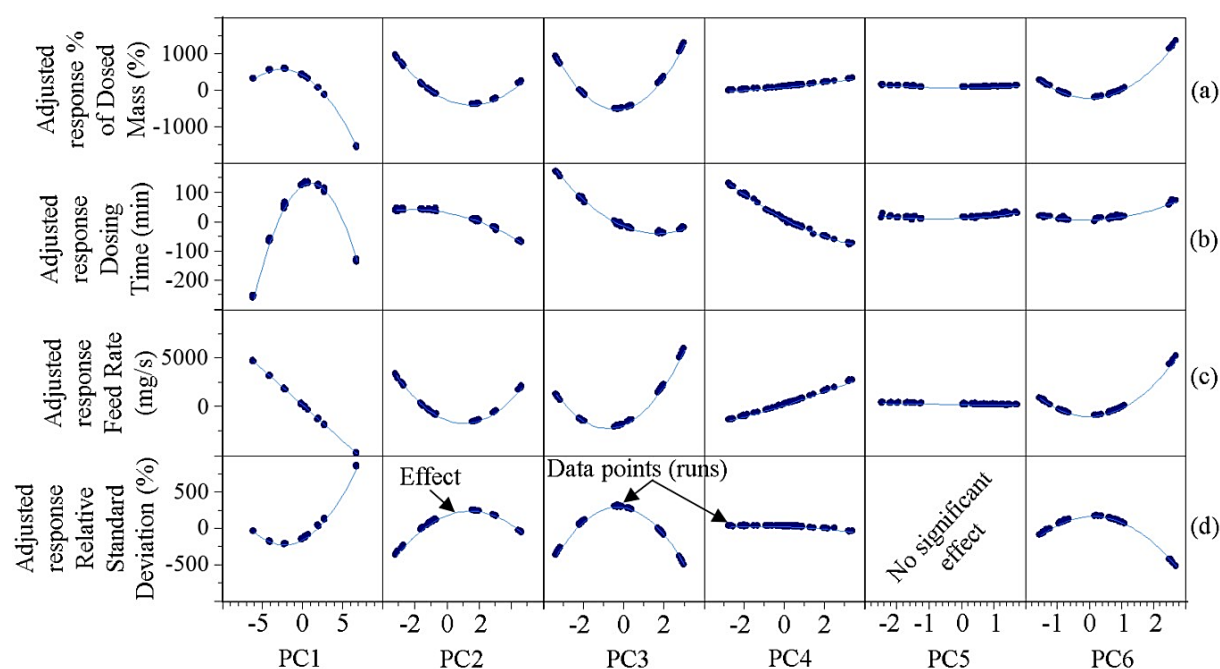


Figure 3.20: Adjusted response graph of the (a) PoDM, (b) DT, (c) $\bar{m}$ and (d) RSD as a function of the six selected PCs.

3.5.2.1 Factors affecting the feed rate

Figure 3.21 illustrates the Pareto graph of effects on the $\bar{m}$. The bars being above the 0-axis signify the positive effect (proportional) of the corresponding parameters, and the ones being below it signifies negative effect (anti-proportional). The factors are distributed in descending order according to the level

of positive or negative effects from left and right, respectively. Therefore, the referred figure indicates that PC1 has a strong negative impact on the $\dot{m}$, where it is evident that the material flow related attributes affect the flow output. For instance, Emcompress, with its big particle sizes, density values and ff_c along with its low C, CPS and AoR values, has good flowability and thus leading to high $\dot{m}$. In addition, the interaction between PC1 and PC2 has a negative impact on the value of $\dot{m}$, where the shape and friction-related attributes influence the mass output. On the other side, PC3 and the interaction between PC3 and PC6 have a positive impact on the value of $\dot{m}$. This adds up that the fluidization-related attributes along with, e.g., the moisture content, play a significant role in influencing the speed of powder discharge. According to Figure 3.20(c), PC1 has a strong linear negative impact on the adjusted response $\dot{m}$, whereas PC2, PC3 and PC6 have a positive quadratic (parabola) effect on it. Note that the terms in the middle of the figure are not shown since they have relatively low impact on the results compared to the terms on the right and left of the figure. Thus, the readability of the figures would be better.

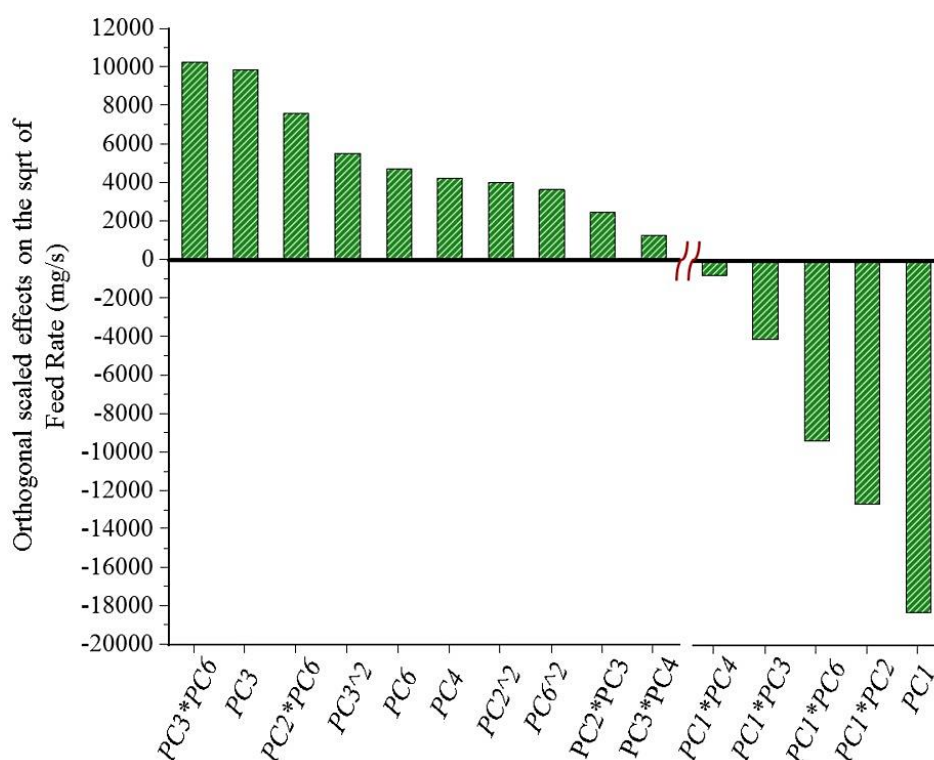


Figure 3.21: Pareto graph of effects of the model factors on the $\dot{m}$.

3.5.2.2 Factors affecting the feed rate variability (RSD)

The Pareto graph of effect on the feed rate variability (RSD) is depicted in Figure 3.22. PC1 has a strong positive impact on the RSD followed by its interaction with PC2. Therefore, the flow-related attributes affect the quality of dosing, which describes how constant and continuous the discharge of the powder is. On the contrary, the interaction between PC3 and PC6 has a significant negative impact on the RSD. It is therefore obvious how different PCs have opposite effects on the $\dot{m}$ and the RSD. This can be

explained by checking, for instance, how Emcompress has a very low RSD and at the same time has very high $\bar{m}$ (check Table 3.2).

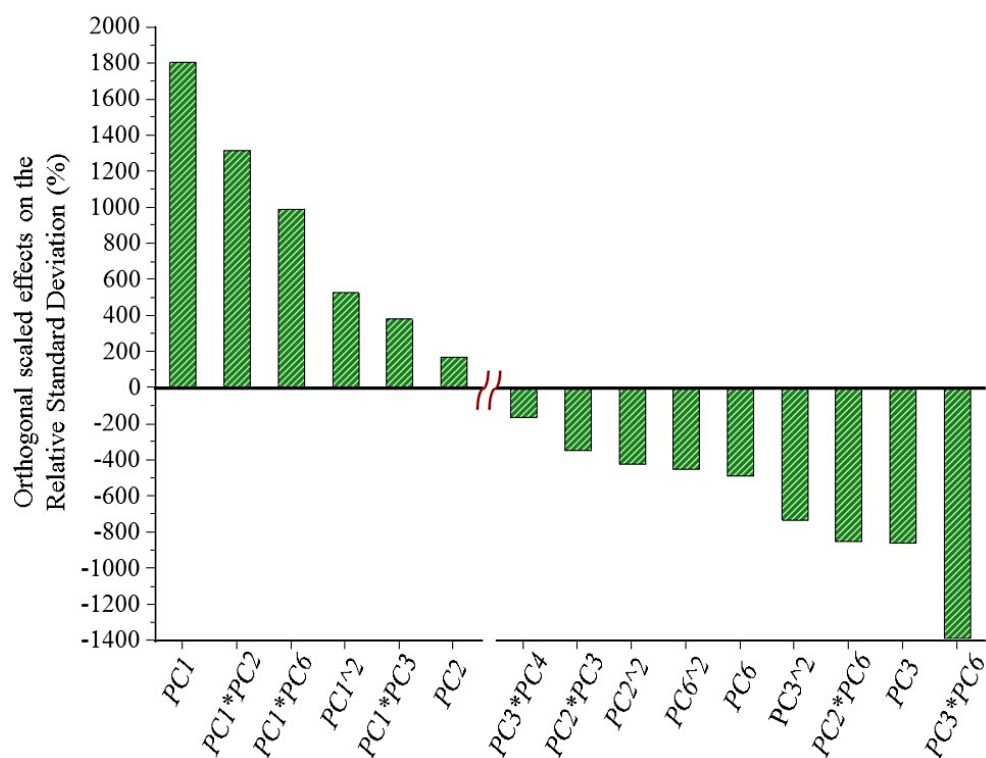


Figure 3.22: Pareto graph of effects of the model factors on the RSD.

3.5.2.3 Factors affecting the dosed mass

Figure 3.23 shows the Pareto graph of effects on the PoDM. PC1 has a strong negative impact on the PoDM followed by the interactions between PC1 and PC2 and between PC1 and PC6, respectively. Apparently, as the cohesion related attributes increases (right side of PC1 in Figure 3.11), the PoDM decreases. In contrast, the interaction between PC3 and PC6 has a strong positive effect on the PoDM followed by PC3.

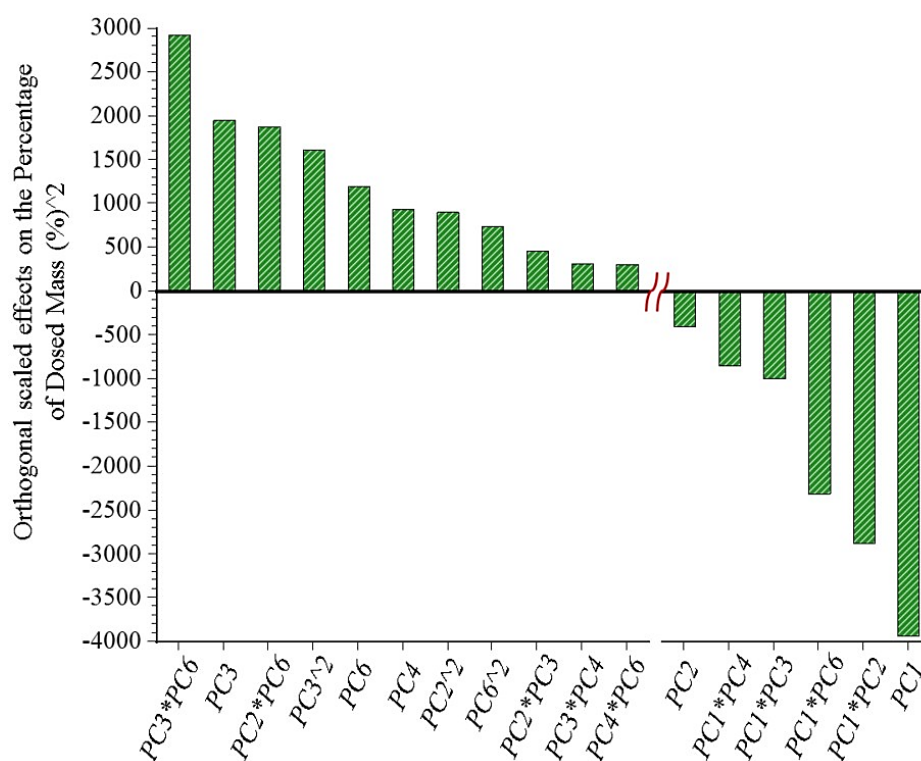


Figure 3.23: Pareto graph of effects of the model factors on the PoDM.

3.5.2.4 Factors affecting the dosing time

The DT is highly negatively influenced by the quadratic term of PC1 followed by a negative impact from PC4. As the pitch becomes bigger, the dosing time of the powder decreases and thus leading to a faster dosing process. This is proven as represented in Table 3.2 and Table 3.3, where auger dosing tests with auger type C (pitch = 4.5 mm) were faster than the dosing tests performed with auger type B (pitch = 3.5 mm). In addition, as the dosing speed has some weight impact on PC4, it is then evident that as the speed increases, the dosing time decreases. On the contrary, PC1 and the interaction between PC3 and PC6 have a positive effect on the DT. The flow-related material attributes (PC1) cause longer DT. It means that as the C, CPS and AoR increase (positive contribution to PC1), the DT increases as well, which took place primarily in case of cohesive powders. The Pareto graph of effect on the DT is represented in Figure 3.24.

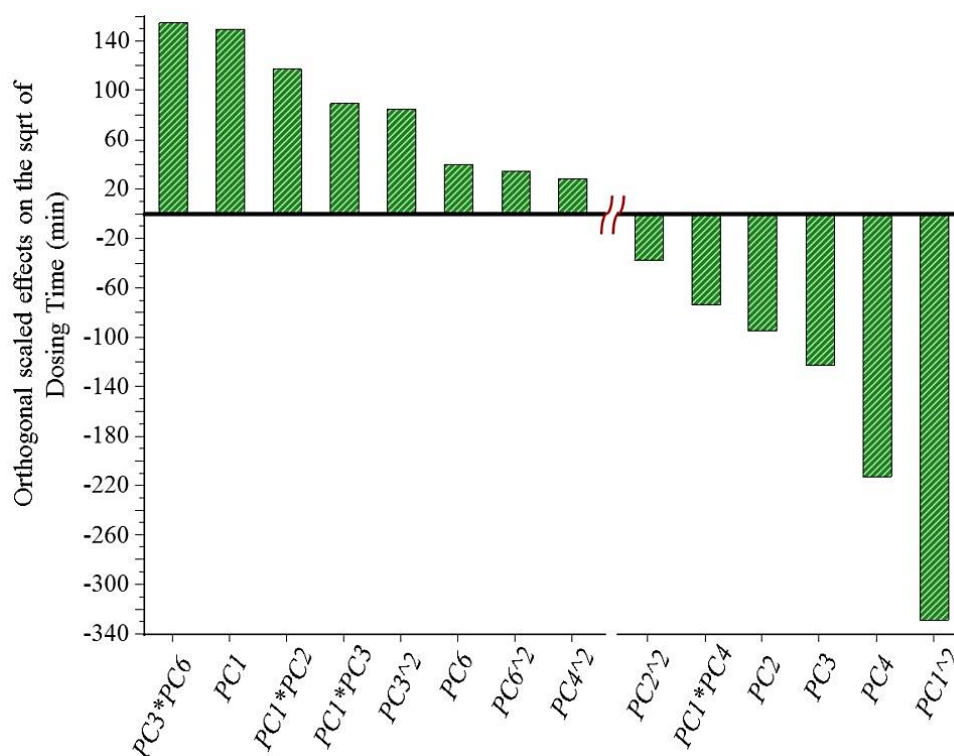


Figure 3.24: Pareto graph of effects of the model factors on the DT.

To be indicated, the underlying target after performing the PCA and regression analyses was to assess the feasibility of predicting the performance of auger feeders from powder attributes to accelerate the identification of an optimal auger dosing solution for any given material, especially the cohesive ones. Unfortunately, the prediction model did not work well here, where the prediction errors were relatively high. This is due to the insufficient data set, where the formed clusters, as represented previously in Figure 3.15 and Figure 3.16, did not include enough materials clustered more together [25]. This makes the mission of predicting the performance of a new material a challenging task to be accomplished. Therefore, one of the proposed solution approaches for future work is to increase the number of studied powders, which would ease the regression analysis by trying to predict the behavior of a new powder by comparing it with similar material attributes already characterized and set in the database [8,25].

3.6 Summary of the correlation study

In this chapter, different methodologies were applied to identify possible correlations between material attributes, process parameters, and volumetric auger dosing performance. The methods included powder material attributes characterization and examination of critical quality attributes of auger dosing application based on PCA followed by multivariate regression analysis. Initially, a 4PC model was identified, where each PC described specific critical material attributes. The (dis)similarities among the tested powders were then revealed using the PCA based on their variation in overarching material attributes. The PCA contributed to identifying the (anti-)correlated as well as the significant attributes, where three clusters of powders were recognized by their critical material attributes. Since the 30 material attributes

were identified to be significant within the initial 4PC model, they were then added along with the four studied process parameters into the advanced PCA model. As a result, the 34 material attributes and process parameters were reduced to six independent principal components that described 85.8 % of the overall explained variance in the data set. As PC1 includes the largest data variability in the original variables, the following attributes revealed their high significance compared to the other studied attributes, which are the particle sizes, densities, flow energy, flowability, cohesion and compressibility.

The regression analysis revealed that most of the investigated material attributes have different range of effects on the studied responses, namely the feed rate, the relative standard deviation of feed rate variability, the percentage of dosed mass, and the dosing time. The results indicated linear and quadratic correlations between them. The auger geometry had a little significant impact on the studied output responses. As the conical auger geometry was not identified in the DoE as an additional varied attribute, the results from the auger dosing experiments acknowledged the positive influence of this distinct geometry form over the straight auger form. It was also interesting to identify the almost insignificant impact of the rotational speed on the four studied responses, where it showed only a low effect on the feed rate and the dosing time. Overall, the material attributes dominated the process parameters in affecting the auger dosing accuracy, and thus had significant impacts on the process performance.

4 Numerical Treatement - DEM

Discrete Element Method (DEM) proved to be an essential tool to optimize the industrial auger dosing process for pharmaceutical powders. One of the key challenges in the implementation of DEM to model powder's flow is the appropriate selection of material parameters, where empirical approaches are mostly applied. During the DEM parameter calibration process of a certain powder, several parameter combinations might lead to a similar bulk response, which could also vary for other bulk responses. Therefore, a methodology is needed to narrow down the number of combinations and be at once close to reality representation. In this chapter, after thoroughly discussing the DEM numerical fundamentals, two case studies are represented. Case study 1 focuses on the development of an alternative systematic numerical approach using AoR test that can efficiently and accurately predict the influence of different DEM parameters on various sought macroscopic responses, where, accordingly, model validation based on experimental data is applied. Design of experiment and multivariate regression analysis (MVRA), using an optimized quadratic D-optimal design model and new analysis tools, i.e., adjusted response and Pareto graphs, are applied. Moreover, an advanced investigation based on auger dosing experiment is discussed in case study 2. A vertical auger dosing setup is used as a standard calibration device to extract three different bulk responses.

4.1 The Discrete Element Method

In DEM, grains, referred to in the following as particles, are treated as rigid bodies, which have translational and rotational degrees of freedom assigned to their center of mass. In this work, the focus is on the behavior of the free-flowing and the dry cohesive powder particles. In the DEM, particle interactions are modeled using the well-established Newton's equations of motion, contact laws, and overlap relationships. The soft-particle approach, originally developed by Cundall and Strack [10], is followed in this work, where the particles are allowed to undergo small deformations, i.e., overlaps. Thereafter, these deformations are used to calculate the elastic and plastic deformations besides the frictional forces between the particles [33]. As two particles i and j come in contact with each other, different interactions, i.e., forces and torques due to gravity, deformations under collisions, as well as static and rolling frictions, might occur. Translational and rotational motion of particle i with a mass m_i and a moment of inertia I_i are represented by the following equations:

$$m_i \frac{dv_i}{dt} = \sum (\mathbf{F}_{ij}^n + \mathbf{F}_{ij}^t) + m_i \mathbf{g} , \quad (4.1)$$

$$I_i \frac{d\omega_i}{dt} = \sum (\mathbf{R}_i \times \mathbf{F}_{ij}^t - \boldsymbol{\tau}_{ij}^r) , \quad (4.2)$$

where $\mathbf{g}$, $\mathbf{v}_i$ and $\boldsymbol{\omega}_i$ are the gravity vector, translational velocity, and rotational velocity of particle i , respectively. The interaction between particles i and j at a defined time step leads to normal and tangential forces, i.e., $\mathbf{F}_{ij}^n$ and $\mathbf{F}_{ij}^t$, respectively. $\mathbf{R}_i$ is the vector between the center of particle i and the contact point, where the tangential force $\mathbf{F}_{ij}^t$ is applied. $\boldsymbol{\tau}_{ij}^r$ is the torque due to rolling friction between the two particles.

4.1.1 Hertz-Mindlin Contact Model

Hertz–Mindlin [121,122] contact law is the most common applied approach in modeling contacts between particles. It is used in calculating the contact forces introduced due to the slight considered overlap between particles. The frictional contact force between two granular particles i and j comprises normal $\mathbf{F}_{ij}^n$ and tangential $\mathbf{F}_{ij}^t$ nonlinear contact forces given as

$$\mathbf{F} = \mathbf{F}_{ij}^n + \mathbf{F}_{ij}^t . \quad (4.3)$$

Each contact force consists of two terms, where the first term stands for the nonlinear elastic Hertz model in the normal direction (spring force) and the linear elastic Mindlin model in the tangential direction (shear force). The second term stands for the damping force (modeled as a dashpot), where a dissipative term is added for both normal and tangential directions to account for energy dissipation during collisions through inelastic deformation and friction. The normal and tangential forces between two spheres i and j during a collision are given by

$$\mathbf{F}_{ij}^n = k_n \boldsymbol{\delta}_{ij}^n - \gamma_n \mathbf{v}_{ij}^n , \quad (4.4)$$

$$\mathbf{F}_{ij}^t = k_t \boldsymbol{\delta}_{ij}^t - \gamma_t \mathbf{v}_{ij}^t , \quad (4.5)$$

where k_n and k_t are the elastic constants for the normal and tangential contact, $\boldsymbol{\delta}_{ij}^n$ is the normal contact overlap, given by

$$|\boldsymbol{\delta}_{ij}^n| = R_i + R_j - l_{ij} , \quad (4.6)$$

where l_{ij} is the distance between the centers of the two particles, $\boldsymbol{\delta}_{ij}^t$ is the tangential contact overlap or displacement vector which is truncated to satisfy a frictional yield criterion, given by the integral of the tangential relative velocity through the collision time they are in contact, i.e.,

$$|\boldsymbol{\delta}_{ij}^t| = \int_0^t |\mathbf{v}_{ij}^t| dt , \quad (4.7)$$

$\mathbf{v}_{ij}^n$ and $\mathbf{v}_{ij}^t$ are the normal and tangential components of the relative velocity of the two particles at the contact, γ_n and γ_t are the viscoelastic damping constant for normal and tangential contact. Figure 4.1

illustrates the mechanical system of the interactions of particles in the Hertz–Mindlin contact model.

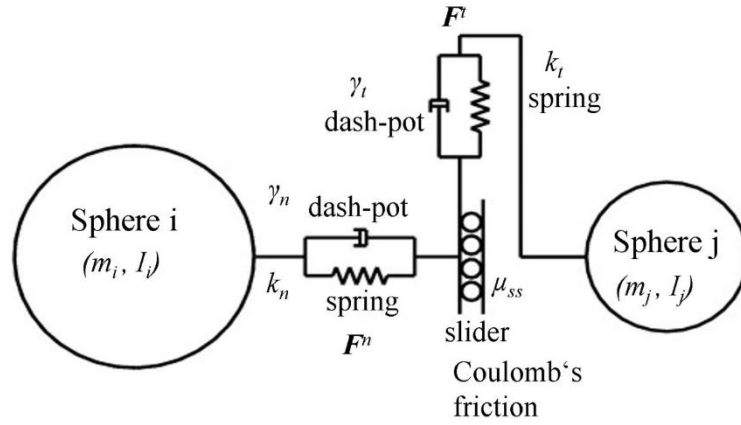


Figure 4.1: Schematic representation of the Hertz–Mindlin contact model between particle i and j [123].

Under sufficient tangential forces, particles will slip relative to each other or relative to surfaces they are in contact with. For noncohesive particles subjected to a constant normal force, the extent of slippage under tangential force is determined by

$$F_{ij}^t < \mu_{ss} |F_{ij}^n|, \quad (4.8)$$

where μ_{ss} , represented by a shear slider, is the static friction coefficient between the two particles in the tangential direction. Thus, the tangential force described by a spring and a damper is limited to the Coulomb friction.

Consequently, based on Equations (4.4) and (4.5), the normal and tangential forces, F_{ij}^n and F_{ij}^t can be expressed in detail as

$$F_{ij}^n = -\frac{4}{3}Y^* \sqrt{R^* \delta_{ij}^n \delta_{ij}^n} - 2\sqrt{\frac{5}{6}}\beta \sqrt{S_n m^*} v_{ij}^n, \quad (4.9)$$

$$F_{ij}^t = -8G^* \sqrt{R^* \delta_{ij}^n \delta_{ij}^t} - 2\sqrt{\frac{5}{6}}\beta \sqrt{S_t m^*} v_{ij}^t, \quad (4.10)$$

where Y^* is the equivalent Young's modulus of the two colliding particles. This is defined by

$$\frac{1}{Y^*} = \frac{1-v_i^2}{Y_i} + \frac{1-v_j^2}{Y_j}, \quad (4.11)$$

where v_i and v_j are Poisson's ratios, R^* is the equivalent radius, defined by

$$\frac{1}{R^*} = \frac{1}{R_i} + \frac{1}{R_j}, \quad (4.12)$$

and m^* is the equivalent mass, defined by

$$\frac{1}{m^*} = \frac{1}{m_i} + \frac{1}{m_j}. \quad (4.13)$$

Moreover, $S_n = 2Y^* \sqrt{R^* \delta_{ij}^n}$ and $S_t = 8G^* \sqrt{R^* \delta_{ij}^n}$ are the normal and tangential contact stiffness with G^* being the equivalent shear modulus, defined as

$$\frac{1}{G^*} = \frac{2(2-\nu_i)(1+\nu_i)}{Y_i} + \frac{2(2-\nu_j)(1+\nu_j)}{Y_j}. \quad (4.14)$$

Besides, β is the damping ratio coefficient, which is a function of the coefficient of restitution, e , and given by $\beta = \ln(e)/\sqrt{\ln^2(e) + \pi^2}$. The coefficient of restitution e is the ratio of relative velocity after collision to the relative velocity before collision. A value of $e = 1$ refers to a perfectly elastic collision, whereas $e = 0$ would mostly be an inelastic collision.

4.1.2 Rolling Friction Model

In addition to sliding friction, rolling friction is present as a contact-dependent force [124]. Rolling friction is a major parameter that serves as a correction factor by compensating for the use of spherical particle shapes in DEM [125], instead of representing their actual shapes in reality. In the literature, different rolling resistance models have been developed and reviewed [125]. In this work, the alternative elastic-plastic spring-dashpot (EPSD2) rolling resistance model [126], implemented in LIGGGHTS open-source code, is applied to calculate the torque $\tau_{ij,t+\Delta t}^r$ at an incremental time step as

$$|\tau_{ij,t+\Delta t}^r| \leq \mu_r R^* |\mathbf{F}_{ij}^n|, \quad (4.15)$$

where μ_r is the rolling coefficient. In this model, the damping torque is disabled for simplicity.

4.1.3 Cohesion Model

The modified simplified Johnson-Kendall-Roberts cohesion model (SJKR), implemented in LIGGGHTS, is used to model the cohesion of particles. This model originates from the JKR model [127], where it represents the influence of various cohesive forces, such as Van der Waals within the contact zone. It can model high adhesive systems, such as fine and dry powders or wet materials. It adds an additional normal force component F_{coh} to the Hertzian contact law tending to maintain a larger contact area (Figure 4.2), i.e.,

$$F_{coh} = Ca, \quad (4.16)$$

where C is the cohesion energy density in J m^{-3} and a is the particle contact area, given as

$$a = \frac{\pi (l_{ij}-R_i-R_j)*(l_{ij}+R_i-R_j)*(l_{ij}-R_i+R_j)*(l_{ij}+R_i+R_j)}{4 l_{ij}^3}, \quad (4.17)$$

where R_i and R_j are the radii of sphere i and j , respectively, and l_{ij} is the distance between the particle centers.

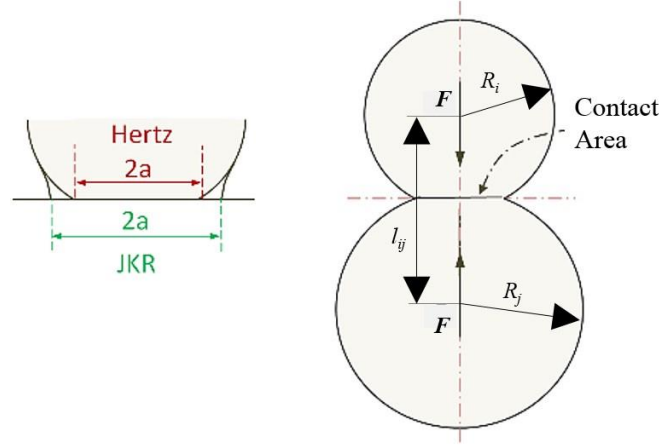


Figure 4.2: Schematic representation of the contact area for the JKR model, compared to that of the Hertz model [127].

4.1.4 Non-contact Force

In addition to the contact forces, the gravitational acceleration g in the z -direction (with $\hat{z}$ as the unit vector), yields the gravitational force of particle i , $\mathbf{F}_i^{grav}$, which can be expressed as

$$\mathbf{F}_i^{grav} = -m_i g \hat{z}. \quad (4.18)$$

4.1.5 Time Step in DEM Simulations

In the DEM simulations of particles flow in this work, the time step (Δt) has been selected to be less than 20 % of the critical time step (Rayleigh time) T_R in order to ensure the stability of the integration scheme. T_R is calculated based on the Rayleigh wave propagation, which represents the elastic wave propagation along the surface of one particle to the adjacent contacting particle according to the following equation [128]:

$$\Delta t < T_R = \frac{\pi R \sqrt{\frac{\rho}{G}}}{0.1631\nu + 0.8766}, \quad (4.19)$$

where ρ is the particle density, G is the shear modulus and ν is the Poisson's ratio.

4.2 Coarse-graining Method

To find a trade-off between computational time and accuracy, it is reasonable to limit the number of particles in the DEM simulation. According to Bierwisch et al. [70], coarse-graining (CG) method is a good approach to save significant computational time but DEM simulations with good accuracy are performed in comparison with real tests. In the coarse-graining technique, the original particles with radius R are substituted by larger grains with radius R' . The scaled-up radius R' can be represented as a multiplication of the original grain radius R and a scaling factor S_f . Moreover, CG method is based on an appropriate adjustment of interaction laws and equations of motion according to scaling rules for the material parameters in such a way that the energy densities, i.e., gravitational potential and kinetic energy, are preserved in the scaled-up system as the original one. As the potential density does not depend on the particle's radius while maintaining a constant volume fraction and particle's density, the particle's density is ought to be fixed upon applying the CG method. According to the coarse-graining proposed by Bierwisch et al. [70], the volume fraction is not affected and, therefore, the potential energy of the scaled system is comparable to the original system. Consequently, the mass of each particle introduced in the system is scaled up by a factor of S_f^3 according to Equation (4.20), but as the total number of particles is decreased also by the same factor, the total mass of the particles is conserved:

$$m' = \frac{4}{3}\pi R'^3 \rho' = S_f^3 m. \quad (4.20)$$

Bierwisch et al. [70] also proved that the energy dissipation per volume and time is preserved if the coefficient of restitution remains the same after applying the coarse-graining method. Thus, the kinetic energy density is preserved because the scaling does not affect the particle velocities. In connection with the applied test in this work, i.e., AoR, it is shown that coarse graining has no considerable effect on the resulting angles. In addition, it was observed that the shape of the heap is changing with increasing the particles sizes, where a noticeable tail of the ground heap and a more procumbent peak could be seen for larger particles. Thus, although the CG method leads to highly acceptable results in terms of bulk properties, it is hard to get exactly the same shape of the heap in the DEM simulation after applying the CG method.

Therefore, in order to decrease the computation time in all the DEM simulations, a scaling-up principle based on coarse-graining method is implemented by increasing the size of particles introduced in the system while maintaining the size of the system geometry [129–132]. This leads in reducing the computational costs of the DEM simulations. In this, the number of particles is decreased dramatically, i.e., by a scaling factor of S^3 . It is important to notice that the upscaling should not affect the velocity of the moving geometries and the gravitational acceleration.

4.3 Multivariate Regression Analysis (MVRA)

MVRA is a statistical technique applied to analyze the variation of more than one outcome variable and thus estimates a regression model. This technique was utilized to observe and analyze the variation and impact of different DEM input parameters on the output parameters. Cornerstone software was used to design the experiments (simulations) and to evaluate the results. By employing Cornerstone, the collected data values from the experiments are standardized; in such a way that the original variables are scaled to have a mean of 0 and a variance of 1 as each attribute has a different unit of measurement. The goodness of fit of the model was described by the adjusted coefficient of determination (R^2). This coefficient describes the proportion of the variability of the data explained by the model, which is adjusted for the number of factors in the model. The confidence level used in the DoE analysis was $\pm 95\%$. An adjusted response graph was used to show the effect of each input parameter on the responses besides showing all data points of the simulation runs, which are the adjusted values and not the real measured values. The response values are adjusted to average out the effects of the other input parameters. Furthermore, a Pareto graph of effects analysis was employed, where it expresses the relative size of the effect of each factor studied in the model. It represents the orthogonally-scaled effects of each investigated factor on the studied responses.

Figure 4.3 illustrates the general flowchart followed in the systematic calibration approach. From the auger dosing experiments, the results of the bulk responses to be studied are considered as the reference targets. The DEM input parameters are varied according to the DoE design model and studied using multivariate quadratic regression analysis. The prediction model is then used to calibrate the parameters by selecting an optimized parameter combination, where a good fit is obtained between the simulation results and the experimental data.

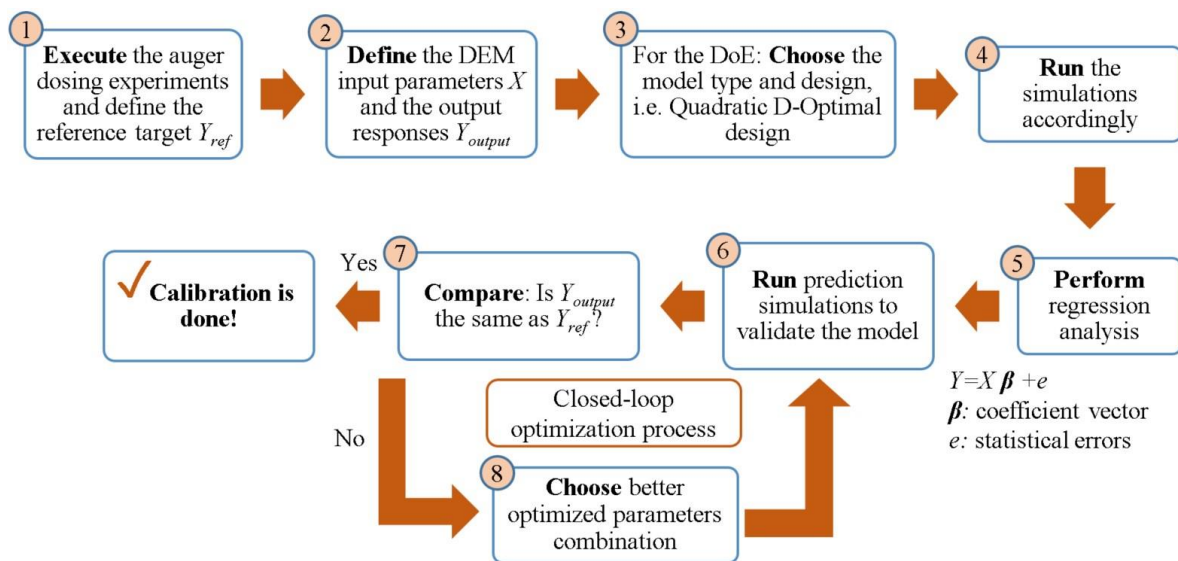


Figure 4.3: The general flowchart of the calibration process.

4.4 Simulation Steup

The DEM simulations were performed using the open-source software LIGGGHTS® -Premium 4.X from DCS Computing GmbH (Linz, Austria). The simulations were carried out on an eight-node high-performance cluster using 32 CPUs (Intel Xeon 2.60 GHz). The postprocessing of the simulations was performed using ParaView, version 5.4.1 64-bit from Kitware Inc. (New York, NY, USA). The three-dimensional (3D) CAD models used in the DEM simulations were modeled in SolidWorks version Premium 2016 from Dassault Systèmes (Vélizy-Villacoublay, France). After that, the 3D models were meshed using Gmsh, version 3.0.6 [133].

4.5 Case Study 1: Angle of Repose and Porosity

In the first case study, a novel simplified semi-automated parametric study and calibration approach is developed using an optimized MVRA. Initially, experimental powder characterization and AoR tests are performed. DoE method, using quadratic D-optimal design model, are applied for the sake of minimizing the number of simulations in an optimized way that retains high significance by recommending suitable possible combinations between DEM microscopic input parameters that should be tested together. The D-optimal design proved its efficiency by building a robust model by saving a lot of time and efforts compared to other models that require much more data points such as full factorial design. The proposed simulation runs from the DoE were performed in a parallel manner by implementing a code programming through the open-source software LIGGGHTS® [134]. The studied microscopic input parameters are the coefficients of static and rolling friction, coefficient of restitution, particle size, Young's modulus and cohesion energy density. The effects of these input parameters on five macroscopic output responses (targets), i.e., AoR, porosity, mass flow rate, translational kinetic energy and computation time, are studied through MVRA, using Cornerstone software, to find out which parameters and in what sense do they influence these targets. This systematic approach was done for a free-flowing powder and a cohesive powder, namely SpheroLac 100 and Inhalac 251, respectively. This method gives a more robust understanding of the impact of the interparticle input parameters on the flowability of the powders at different ranges (low, center, and high) by tuning them in a methodical way rather than trial-and-error. Since it is possible to have the same AoR with various DEM input parameter combinations [135,136], the porosity is measured along with the AoR in DEM simulations in order to determine a more reliable and practical value for the studied input parameters. Accordingly, a prediction model is built to validate the applied regression analysis and to calibrate the experimental AoR and porosity values by selecting one optimized parameter combination.

4.5.1 Experimental Angle of Repose

The AoR is defined as the slope of a poured conical pile of loose uncompacted bulk solid material [46]. It is a characteristic related to the cohesiveness of the powder [2] and used as a common method to characterize powders according to their flowability [101,103,137,138]. It is a common method to calibrate DEM simulation parameters, where the angle at which a heap of powder settles depends on the strength of the inter-particulate friction. This simple test case was chosen to perform a parametric study of the DEM microscopic input parameters. As depicted in Figure 4.4, a funnel made of 1.4404 stainless steel was placed above a catching container made of borosilicate 3.3 glass. Initially, the funnel was loosely filled with 5 g of powder where the lower opening of the funnel was closed. Then, the funnel was opened to let the powder discharge under gravity forming a pile on the bottom of the catching container. The test for each powder was repeated five times, and an average of the obtained angles was considered as a value for the AoR for each powder.

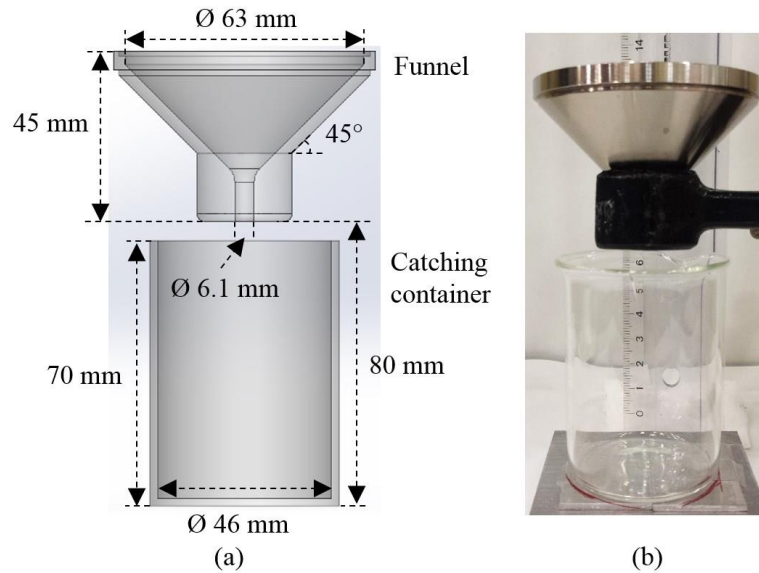


Figure 4.4: AoR test in (a) two-dimensional CAD view and (b) initial laboratory equipment setting.

4.5.2 Simulation Procedure and Parameters

4.5.2.1 Fixed Parameters throughout the Study

In the DEM simulations, the Hertz–Mindlin contact model and the rolling friction model (EPSD2) were applied for both the free-flowing and the dry cohesive studies. The adhesion contact model (SJKR) was added to the cohesive study. The values of the Young's modulus (Y) and the Poisson's ratio (ν) for the funnel and the catching container were selected from the literature [139]. The boundary conditions such as restitution and friction between particle and walls (funnel and glass in this case) were kept constant to reduce the number of simulations in the DoE. Table 4.1 lists the input parameters in the applied two DEM AoR studies.

Material parameter/contact model	Symbol	Unit	Value
Contact model			Hertz–Mindlin
Rolling friction model	EPSD2		Alternative elasto–plastic spring-dashpot model
Cohesion contact model ²	SJKR		Modified simplified Johnson–Kendall–Roberts model
Young’s modulus funnel ¹	Y_F	GPa	200
Young’s modulus catching container ¹	Y_C	GPa	67.8
Poisson’s ratio funnel ¹	ν		0.28
Poisson’s ratio catching container ¹	ν		0.21
Poisson’s ratio particle	ν		0.3
Coefficient of restitution particle–wall	e_{pw}		0.2
Coefficient of rolling friction particle–wall	$\mu_{r,pw}$		0.2
Coefficients of restitution and friction wall–wall			0.2
Acceleration due to gravity	g	m/s ²	9.81
Cohesion energy density particle–wall ^b	C_{pw}	J/m ³	1000
Cohesion energy density wall-wall ^b	C_{ww}	J/m ³	100

¹ [139]

² Set for the cohesion study

Table 4.1: DEM input parameters considered throughout the two studies.

4.5.2.2 Parameters Varied in the Design of Experiments

In the free-flowing study, five particle properties were varied as part of the DoE which were as follows: particle’s Young’s modulus (Y), particle size (d), coefficient of restitution (e_{pp}), coefficient of static friction ($\mu_{s,pp}$) and coefficient of rolling friction ($\mu_{r,pp}$). In addition to these parameters, cohesion energy density (C_{pp}) was varied in the dry cohesion study. The latter four parameters are related to the particle-to-particle interactions.

In the calibration process, the variation study of the parameters aimed to give the best fit between the DEM simulation and the corresponding reference data. To this end, the $\mu_{s,pp}$ was determined using the FT4 powder rheometer, but because of reducing the Y parameter due to computational limitations, this coefficient was calibrated to get the same bulk solid behavior [135,140]. e_{pp} and $\mu_{r,pp}$ were also tuned because they are difficult to be determined for microscale powders. When using spherical particle shape, the rolling resistance is considered to be less dominant over the static friction as the contact area is smaller. The range value of the cohesion energy density was roughly estimated by performing presimulation runs or so-called initial screening investigations.

Table 4.2 lists the values of the six particle properties that were varied in our studies. Three values, which constitute a low, center and high levels for each parameter, were selected. The center values of e_{pp} , $\mu_{s,pp}$, $\mu_{r,pp}$ and C_{pp} are linearly spaced between low and high values, whereas for Y , the center value is spaced on a logarithmic scale. The values of Y in the DoE are logarithmic where the simulation values 0.026, 0.26 and 2.6 GPa are set to 7.41, 8.41 and 9.41 log GPa to ensure a center point in the analysis.

Material parameter	Symbol	Unit	Value
Young's modulus of particles	Y	GPa	0.026, 0.26, 2.6
Particle diameter of SpheroLac 100	d_S	μm	460, 575, 690
Particle diameter of InhaLac 251	d_I	μm	480.6, 587.4, 694.2
Coefficient of restitution particle–particle	e_{pp}		0.051, 0.3255, 0.6
Coefficient of restitution particle–particle ²	e_{pp}		0.051, 0.2755, 0.5
Coefficient of static friction particle–particle	$\mu_{s,pp}$		0.1, 0.48, 0.85
Coefficient of static friction particle–particle ²	$\mu_{s,pp}$		0.2, 0.5, 0.8
Coefficient of rolling friction particle–particle	$\mu_{r,pp}$		0.05, 0.4, 0.75
Coefficient of rolling friction particle–particle ²	$\mu_{r,pp}$		0.1, 0.4, 0.7
Cohesion energy density particle–particle ^b	C_{pp}	J/m ³	30,000, 50,000, 70,000

² Set for the cohesion study

Table 4.2: List of DEM particle input parameters varied in the DoE of the two studies.

Three-level design was used where each factor was considered at three levels. For the free-flowing powder, 40 simulation runs were executed whereas 53 runs for the dry cohesive one.

4.5.3 DEM Studied Responses

In preparation of the mesh for the simulations, all the faces that have no contact with the particles were deleted in SolidWorks in order to optimize the computation time. Figure 4.5 shows the modified mesh file of the funnel in ParaView filled with spherical particles as an initial state before removing the stopper and allowing the particles to flow out.

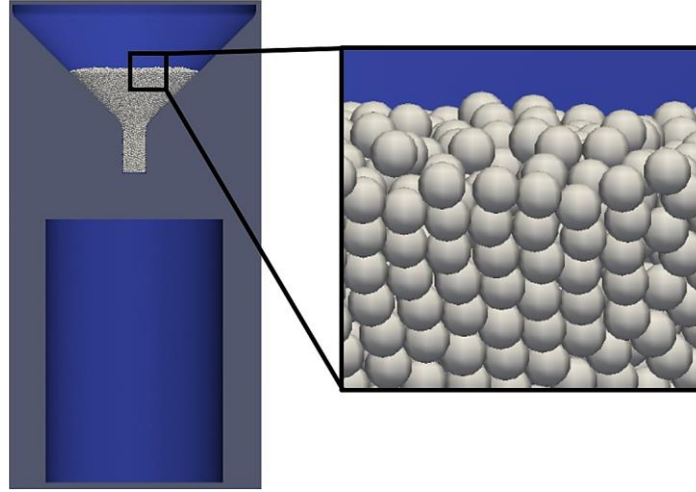


Figure 4.5: Initial state of particles filled inside the modified mesh funnel in ParaView software.

Five macroscopic output responses, namely AoR, porosity (ϵ), mass flow rate ($\dot{m}$), overall translational kinetic energy (TKE) and the computation time (CT), were studied on the two powders. The porosity was measured in the funnel due to the simplicity in defining a common volume since no heap (zero AoR) was formed in some simulations. The $\dot{m}$ at the discharge of the funnel was studied, where the mass values were measured every 25 ms. The overall translational kinetic energy (TKE) of the system was calculated over time to check how the dissipation of energy varies according to different simulation input combinations. The average of the TKE, after opening the funnel, was calculated and then normalized (NA-TKE) with the total discharge time of the particles. Table 4.3 reports the DoE followed and its simulation results of five output responses under the variation in five input parameters for the free-flowing powder.

Run	$\mu_{r,pp}$	$\mu_{s,pp}$	e_{pp}	Y (log GPa)	d_s (μm)	AoR ($^\circ$)	ϵ (%)	$\dot{m}$ (g/25ms)	NA-TKE (mJ/s)	CT (hr)
1	0.05	0.1	0.051	7.41	460	15.7	43.37	0.1562	0.28395	3.417
2	0.75	0.1	0.051	7.41	460	18.3	43.65	0.1562	0.27038	3.633
3	0.05	0.85	0.051	7.41	460	17.9	47.47	0.1282	0.19147	3.083
4	0.4	0.48	0.3255	7.41	460	34.2	48.61	0.119	0.15499	3.067
5	0.75	0.1	0.6	7.41	460	0	43.56	0.1562	0.27767	3.117
6	0.05	0.85	0.6	7.41	460	25.5	46.25	0.1316	0.20199	2.486
7	0.75	0.85	0.6	7.41	460	30	49.83	0.1111	0.13257	3.1
8	0.75	0.85	0.051	8.41	460	37.8	52.31	0.1	0.10730	9.717
9	0.05	0.1	0.6	8.41	460	0	43.19	0.1562	0.28704	6.967
10	0.75	0.1	0.051	9.41	460	18	44.32	0.1515	0.26398	21.9
11	0.05	0.85	0.051	9.41	460	18.8	46.99	0.1282	0.18970	31.483
12	0.75	0.85	0.051	9.41	460	37.3	52.35	0.1	0.10924	31.883

13	0.05	0.1	0.3255	9.41	460	11.3	43.42	0.1562	0.28595	27.05
14	0.4	0.1	0.6	9.41	460	0	43.88	0.1562	0.27757	30.367
15	0.75	0.48	0.6	9.41	460	32.8	48.97	0.1163	0.15102	22.583
16	0.05	0.85	0.6	9.41	460	24	46.64	0.1316	0.20092	26.786
17	0.75	0.48	0.051	7.41	575	39	50.99	0.102	0.11446	1.017
18	0.4	0.85	0.3255	7.41	575	35	50.42	0.1064	0.12612	1.05
19	0.4	0.1	0.6	7.41	575	0	44.01	0.1471	0.24504	1.167
20	0.05	0.48	0.3255	8.41	575	24.5	46.65	0.1282	0.18215	4.333
21	0.75	0.1	0.6	8.41	575	0	44.03	0.1471	0.24895	3.53
22	0.05	0.1	0.051	9.41	575	17	44.36	0.147	0.24747	14.733
23	0.75	0.85	0.6	9.41	575	30.2	50.61	0.1064	0.12360	12.983
24	0.05	0.1	0.051	7.41	690	15.8	44.43	0.1429	0.22569	0.645
25	0.05	0.85	0.051	7.41	690	26.5	48.02	0.1163	0.14866	0.65
26	0.75	0.85	0.051	7.41	690	37.5	53.53	0.0887	0.08506	0.883
27	0.75	0.1	0.3255	7.41	690	17	44.39	0.1389	0.22449	0.633
28	0.05	0.1	0.6	7.41	690	0	44.29	0.1429	0.23232	0.486
29	0.75	0.48	0.6	7.41	690	33.5	48.25	0.1064	0.12525	0.5
30	0.05	0.85	0.6	7.41	690	27	46.70	0.119	0.15992	0.65
31	0.75	0.85	0.6	7.41	690	32.3	49.95	0.1	0.10861	0.533
32	0.4	0.1	0.051	8.41	690	18	45.24	0.1776	0.27474	2.283
33	0.4	0.85	0.6	8.41	690	33	50	0.1346	0.15721	2.5
34	0.75	0.1	0.051	9.41	690	18.5	45.42	0.1351	0.20840	5.817
35	0.4	0.48	0.051	9.41	690	34.3	50.47	0.102	0.11600	7.083
36	0.05	0.85	0.051	9.41	690	27.5	48.96	0.1163	0.14799	6.3
37	0.75	0.85	0.3255	9.41	690	35.8	52.59	0.0926	0.09369	5.483
38	0.05	0.1	0.6	9.41	690	0	44.39	0.1388	0.22906	4.967
39	0.75	0.1	0.6	9.41	690	0	45	0.1388	0.22104	5.733
40	0.05	0.85	0.6	9.41	690	25.5	47.40	0.119	0.15692	5.367

Table 4.3: Measurements of the five responses, namely AoR, ε , $\dot{m}$, NA-TKE and CT, from the free-flowing DEM simulations.

Table 4.4 shows the DoE used and the corresponding results of five output responses under the variation in six input parameters for the cohesive powder.

Run	$\mu_{r,pp}$	$\mu_{s,pp}$	e_{pp}	Y (log GPa)	d_l (μm)	C_{pp} (J/m ³)	AoR (°)	ε (%)	$\dot{m}$ (g/25m s)	NA- TKE (mJ/s)	CT (hr)
1	0.7	0.2	0.051	7.41	480.6	70,000	46.7	52.91	0.1263	0.17687	2.483
2	0.1	0.5	0.051	7.41	480.6	70,000	33	53.97	0.1174	0.15372	2.317
3	0.1	0.8	0.051	7.41	480.6	30,000	33.5	53.67	0.1202	0.16813	3.133
4	0.4	0.8	0.2755	7.41	480.6	70,000	43	57.05	0.0965	0.10367	3.133
5	0.7	0.8	0.2755	7.41	480.6	30,000	44	56.79	0.0964	0.10436	3.083
6	0.1	0.2	0.5	7.41	480.6	70,000	35.8	51.20	0.1264	0.18173	3.233
7	0.7	0.2	0.5	7.41	480.6	30,000	41	51	0.1263	0.18733	2.483
8	0.1	0.2	0.051	8.41	480.6	30,000	24.7	50.43	0.1429	0.22378	8.542
9	0.7	0.8	0.051	8.41	480.6	50,000	41.5	57	0.1	0.10710	8.633
10	0.4	0.2	0.5	8.41	480.6	50,000	29.8	50.95	0.1351	0.20982	7.317
11	0.7	0.5	0.5	8.41	480.6	70,000	39	54.51	0.1111	0.13691	7.231
12	0.1	0.8	0.5	8.41	480.6	30,000	31	52.50	0.1282	0.18427	8.455
13	0.1	0.2	0.051	9.41	480.6	70,000	23.5	51.21	0.1413	0.22382	21.85
14	0.7	0.2	0.051	9.41	480.6	30,000	35	51.72	0.1299	0.19235	21.985
15	0.1	0.5	0.051	9.41	480.6	50,000	29.5	52.43	0.1266	0.18295	23.1
16	0.7	0.8	0.051	9.41	480.6	70,000	39	57.06	0.0985	0.10935	22.95
17	0.1	0.8	0.2755	9.41	480.6	30,000	29.2	52.68	0.1267	0.17971	23.217
18	0.1	0.2	0.5	9.41	480.6	30,000	21.7	50.08	0.1412	0.23540	22.833
19	0.7	0.2	0.5	9.41	480.6	70,000	31.5	50.85	0.1232	0.20773	27.05
20	0.1	0.8	0.5	9.41	480.6	70,000	30.5	52.22	0.1266	0.18538	23.35
21	0.7	0.8	0.5	9.41	480.6	30,000	40.5	55.24	0.1072	0.12701	23.12
22	0.7	0.2	0.051	7.41	587.4	30,000	41.5	52.57	0.119	0.15362	1.433
23	0.1	0.5	0.5	7.41	587.4	30,000	31	52.66	0.1219	0.16557	1.1
24	0.4	0.8	0.5	7.41	587.4	50,000	40	56.08	0.1	0.11196	1.267
25	0.7	0.8	0.5	7.41	587.4	70,000	Blocked	—	—	—	—
26	0.7	0.8	0.5	7.41	587.4	30,000	38	56.57	0.098	0.10267	1.283
27	0.1	0.8	0.051	8.41	587.4	70,000	30.8	53.64	0.119	0.15489	3.514
28	0.1	0.2	0.2755	8.41	587.4	70,000	26	51.29	0.1351	0.20434	3.717
29	0.4	0.8	0.051	9.41	587.4	30,000	40	56.50	0.1064	0.12147	14.217
30	0.7	0.5	0.2755	9.41	587.4	50,000	41.5	55.99	0.1042	0.11988	15.767
31	0.1	0.2	0.051	7.41	694.2	50,000	37.5	52.30	0.119	0.15714	0.633
32	0.4	0.5	0.051	7.41	694.2	30,000	40.3	56.53	0.098	0.10468	0.717
33	0.1	0.8	0.051	7.41	694.2	50,000	39.3	54.62	0.1087	0.12998	0.6

34	0.1	0.2	0.051	7.41	694.2	30,000	30	51.62	0.125	0.15372	0.717
35	0.7	0.8	0.051	7.41	694.2	70,000	Block ed	—	—	—	—
36	0.7	0.8	0.051	7.41	694.2	30,000	42	58.89	0.0847	0.07738	0.767
37	0.1	0.2	0.5	7.41	694.2	30,000	27.5	51.05	0.125	0.18082	0.7
38	0.7	0.2	0.5	7.41	694.2	70,000	46	52.35	0.1042	0.11591	0.75
39	0.1	0.8	0.5	7.41	694.2	70,000	36.8	54.01	0.1087	0.13338	0.767
40	0.4	0.2	0.051	8.41	694.2	70,000	34.5	52.66	0.119	0.15716	1.95
41	0.7	0.2	0.2755	8.41	694.2	30,000	37.5	51.50	0.1219	0.16729	1.933
42	0.1	0.8	0.2755	8.41	694.2	30,000	31.5	53.36	0.1136	0.14516	1.967
43	0.1	0.5	0.5	8.41	694.2	70,000	29	52.86	0.119	0.15604	2.03
44	0.7	0.8	0.5	8.41	694.2	30,000	38	55.98	0.0962	0.10210	2.117
45	0.1	0.2	0.051	9.41	694.2	30,000	27	51.40	0.1282	0.17994	8.383
46	0.7	0.2	0.051	9.41	694.2	70,000	32.7	52.51	0.119	0.15717	4.7
47	0.1	0.8	0.051	9.41	694.2	70,000	30.5	54.06	0.1136	0.14289	6.617
48	0.7	0.8	0.051	9.41	694.2	30,000	41	57.84	0.0909	0.09058	5.9
49	0.4	0.5	0.2755	9.41	694.2	70,000	37	54.82	0.1064	0.12178	6.633
50	0.1	0.2	0.5	9.41	694.2	70,000	22.5	51	0.1316	0.19333	6.65
51	0.7	0.2	0.5	9.41	694.2	30,000	32.5	51.85	0.125	0.17160	6.883
52	0.1	0.8	0.5	9.41	694.2	30,000	29.5	53.26	0.1163	0.15217	4.983
53	0.7	0.8	0.5	9.41	694.2	70,000	37.5	55.83	0.1	0.10724	6.85

Table 4.4: Measurements of the five responses, namely AoR, ε , $\dot{m}$, NA-TKE and CT, from the cohesive DEM simulations.

4.6 Case Study 1: Results and Discussion

4.6.1 Angle of Repose

The AoR for each experiment was calculated by taking the average value between the left and right angles, θ_1 and θ_2 , respectively. The obtained results of the AoR for both powders were set as reference values in the validation of the regression analysis in the prediction model and thus in the calibration approach. Figure 4.6 shows the heap formed by the AoR test of the free-flowing powder (SpheroLac 100) in the glass-catching container where the average angle was 33° .



Figure 4.6: Experimental AoR of SpheroLac 100.

On the contrary, the average AoR of the cohesive powder (InhaLac 251) was 40° where Figure 4.7 shows the heap formed in the experiment.



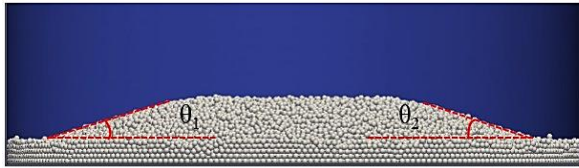
Figure 4.7: Experimental AoR of InhaLac 251.

4.6.2 Parametric Study

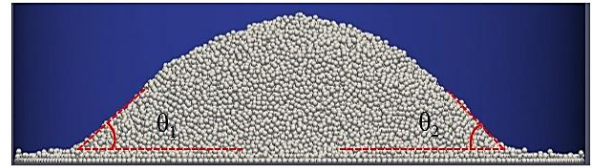
4.6.2.1 Factors affecting angle of repose

4.6.2.1.1 Free-flowing

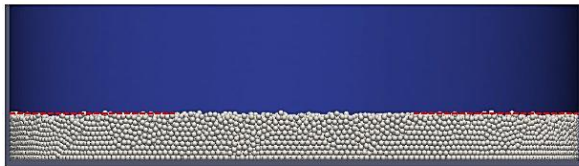
The average AoR for each simulation test was calculated. The values of AoR ranged between 0° and 39° , where in eight cases the AoR was equal to zero due to low $\mu_{s,pp}$ as listed in Table 4.3. Figure 4.8 presents numerical examples of six AoR DEM simulations for the free-flowing powder formed in the catching container.



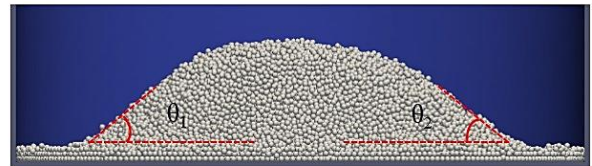
Run 1: $Y = 7.41 \log \text{ GPa}$, $d_s = 460 \mu\text{m}$, $e_{pp} = 0.051$, $\mu_{s,pp} = 0.1$, $\mu_{r,pp} = 0.05$, AoR = 16° .



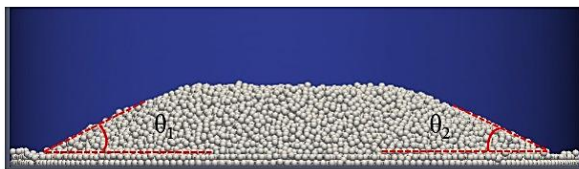
Run 8: $Y = 8.41 \log \text{ GPa}$, $d_s = 460 \mu\text{m}$, $e_{pp} = 0.051$, $\mu_{s,pp} = 0.85$, $\mu_{r,pp} = 0.75$, AoR = 38° .



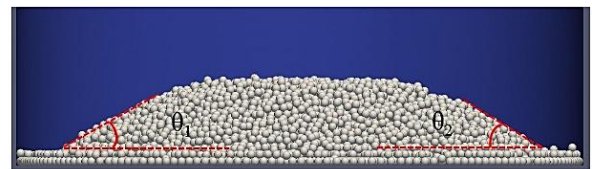
Run 9: $Y = 8.41 \log \text{ GPa}$, $d_s = 460 \mu\text{m}$, $e_{pp} = 0.6$, $\mu_{s,pp} = 0.1$, $\mu_{r,pp} = 0.05$, AoR = 0° .



Run 15: $Y = 9.41 \log \text{ GPa}$, $d_s = 460 \mu\text{m}$, $e_{pp} = 0.6$, $\mu_{s,pp} = 0.48$, $\mu_{r,pp} = 0.75$, AoR = 32.5° .



Run 20: $Y = 8.41 \log \text{ GPa}$, $d_s = 575 \mu\text{m}$, $e_{pp} = 0.3255$, $\mu_{s,pp} = 0.48$, $\mu_{r,pp} = 0.05$, AoR = 24.5° .



Run 36: $Y = 9.41 \log \text{ GPa}$, $d_s = 690 \mu\text{m}$, $e_{pp} = 0.051$, $\mu_{s,pp} = 0.85$, $\mu_{r,pp} = 0.05$, AoR = 27.5° .

Figure 4.8: Illustration of AoR of six different DEM run conditions of the free-flowing powder.

The model of the AoR for the free-flowing powder showed an adjusted $R^2 = 0.954$, indicating a very good fit to the data points. Figure 4.9a shows the adjusted response graph for the AoR. Each of the presented sections (column-wise) corresponds to an input parameter. In each section, all the data points

obtained from the simulations are presented in addition to the effect drawn from these points. For a data point, the effects of the other factors included in the model were averaged; therefore, the data points were adjusted in such a way that the effect of the corresponding factor on the response is only presented (adjusted response graph). As the data points were recalculated, it can be seen that the actual values of the responses listed in Table 4.3 do not match with the adjusted values, yet the effect of each parameter can still be modeled. Finally, scattering points signify data points that cannot be described by the model due to some noise effects.

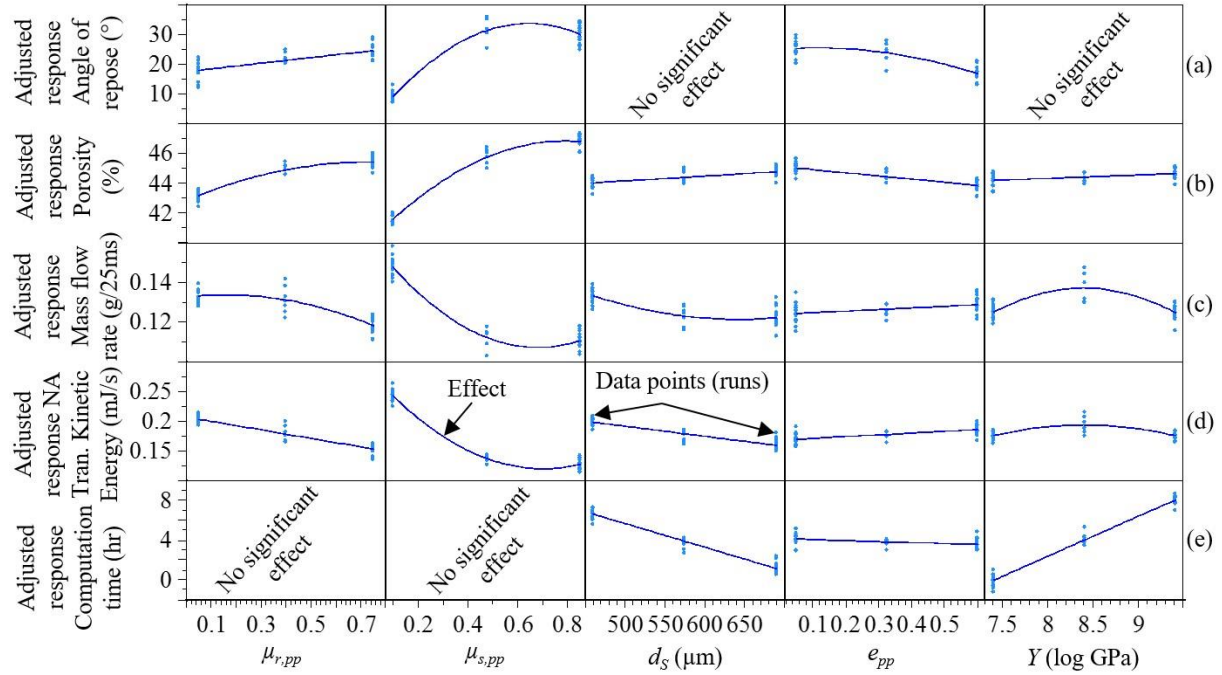


Figure 4.9: Adjusted response graph of the (a) AoR, (b) ε , (c) $\dot{m}$, (d) NA-TKE and (e) CT for the free-flowing as in relation to the five varied DEM input parameters.

From Figure 4.9a, the following trends can be observed regarding AoR: Y and d_s have no significant effects on the AoR, e_{pp} and $\mu_{r,pp}$ have a slight effect and $\mu_{s,pp}$ has the major effect. The absence of d_s as an effective factor to the AoR ensures the correct implementation of the coarse-graining scheme proposed by Bierwisch in our system. Moreover, the results obtained here contradict many studies which proved that the particle size has an impact on the AoR [2,50,53].

Figure 4.10 illustrates the Pareto graph of effects on the AoR of the free-flowing powder where the largest positive effects are shown on the left and the largest negative effects on the right, with factors of smaller estimated effects in the middle of the graph. It can be seen that $\mu_{s,pp}$ has the highest positive effect (proportional) on the AoR, where, as $\mu_{s,pp}$ increases, the AoR increases as well. The rest of the affecting factors, i.e., $\mu_{r,pp}$ and e_{pp} , interactions (e.g., $\mu_{s,pp} * \mu_{r,pp}$), and quadratic terms (e.g., e_{pp}^2) have lower effect compared to that of $\mu_{s,pp}$. To be stated, as e_{pp} increases, the AoR decreases, which is previously confirmed by Wei et al. [63]. As e_{pp} is the ratio of the relative velocity after collision to the relative velocity before collision of an object with a surface, the results can be explained by the fact that due to

the high elasticity of the spherical particles, they can more likely bounce than to settle down along a free surface.

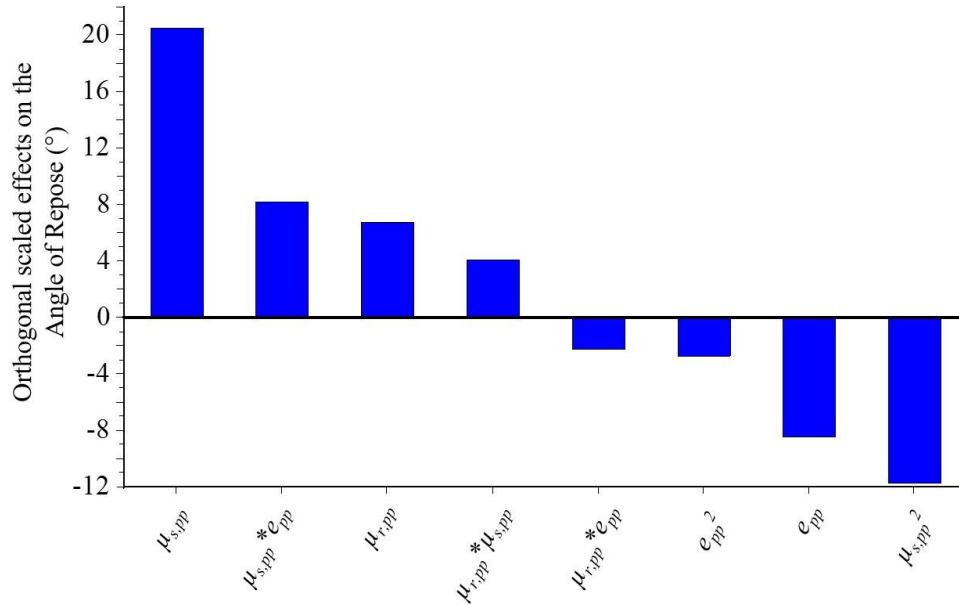


Figure 4.10: Pareto graph of effects of the model factors on the AoR of the free-flowing powder.

4.6.2.1.2 Cohesive

Figure 4.11 illustrates numerical examples of six AoR DEM simulations for the cohesive powder formed in the catching container.

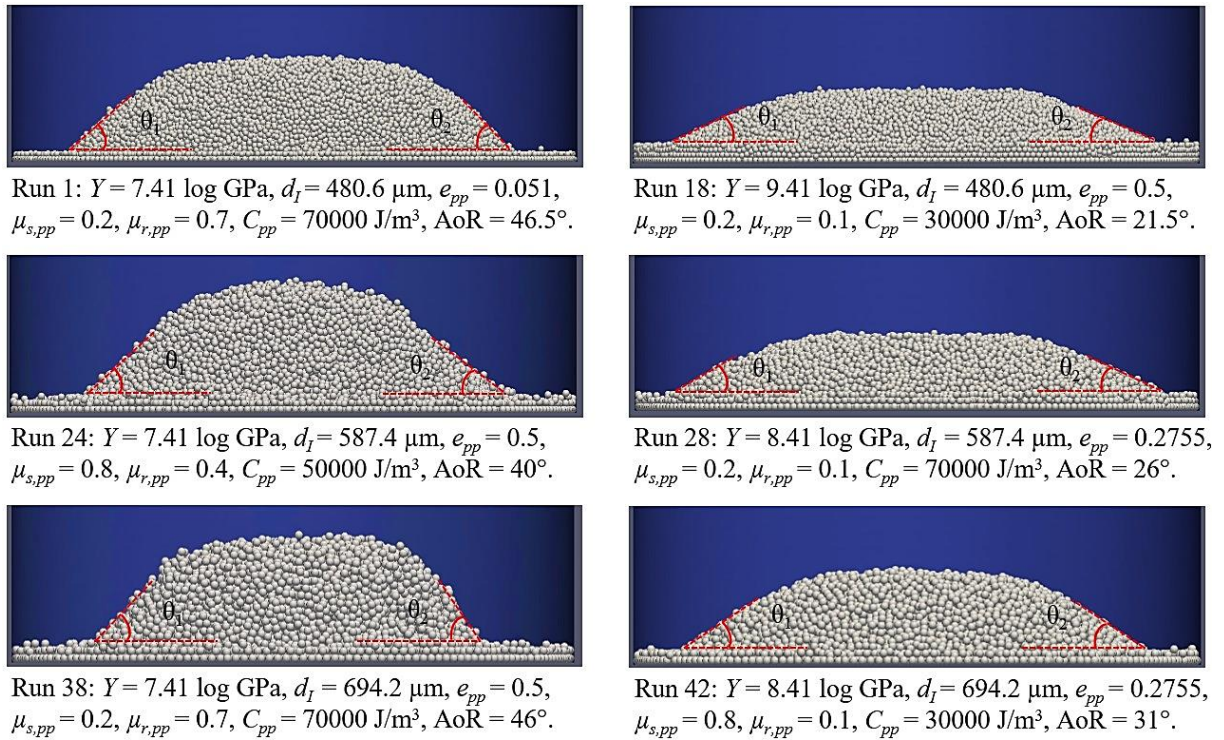


Figure 4.11: Illustration of AoR of six different DEM run conditions of the cohesive powder.

The adjusted R^2 of the AoR of the cohesive powder model has a very good value of 0.915. Figure 4.12a shows the adjusted response graph for the AoR. It indicates that the introduction of cohesion in the system through the SJKR model added a new affecting parameter on the AoR, which is Y . Unlike the free-flowing case, it is proved that Y has a significant negative effect on AoR when C_{pp} is involved. This ensures the importance of such a parametric study, where Y is not included in many other calibration processes due to its computational cost impact, though it has influence. From Figure 4.12a, it can be seen that the angle of repose increases with an increase in the sliding and rolling friction coefficients, which is proven in other studies as well [53,136,141–144]. This positive correlation is similar in both free-flowing and cohesive powders.

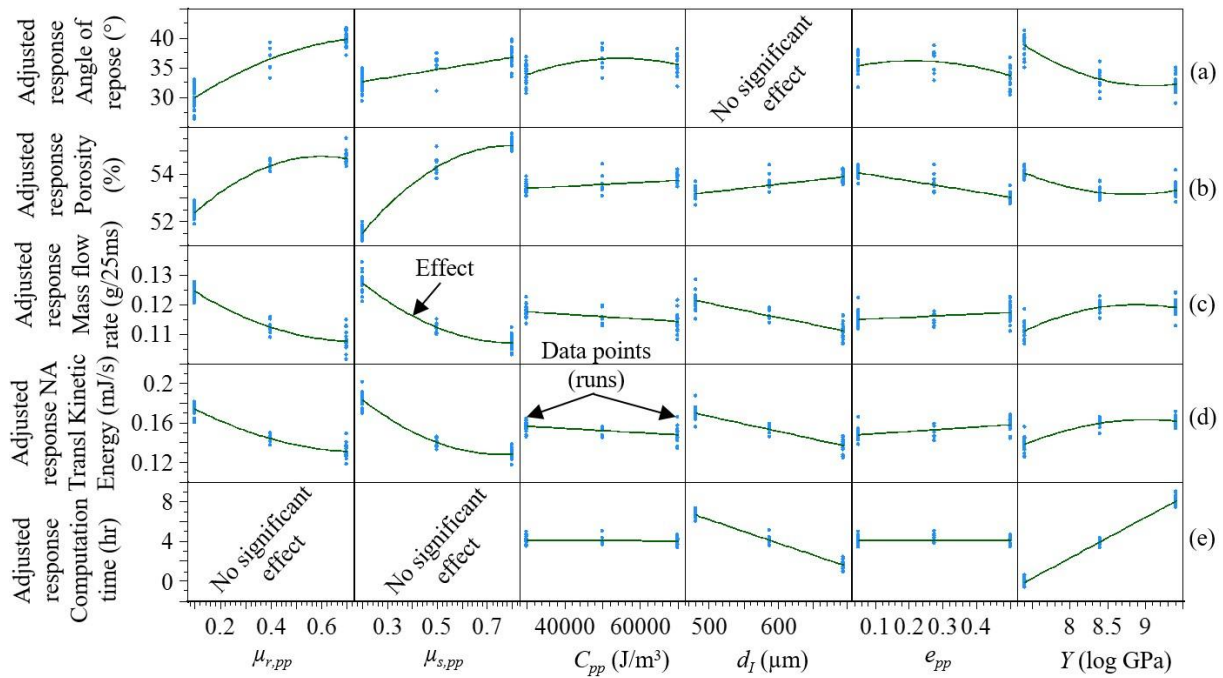


Figure 4.12: Adjusted response graph of the (a) AoR, (b) ϵ , (c) $\dot{m}$, (d) NA-TKE and (e) CT for the cohesive as in a relation to the six varied DEM input parameters.

Figure 4.13 demonstrates the Pareto graph of effects on the AoR of the cohesive powder. It shows that $\mu_{r,pp}$ has the highest positive effect on the AoR, which is contrary to most of the studies that proved the superiority of $\mu_{s,pp}$ in terms of effectivity. Therefore, it is very critical to always relate the effectiveness of the parameters to the circumference of the case study. Moreover, Y has a bigger impact on the AoR than $\mu_{s,pp}$.

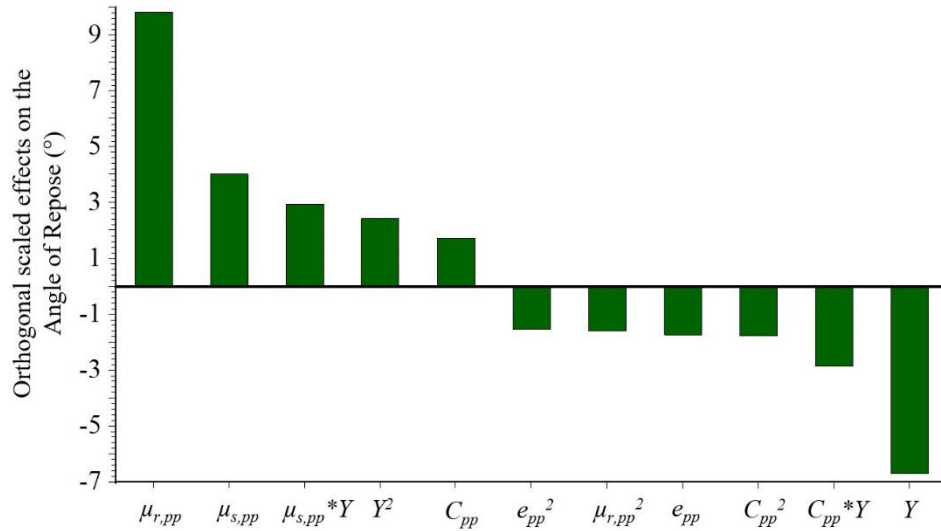


Figure 4.13: Pareto graph of effects of the model factors on the AoR of the cohesive powder.

4.6.2.2 Factors affecting porosity

4.6.2.2.1 Free-flowing

The regression analysis of the ε of the free-flowing powder resulted in a very good adjusted R^2 value of 0.981. The adjusted response graph of ε in Figure 4.9b demonstrates the high significance positive correlation of $\mu_{s,pp}$ on ε followed by $\mu_{r,pp}$, which was also proven by Wei et al. [63]. The other studied microscopic parameters indicated less significance, where the relative size of effects of these parameters is presented in Figure 4.14.

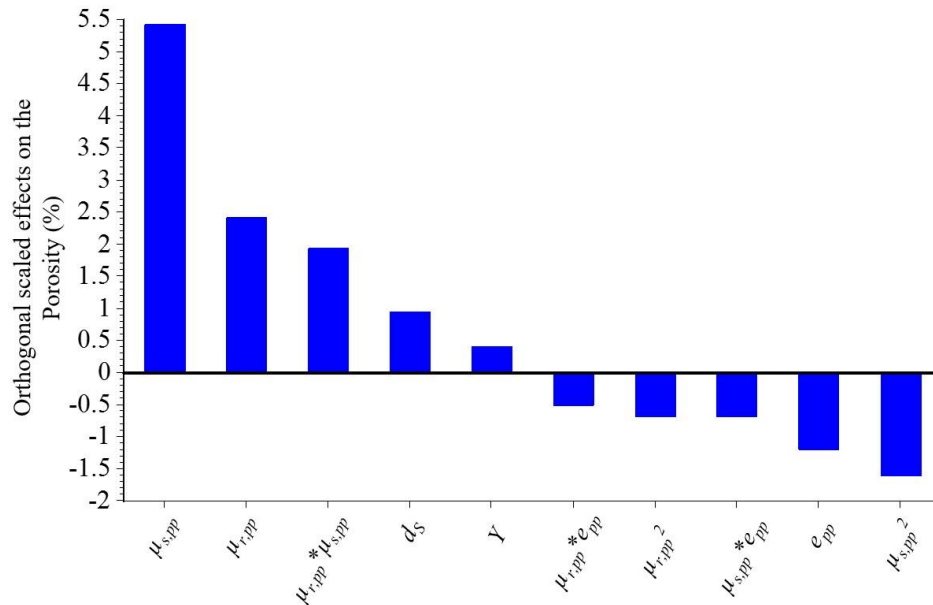


Figure 4.14: Pareto graph of effects of the model factors on the ε of the free-flowing powder.

4.6.2.2.2 Cohesive

The adjusted R^2 of the ε of the cohesive powder was very good as well with a value of 0.983. Similarly to the free-flowing powder, $\mu_{s,pp}$ had the highest positive influence on ε followed by $\mu_{r,pp}$ as revealed in Figure 4.12b. In addition, the Pareto graph of effect in Figure 4.15 shows the superiority positive relative effect of $\mu_{s,pp}$ on ε compared to the other parameters.

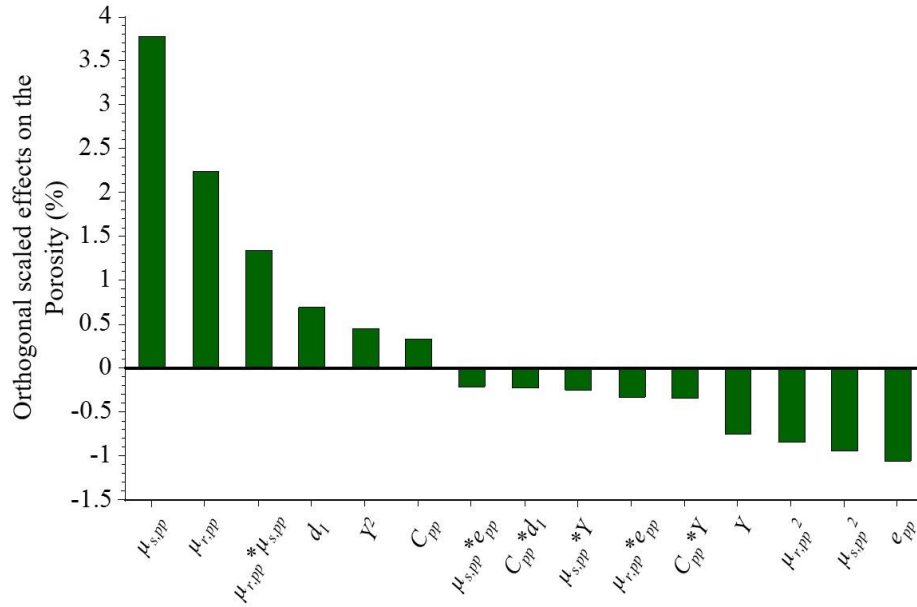


Figure 4.15: Pareto graph of effects of the model factors on the ε of the cohesive powder.

4.6.2.3 Factors affecting mass flow rate

4.6.2.3.1 Free-flowing

The adjusted R^2 of the $\dot{m}$ model of the free-flowing powder was recorded to be 0.95. The adjusted response graph for the $\dot{m}$ is shown in Figure 4.9c. As observed, all the studied parameters affect the $\dot{m}$, but each in a different way which is also presented in the Pareto graph (Figure 4.16). The Pareto graph shows that $\mu_{s,pp}$ has the highest negative effect on the $\dot{m}$. This indicates that as $\mu_{s,pp}$ increases, the particles' flow out of the funnel gets delayed due to the sliding friction between the particles. The same holds for $\mu_{r,pp}$ and d_s . Unlike e_{pp} , as it increases, $\dot{m}$ increases slightly.

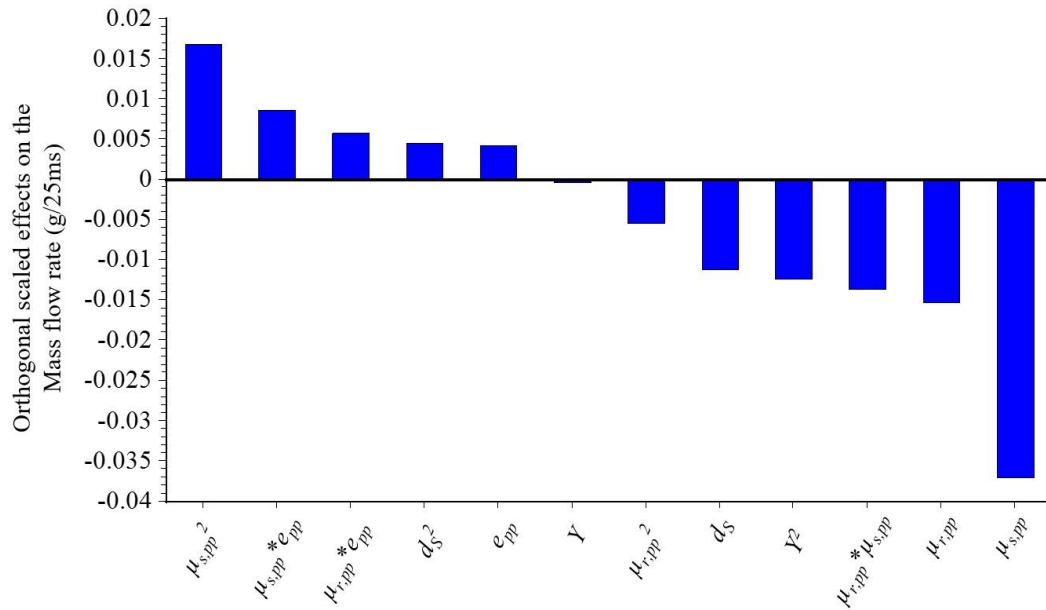


Figure 4.16: Pareto graph of effects of the model factors on the $\dot{m}$ of the free-flowing powder.

Following this, the mass and $\dot{m}$ of the six example simulations shown in Figure 4.11 were plotted over time in one graph (Figure 4.17) to compare them with each other. The parameter combinations and the mean $\dot{m}$ values of each run are presented in Table 4.3.

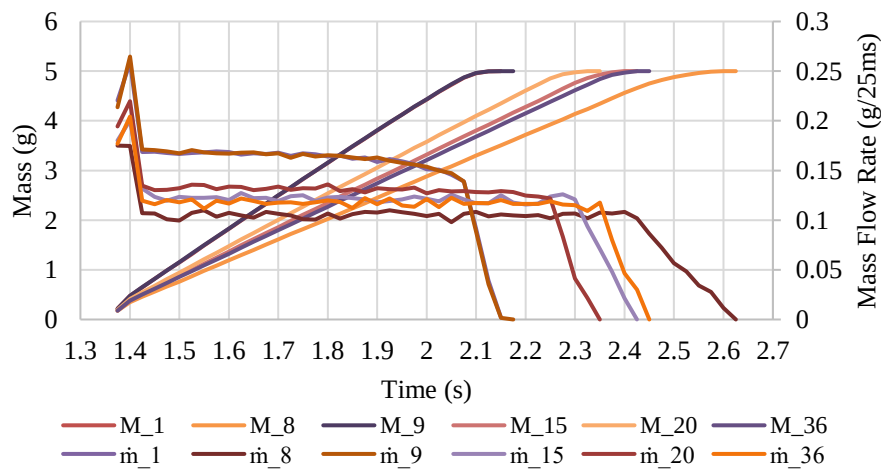


Figure 4.17: Mass and $\dot{m}$ over time of six DEM simulations of the free-flowing powder.

Figure 4.17 shows that the slopes of mass graphs M_1 and M_9 , corresponding to the first and ninth DEM simulations, have the highest $\dot{m}$ while having the lowest values of $\mu_{s,pp}$ (0.1). M_8 is showing the lowest slope ($\dot{m}$) with the assignment of having a high value of $\mu_{s,pp}$ (0.85).

4.6.2.3.2 Cohesive

The model fit of the $\dot{m}$ in the cohesive study was very good with an adjusted $R^2 = 0.957$. Figure 4.12c presents the adjusted response graph for the $\dot{m}$ and the Pareto graph (Figure 4.18) shows the weight value effect of the input parameters on the $\dot{m}$. It is apparently shown that $\dot{m}$ has been highly negatively

affected by $\mu_{s,pp}$, $\mu_{r,pp}$ and d_I in decreasing order of effectivity. A lower but significant positive effect has been witnessed for Y . This is due to the fact that particles with high stiffness tend to flow easier under the effect of gravity.

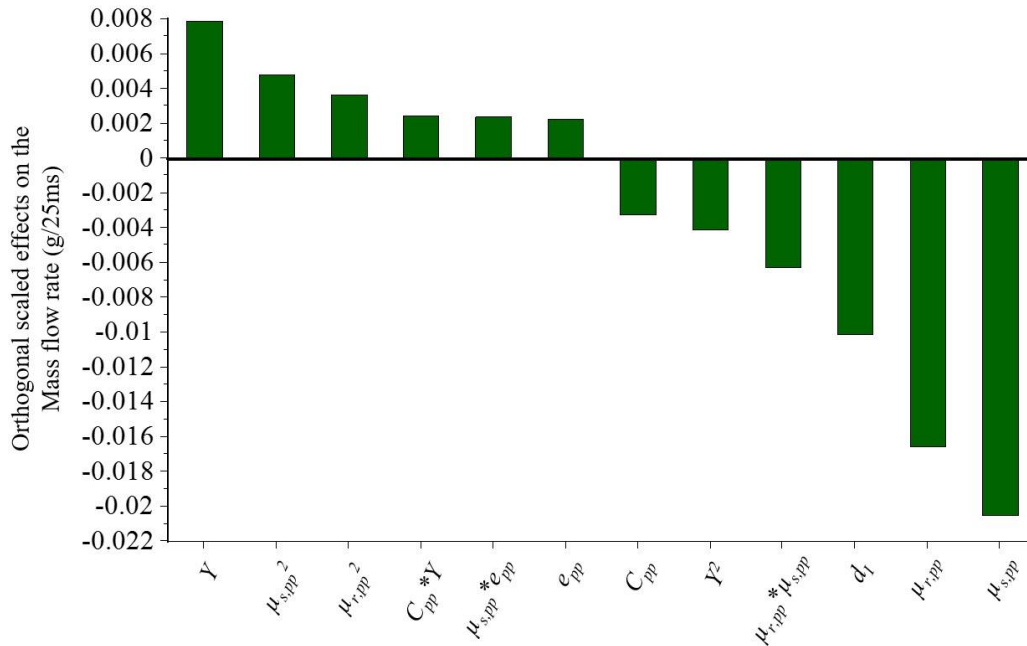


Figure 4.18 Pareto graph of effects of the model factors on the $\dot{m}$ of the cohesive powder.

More clear visualization of how $\dot{m}$ is varying with respect to the parameters can be seen in Figure 4.19. The figure shows an example of the mass and $\dot{m}$ of six simulation cases plotted over time. The varied input parameters and the mean $\dot{m}$ values of each run are presented in Figure 4.19.

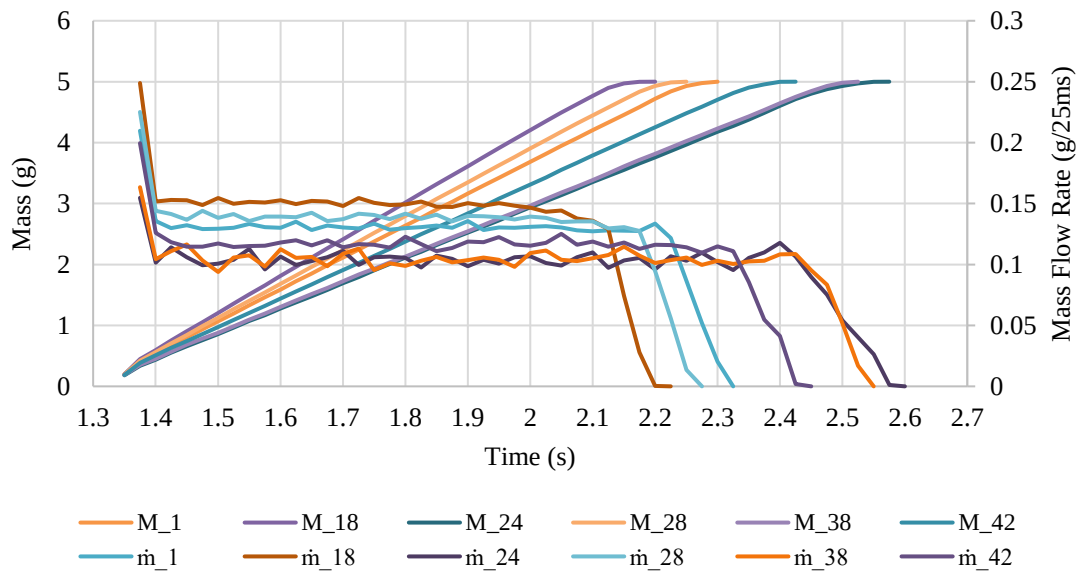


Figure 4.19: Mass and $\dot{m}$ over time of six DEM simulations of the cohesive powder.

4.6.2.4 Factors affecting kinetic energy

4.6.2.4.1 Free-flowing

The overall translational kinetic energy (TKE) was recorded every 1 ms starting from the moment of particle insertion inside the funnel till the discharge of the last particle inside the catching container. The results showed that $\mu_{s,pp}$ has a negative impact on the TKE as illustrated in an example of three runs in Figure 4.20.

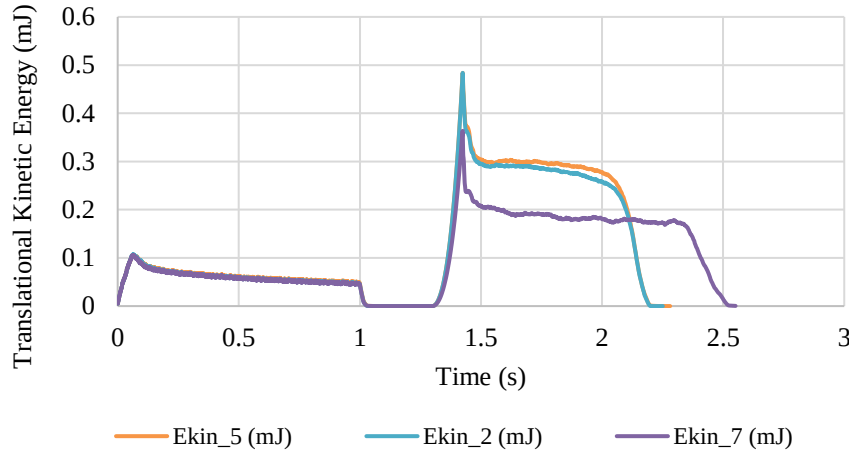


Figure 4.20: Overall TKE graph plot over time for simulation runs 2, 5 and 7 for the free-flowing powder. Conditions for these plots: $\mu_{r,pp} = 0.75$, $Y = 7.41 \log \text{ GPa}$, $d_s = 460 \mu\text{m}$, $e_{pp} = 0.6$ (E_{kin_5} and E_{kin_7}), $e_{pp} = 0.051$ (E_{kin_2}), $\mu_{s,pp} = 0.1$ (E_{kin_2} and E_{kin_5}) and $\mu_{s,pp} = 0.85$ (E_{kin_7}).

From 0 s to 1 s, the particles were inserted inside the funnel, then there was 0.3 s settlement time for the particles before removing the stopper at 1.3 s. This resulted in having an almost identical character of variation of TKE from 0 to 1.3 s. The rapid increase of TKE within 0–0.1 s is explained by the falling (insertion) of the particles into the funnel. Within 0.1–1 s, the distance between the insertion region and the first contact of the particles in the vertical direction decreased as the funnel was being filled with particles. Then, as the particles were settling down, the TKE started decreasing. In the time between 1–1.3 s, a rapid decrease in the TKE was observed as no more particles are inserted and most of the particles are got in rest (settled down). It can be seen that upon removing the stopper (at time 1.3 s), the peak of E_{kin_7} was smaller than that of E_{kin_5} and then E_{kin_7} remained smaller and the total time for full discharge was longer compared to E_{kin_5} . The only difference between these two runs is the value of the $\mu_{s,pp}$ where it was 0.1 in run 5 and 0.85 in run 7. So, the high value of $\mu_{s,pp}$ decreases the overall TKE of the system.

On the other hand, E_{kin_2} has a smaller e_{pp} than E_{kin_5} , which showed a minimal difference in a small interval of time during the discharge of the particles. Thus, e_{pp} has no significant influence on the overall TKE when $\mu_{s,pp}$ is low. On the contrary, e_{pp} has a significant influence on the overall TKE when $\mu_{s,pp}$ and $\mu_{r,pp}$ are both high as shown in Figure 4.21.

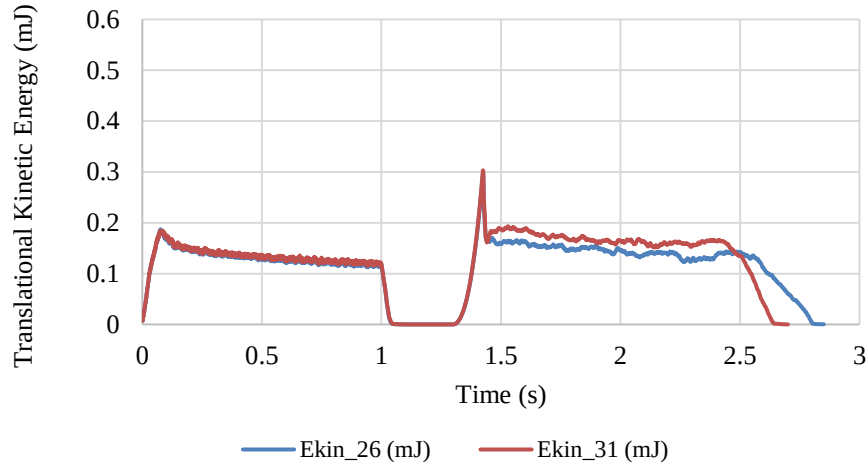


Figure 4.21: Overall TKE graph plot over time for simulation runs 26 and 31 for the free-flowing powder. Conditions for these plots: $\mu_{r,pp} = 0.75$, $\mu_{s,pp} = 0.85$, $Y = 7.41 \log \text{ GPa}$, $d_s = 690 \mu\text{m}$, $e_{pp} = 0.051$ (E_{kin_26}) and $e_{pp} = 0.6$ (E_{kin_31}).

Moreover, the particle size d_s has a major influence on the overall TKE. Figure 4.22 plots the overall TKE graphs of simulation cases 1 and 24 where the d_s is $460 \mu\text{m}$ in run 1 and $690 \mu\text{m}$ in run 24.

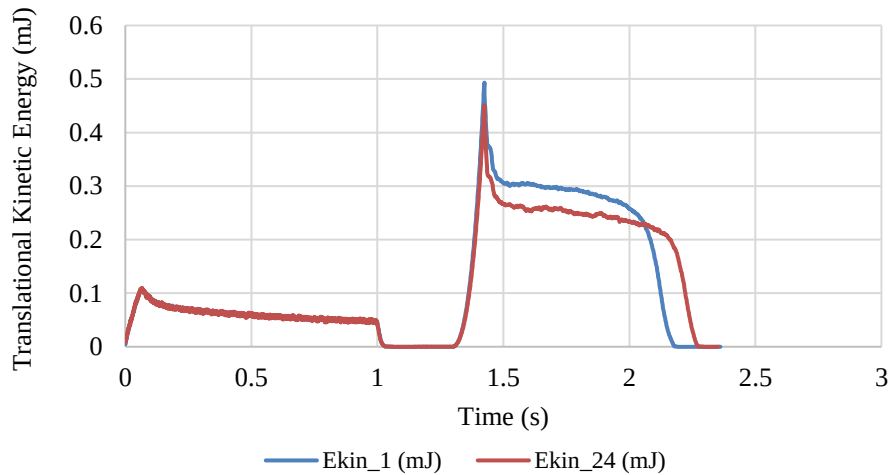


Figure 4.22: Overall TKE graph plot over time for simulation runs 1 and 24 for the free-flowing powder. Conditions for these plots: $\mu_{r,pp} = 0.05$, $\mu_{s,pp} = 0.1$, $e_{pp} = 0.051$, $Y = 7.41 \log \text{ GPa}$, $d_s = 460 \mu\text{m}$, (E_{kin_1}) and $d_s = 690 \mu\text{m}$ (E_{kin_24}).

The graph plot shows that both runs have similar overall TKE during the insertion time. After removing the stopper, the overall TKE of run 1 with lower d_s ($460 \mu\text{m}$) is higher and reaches full discharge faster than run 24 with higher d_s ($690 \mu\text{m}$). This can be explained by that the number of particles decreases with the increase in radii keeping the same mass; therefore, the number of collisions decreases and thus leads to a decrease in the TKE.

For a better understanding of the relative level of effectivity of these parameters, the adjusted response graph and Pareto chart are also designed for this response. The adjusted R^2 of the NA-TKE model of the

free-flowing powder was 0.981. In Figure 4.9d, the adjusted response graph for the NA-TKE is illustrated. The Pareto graph of effects on the NA-TKE of the free-flowing powder is shown in Figure 4.23. It shows that the quadratic term of $\mu_{s,pp}$ has the highest positive effect and $\mu_{s,pp}$ has the highest negative effect on the NA-TKE, followed by $\mu_{r,pp}$. These results harmonize perfectly with the definitions of these two factors, as they represent, more or less, the extent of resistance of a particle to remain in its position upon being exposed to a certain force. Therefore, as these coefficients of friction increase, the particles are more hindered, and thus the TKE decreases. Moreover, since the particles in our system do not tend to move in the rotational direction as in the translational direction, the $\mu_{r,pp}$ is showing less effect than $\mu_{s,pp}$.

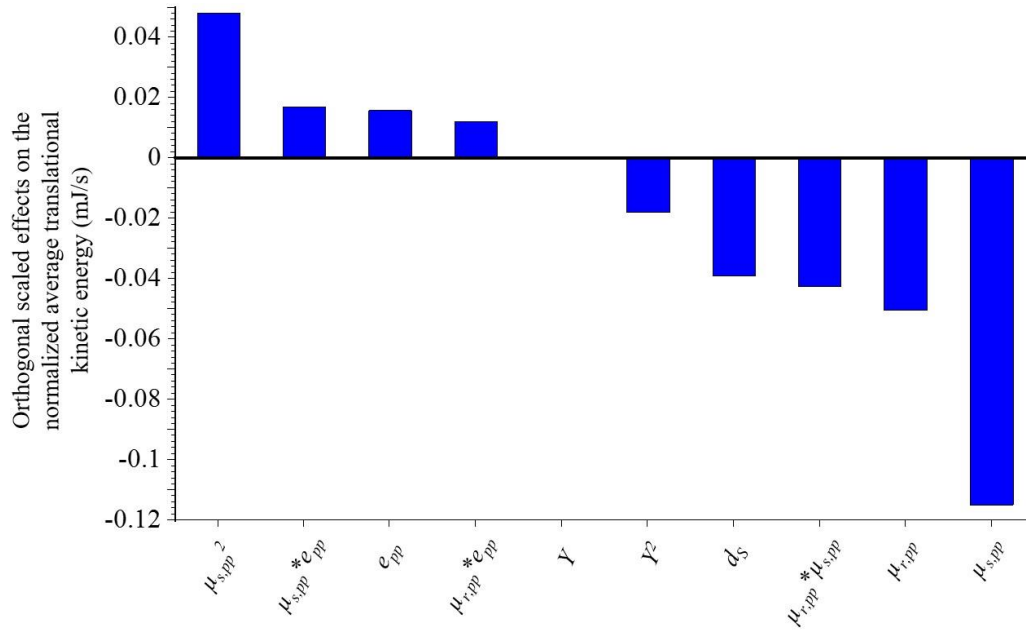


Figure 4.23: Pareto graph of effects of the model factors on the NA-TKE of the free-flowing powder.

4.6.2.4.2 Cohesive

The model fit of the NA-TKE of the cohesive powder was very good with an adjusted $R^2 = 0.97$. The adjusted response graph for the NA-TKE is shown in Figure 4.12d. The Pareto graph of effects on the NA-TKE (Figure 4.24) shows that Y has the highest positive effect and $\mu_{s,pp}$ has the highest negative effect on the NA-TKE. As seen, a new factor, namely Y , compared to the free-flowing case, has been introduced as a significant parameter. The reason for that is the existence of the C_{pp} in the system, in which an additional normal contact force tending to stick the particles more to each other is added. As Y denotes the stiffness of the particles, exerting an additional force on them along with less tendency to deform leads to some repulsion and thus causing the higher velocity of the particles. This interactive effect of these two parameters can also be seen by the term $C_{pp} * Y$ having a positive impact on the NA-TKE. On the contrary, maintaining the same value of Y with increasing the C_{pp} , decreases the NA-TKE, because the additional normal contact force ought to keep the particles in contact. It can also be observed that the particle size has a higher significant effect than that in the free-flowing case. In fact, this can be

justified by the increase in contact area along with the increase in normal contact force, the thing that would cause an increase in the frictional force in order to maintain a constant $\mu_{s,pp}$.

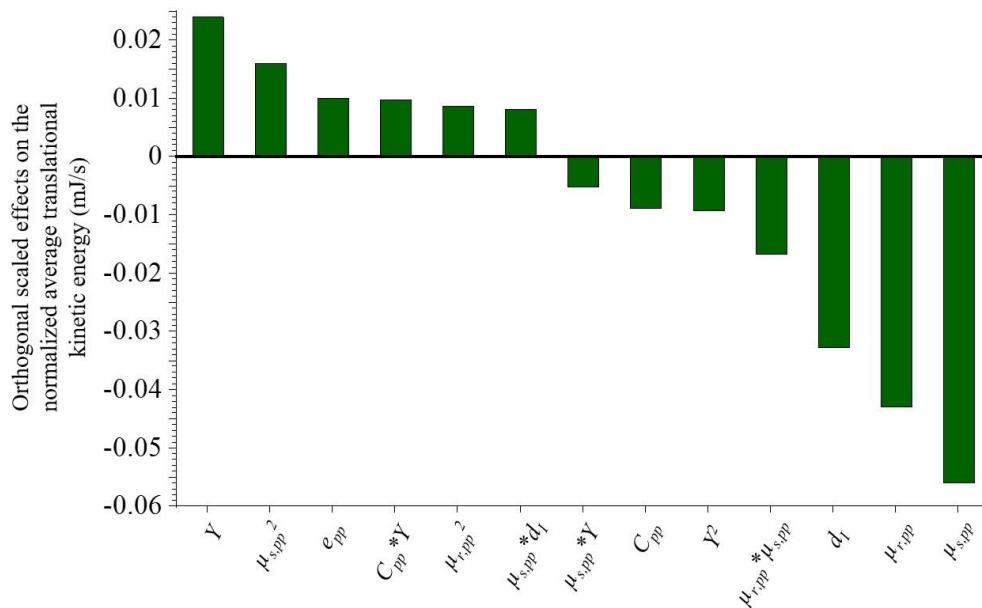


Figure 4.24: Pareto graph of effects of the model factors on the NA-TKE of the cohesive powder.

In Figure 4.25, the overall behavior of the TKE of the system in the cohesive case generally looks similar to the free-flowing case. Some phenomena discussed earlier regarding the effects of some parameters can be observed.

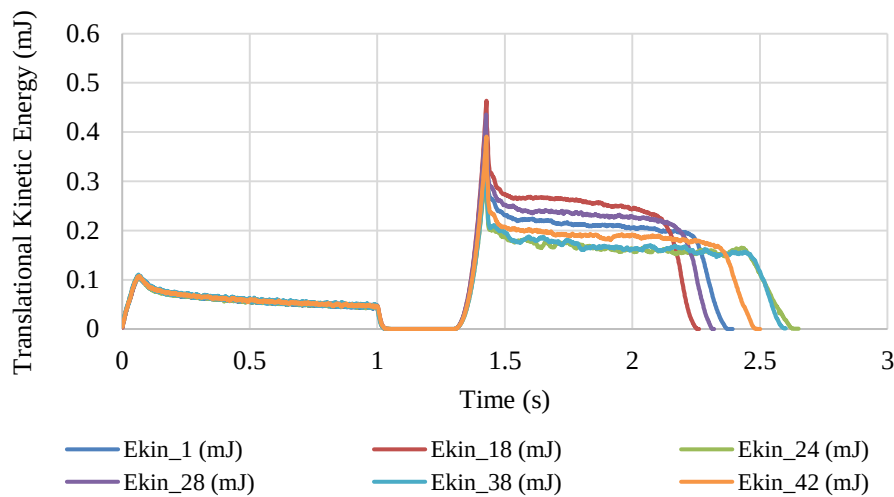


Figure 4.25: Overall TKE graph plot over time for simulation runs 1, 18, 24, 28, 38 and 42 for the cohesive powder.

4.6.2.5 Factors affecting computation time

4.6.2.5.1 Free-flowing

The CT model of the free-flowing powder had a very high adjusted R^2 value of 0.987. It is clearly shown in Table 4.3 how the CT is varying through the different simulation runs. The adjusted response graph for the CT is illustrated in Figure 4.9e. In addition, Figure 4.26 demonstrates the Pareto graph of effects on the CT of the free-flowing powder. It is indicated that Y has a big positive impact on CT, whereas d_s has a smaller negative effect on it. In other words, as Y becomes bigger, the CT increases and as d_s increases, the CT decreases. Although e_{pp} has much less effect on CT compared to Y and d_s , yet it is important to notice that e_{pp} has a negative effect which might be used in another applications consuming relatively high computation time.

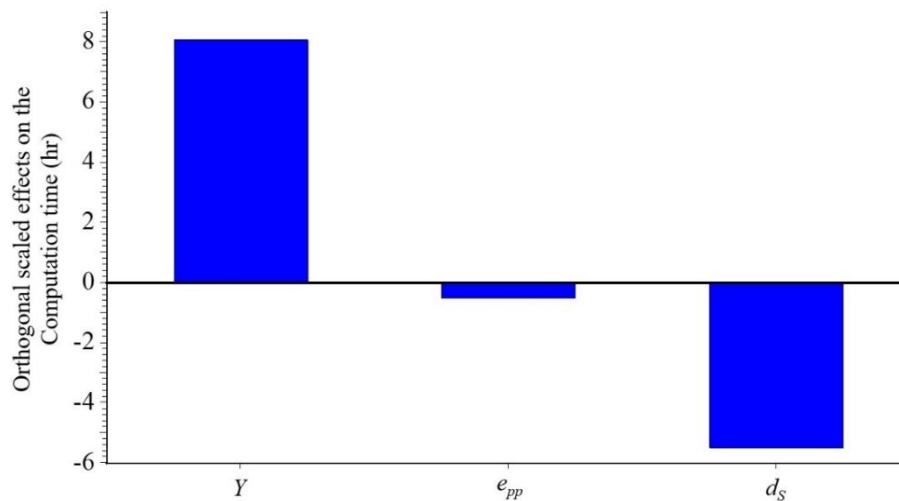


Figure 4.26: Pareto graph of effects of the model factors on the CT of the free-flowing powder.

4.6.2.5.2 Cohesive

The adjusted R^2 of the CT model of the cohesive powder was very high as well (0.991). Figure 4.12e illustrates the adjusted response graph for the CT, and Figure 4.27 presents the Pareto graph of effects on the CT of the cohesive powder. Similarly, the results showed that Y and d_l are the almost only two parameters affecting the CT as in the free-flowing study. In that, using a small Y and bigger d_l via coarse-graining method, would save a lot of computational costs.

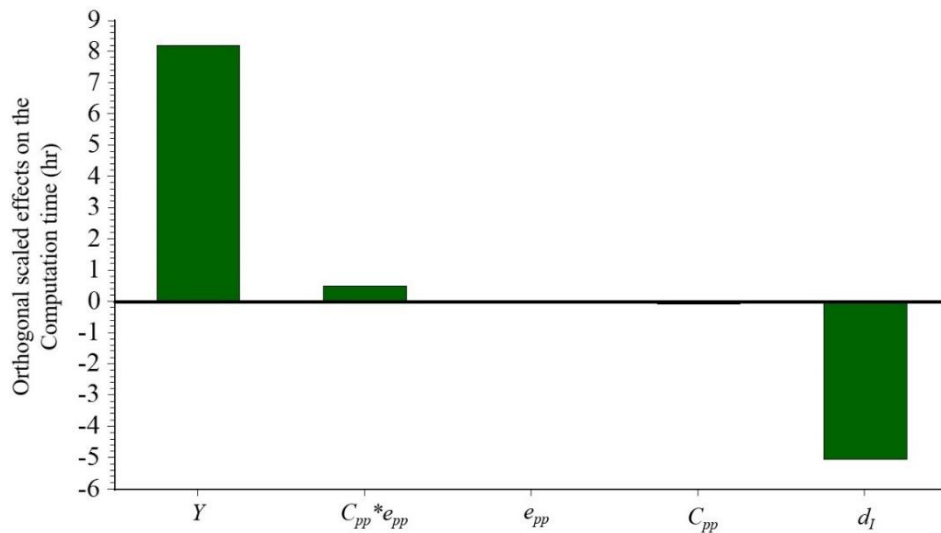


Figure 4.27: Pareto graph of effects of the model factors on the CT of the cohesive powder.

To add up, the count of particles for the free-flowing powder was 18,923, 32,700 and 63,864 when using particle size values of 690 μm , 575 μm and 460 μm , respectively, and that is after applying CG. On the other side, the number of particles for the cohesive powder was 18,540, 30,604, and 55,877 when using particle size values of 694.2 μm , 587.4 μm and 480.6 μm , respectively.

Table 4.5 sums up the adjusted R^2 values of the regression models of the five responses on the free-flowing and cohesive powders.

Free-flowing						Cohesive				
	AoR	ε	$\dot{m}$	NA-TKE	CT	AoR	ε	$\dot{m}$	NA-TKE	CT
Adjusted R^2	0.954	0.981	0.95	0.981	0.987	0.915	0.983	0.957	0.97	0.991

Table 4.5: Summary of the adjusted R^2 values of the ten regression models.

4.6.3 Prediction Model and Experimental Validation

The main aim of this section is to validate the quadratic regression models of the parametric studies obtained above and to validate the numerical model based on experimental data. Therefore, the reference values of the AoR and ε determined from the experiments will be our targets in the following prediction simulations.

4.6.3.1 Free-flowing

Based on the results obtained, a prediction model was built to estimate the values of the AoR and ε upon using an optimized combination between the DEM input parameters. The predicted AoR and ε for the free-flowing powder are calculated according to Equations (4.21) and (4.22), respectively, as follows:

$$\begin{aligned} \text{AoR}_s = & \lambda + \lambda_1 \cdot e_{pp} + \lambda_2 \cdot \mu_{s,pp} + \lambda_3 \cdot \mu_{r,pp} + \lambda_4 \cdot \mu_{r,pp} \cdot \mu_{s,pp} + \lambda_5 \cdot \mu_{r,pp} \cdot e_{pp} + \lambda_6 \\ & \cdot \mu_{s,pp} \cdot e_{pp} + \lambda_7 \cdot \mu_{s,pp}^2 + \lambda_8 \cdot e_{pp}^2 \end{aligned} \quad (4.21)$$

$$\begin{aligned} \varepsilon_s = & \lambda + \lambda_1 \cdot d_s + \lambda_2 \cdot Y + \lambda_3 \cdot e_{pp} + \lambda_4 \cdot \mu_{s,pp} + \lambda_5 \cdot \mu_{r,pp} + \lambda_6 \cdot \mu_{r,pp} \cdot \mu_{s,pp} + \lambda_7 \\ & \cdot \mu_{r,pp} \cdot e_{pp} + \lambda_8 \cdot \mu_{s,pp} \cdot e_{pp} + \lambda_9 \cdot \mu_{r,pp}^2 + \lambda_{10} \cdot \mu_{s,pp}^2 \end{aligned} \quad (4.22)$$

The constants in both equations are presented in Table 4.6.

Variable	Factors of AoR (Free-flowing)	Factors of ε (Free- flowing)	Factors of AoR (Cohesive)	Factors of ε (Cohesive)
λ	6.7266	37.9127	202.481	73.8381
λ_1	- 5.9508	0.0041	30.4258	8.1474
λ_2	87.8684	0.2037	- 34.8184	17.7210
λ_3	6.0857	0.4985	13.1172	- 0.5760
λ_4	15.6309	16.2729	- 43.4370	- 7.4100
λ_5	- 11.8716	5.3753	0.0010	0.0058
λ_6	39.7847	7.38747	4.9399	0.0001
λ_7	- 83.8815	- 2.6774	- 7.1505e-05	7.5073
λ_8	- 36.3797	- 3.4024	- 17.5265	- 2.4609
λ_9	—	- 5.6972	- 30.8083	- 1.5762
λ_{10}	—	- 11.4497	2.4489	- 0.4159
λ_{11}	—	—	- 4.4295e-09	- 8.7405e-06
λ_{12}	—	—	—	- 5.2809e-08
λ_{13}	—	—	—	- 9.3584
λ_{14}	—	—	—	- 10.4861
λ_{15}	—	—	—	0.4565

Table 4.6: Factors of the prediction model equations of the AoR and the ε for the free-flowing and cohesive powders.

The value of $\mu_{s,pp}$ is calculated by finding the tangent of the angle of internal friction (AIF) which was determined using the FT4 rheometer as discussed previously in chapter 2. In that, the experimental value of $\mu_{s,pp}$ of the free-flowing powder is 0.61. Since it is within the tested range of $\mu_{s,pp}$, it was fixed in the prediction model. The experimental values of ε are determined in chapter 2 as well. In addition, according to the obtained results, Y was set to minimum and d_s was set to maximum to reduce the CT. Using an integrated optimization feature algorithm in Cornerstone software, an optimized DEM parameter combination of $\mu_{r,pp}$ and e_{pp} , which aims to match the studied bulk responses (AoR and ε) with the experimental reference values with minimum deviation errors, is selected. Table 4.7 shows the combi-

nation of the DEM input parameters used in validating the prediction model and thus calibrating SpheroLac 100 (the free-flowing) with its experimental AoR and ε values. The values to be predicted were set as the experimental reference values.

Run	$\mu_{r,pp}$	$\mu_{s,pp}$	e_{pp}	Y (log GPa)	d_s (μm)	Predicted AoR ($^\circ$)	DEM AoR ($^\circ$)	Deviation error of AoR (%)
41	0.58	0.61	0.57	7.41	690	33	33.4	1.21
						Predicted ε (%)	DEM ε (%)	Deviation error of ε (%)
						53.5	52.61	-1.66

Table 4.7: Executed simulation run for the validation of the process model of the AoR and the ε of the free-flowing powder.

The results in Table 4.7 validate the robustness of our regression model and thus our powder was calibrated. On this occasion, Figure 4.28 shows the numerical AoR of run 41 validated with the experimental AoR of the free-flowing powder.

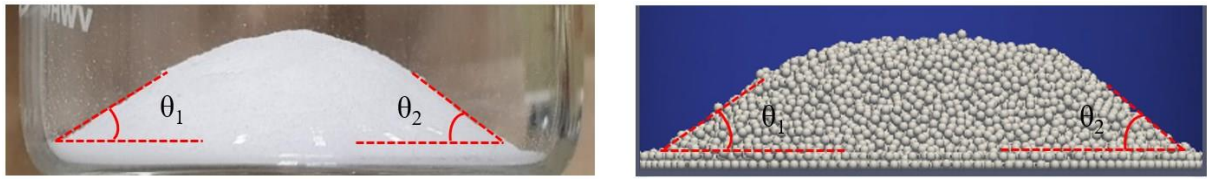


Figure 4.28: DEM AoR from the prediction model validated with the experimental AoR of the free-flowing powder.

4.6.3.2 Cohesive

Similarly, Equations (4.23) and (4.24) represent the prediction equations for the AoR and ε for the cohesive powder, respectively, as follows:

$$\text{AoR}_I = \lambda + \lambda_1 \cdot \mu_{r,pp} + \lambda_2 \cdot \mu_{s,pp} + \lambda_3 \cdot e_{pp} + \lambda_4 \cdot Y + \lambda_5 \cdot C_{pp} + \lambda_6 \cdot \mu_{s,pp} \cdot Y \quad (4.23)$$

$$+ \lambda_7 \cdot C_{pp} \cdot Y + \lambda_8 \cdot \mu_{r,pp}^2 + \lambda_9 \cdot e_{pp}^2 + \lambda_{10} \cdot Y^2 + \lambda_{11} \cdot C_{pp}^2,$$

$$\varepsilon_I = \lambda + \lambda_1 \cdot \mu_{r,pp} + \lambda_2 \cdot \mu_{s,pp} + \lambda_3 \cdot e_{pp} + \lambda_4 \cdot Y + \lambda_5 \cdot d_I + \lambda_6 \cdot C_{pp} + \lambda_7 \quad (4.24)$$

$$\cdot \mu_{r,pp} \cdot \mu_{s,pp} + \lambda_8 \cdot \mu_{r,pp} \cdot e_{pp}$$

$$+ \lambda_9 \cdot \mu_{s,pp} \cdot e_{pp} + \lambda_{10} \cdot \mu_{s,pp} \cdot Y + \lambda_{11} \cdot C_{pp} \cdot Y + \lambda_{12} \cdot C_{pp} \cdot d_I + \lambda_{13} \cdot \mu_{r,pp}^2$$

$$+ \lambda_{14} \cdot \mu_{s,pp}^2 + \lambda_{15} \cdot Y^2.$$

The constants of these latter equations are listed in Table 4.6. To be stated, the prediction equations for both free-flowing and cohesive powders showed a nonlinear relation between the parameters, unlike the study conducted by Boikov [145]. He had no ambiguity in his sought responses as he assumed that the

parameters are linearly dependent. It can be seen in this study that this depends on the application and not only on the type of powder being studied.

The prediction model of the cohesive powder was validated by running a new simulation having $\mu_{s,pp}$ equal to the experimental value (0.64). Similarly, based on selecting the optimized parameter combination, Table 4.8 lists the results of the prediction run simulation of the AoR and ε compared and validated with the experimental data.

Run	$\mu_{r,pp}$	$\mu_{s,pp}$	e_{pp}	Y (log GPa)	d_s (μm)	C_{pp} (J/m^3)	Predicted AoR ($^\circ$)	DEM AoR ($^\circ$)	Deviation error of AoR (%)
54	0.244	0.64	0.051	7.41	694.2	50000	40	40.8	2
							Predicted ε (%)	DEM ε (%)	Deviation error of ε (%)
							59.3	57.84	-2.46

Table 4.8: Executed simulation run for the validation of the process model of the AoR and the ε of the cohesive powder.

The very low deviation errors confirm the validity of the study approach and thus InhaLac 251 (the cohesive powder) was calibrated as well. Moreover, Figure 4.29 shows the AoR of run 54 compared with the experimental AoR of the cohesive powder.

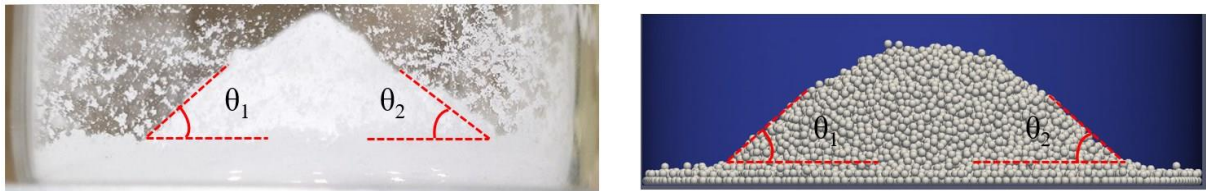


Figure 4.29: DEM AoR from the prediction model validated with the experimental AoR of the cohesive powder.

4.6.4 Summary

In the presented work of case study 1, a semi-automated calibration approach was introduced based on multivariate quadratic regression analysis in which two different powders, i.e., free-flowing and cohesive, were systematically studied and their DEM parameters were numerically calibrated with the experimental data. The work aimed, firstly, to perform a systematic statistical parametric study to understand the impact of six interparticle DEM microscopic input parameters on four macroscopic bulk responses and on the computational time. Secondly, numerical validation of the experimental angle of repose and porosity values using a statistical prediction model was performed. The examined microscopic parameters were Young's modulus, particle size, particle-particle coefficient of restitution, static and rolling friction coefficients, and cohesion energy density. Interaction properties between particle

and wall and between wall and wall were set constant, whereas particle–particle properties were extensively studied. Some of the highlights can be summarized as follows:

- Applying powder characterization using measurement devices decreased the number of needed parameters to be calibrated, which improves the robustness of parameter tuning.
- DoE proved to be an efficient tool in the calibration approach, whereas the regression analysis served in better understanding the impact of different DEM input parameters on the observed responses.
- The quadratic D-optimal design model ascertained its efficiency and reliability in saving computational time by setting up a small-optimized number of simulations to be run.
- For the free-flowing study, it was shown that the static friction coefficient has the most significant impact on almost all the output responses. The particle size and stiffness, however, had the highest impact on the computation costs.
- Regarding the cohesive study, the results were in many cases similar to those of the free-flowing study except that the particle stiffness had a negative impact on the angle of repose, and the cohesion energy density had some impact on all the responses. It has been proven in this study that the involvement of cohesion increased the influence of the particle stiffness on some responses, e.g., the angle of repose. In addition, the underlying analyses show the substantial impact of the rolling friction coefficient and the minor role of the static friction coefficient or the particle size on the angle of repose in the cohesive powder.
- Scaling up the particle size using the coarse-graining method proved to have no effect on the angle of repose in both studies.
- The built statistical prediction model yielded a good quantitative agreement between the DEM simulation and the experimental result values.

4.7 Case Study 2: Auger Dosing

The existence of a large number of parameter combinations outputting the same bulk behavior in simulations is considered a bottleneck for many DEM users. In this study, a novel semi-automated DEM calibration approach that focuses on the ambiguity problem is developed. An auger dosing application is used to output three different bulk responses, namely angle of repose, mass flow rate, and bulk density. Unlike most of the studies involving screw-conveying machines, a vertical-oriented micro auger is used in this work to represent an auger dosing process of free-flowing pharmaceutical powder. The objective of this research work is first to perform a parametric study and then to calibrate free-flowing powder using DoE and MVRA. A quadratic D-Optimal design model is used to setup an optimized number of combinations to be simulated. An analytical mass flow rate test for dynamic powder discharge using an auger is derived and verified by the DEM simulations and experiments. The auger dosing performance is studied in dependence on auger speed, particle-particle coefficients of static and rolling friction, and particle-wall static friction coefficient. Although the coefficient of friction between particles and boundaries is a significant affecting parameter, it is not widely included in the literature. This work extended the scope of previous studies by including this factor into the parametric studies. In this study, the level of effectiveness of these coefficients is investigated to check how they impact four studied responses, i.e., angle of repose, mass flow rate, bulk density, and translational kinetic energy. Meanwhile, as it is almost impossible to measure the energy output on the single-particle scale experimentally, the energy dissipation of the particles is only identified through DEM.

In this work, the DEM parameter combinations are reduced efficiently to reach an almost solitary combination of parameters by superimposing three output prediction contour plots. Calibrations are done based on the bulk behaviors obtained from the experiments and thus the regression analysis model is validated. By this, it is well clarified that using multiple calibration tests lead to more accurate and realistic results and thus reduce the ambiguity. As a result, the simple auger dosing machine can be considered as a standard calibration device for advanced industrial applications.

4.7.1 Experimental Investigation

4.7.1.1 Bulk density

In order to visualize the height of the powder bed and compare the DEM simulation with the experiment, a simple test using a transparent glass tube of an inner diameter 24 mm and a total height of 94 mm was utilized. Initially, a mass of 6 g of SpheroLac 100 was filled inside the tube where then the volume was recorded and thus the bulk density (BD) was obtained.

4.7.1.2 Auger dosing

An auger powder container with an inner diameter of 64 mm and a total height of 224 mm was used in running the mass flow rate ($\dot{m}$) experiments of SpheroLac 100 powder. It consists of a cover, a funnel,

an opaque cylindrical glass tube, a shaft connected to a rotational motor, and an auger. The geometry and the dimensions of the auger, including the inner and outer diameters, the pitch, and the auger filling volume per one revolution, which is used to conduct the experiments are illustrated in Figure 4.30. Theoretically, the auger filling volume represents the maximum volume a material could be filled in one complete rotation.

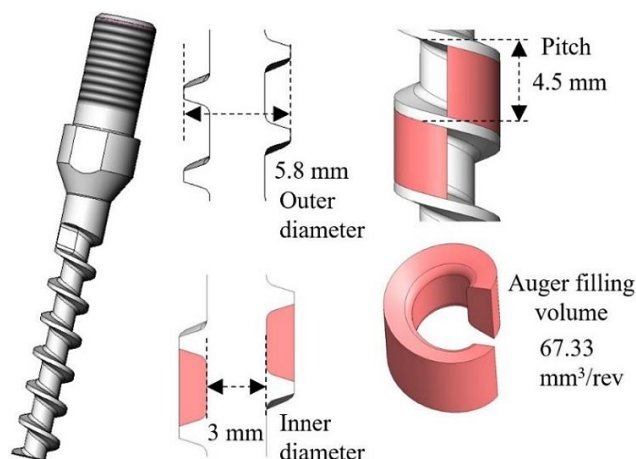


Figure 4.30: Illustration of the auger geometry used in the tests.

The auger dosing container encloses a stainless steel (1.4404) funnel with an upper diameter of 63 mm, a lower opening diameter of 6 mm and an inclination angle of 45°. The auger is connected directly to the drive shaft of an overhead stirrer rotating motor RZR 2102 control from Heidolph Instruments (Germany). A laboratory weighing balance XPE 204 from Mettler Toledo (Germany), used as a catch scale, is placed under the outlet of the auger dosing container to record the powder mass flow every 0.2 s. The discharged powder is collected in a catching container made of borosilicate 3.3 glass (Figure 4.31). The XPE 204 is placed on a Regupol sound 17 isolation pad to avoid the transmission of any external vibration into it. The XPE 204 has readability down to 0.1 mg and a capacity up to 220 g. HTerm software is used on a computer to record the values of weight (in g) over time. In addition, an ionization device Static Line LC from HAUG GmbH & Co. KG. (Germany) is used to neutralize electrostatic charges. The powder container, loosely filled with 6 g of the tested powder, was placed at a distance of 80 mm from the bottom of the catching container.

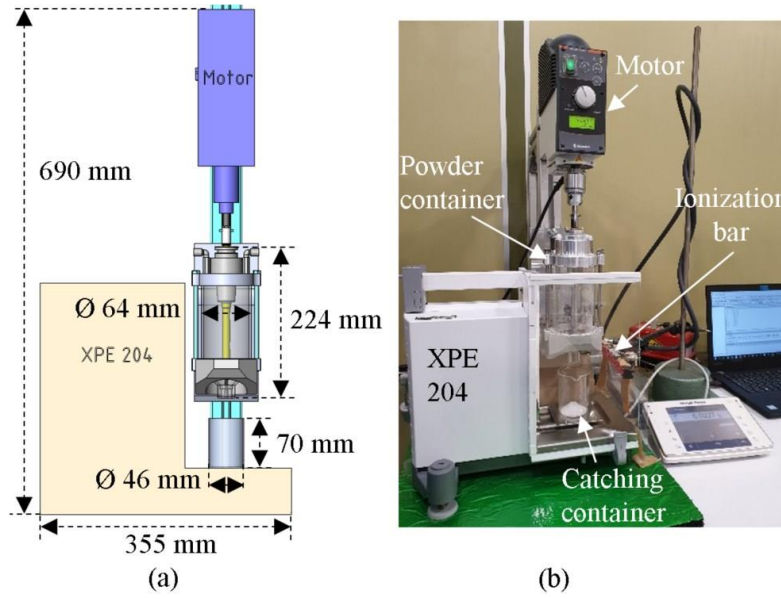


Figure 4.31: Experimental setup of auger dosing tests in (a) front CAD view and (b) laboratory equipment setup.

The volumetric mass flow rate ($\dot{m}$) is calculated using the difference in mass (Δm) obtained from the XPE 204 divided by the difference in time (Δt) between consecutive weighing balance measurements:

$$\dot{m} \left(\frac{g}{s} \right) = \frac{\Delta m (g)}{\Delta t (s)}. \quad (4.25)$$

To ensure repeatability, the test was repeated three times at three different auger rotational speeds, i.e., 400, 600 and 800 rpm, where an average of the obtained $\dot{m}$ was considered as an experimental reference value at each speed. After dosing, the formed AoR inside the catching container was measured.

4.7.2 Simulation Procedure and Parameters

4.7.2.1 Fixed Parameters throughout the Study

The Young's modulus and Poisson's ratio values of the funnel and the auger (stainless steel) and the catching container (borosilicate 3.3 glass) were selected from the literature [133,139]. Following the recommendations [49,146], Young's modulus for the particle was reduced within a certain permissible limit in order to reduce computational costs. The boundary conditions such as coefficients of restitution and rolling friction between the particle and the walls (stainless steel and glass) were set constant throughout the simulations. The particle size was scaled-up six times using the coarse-graining method identified by Bierwisch [70]. Table 4.9 lists the applied DEM input parameters in the auger dosing study.

Material parameter/contact model	Symbol	Unit	Value
Young's modulus funnel and auger ^a	Y_F	GPa	200
Young's modulus catching container ^b	Y_C	GPa	67.8
Young's modulus particle	Y_p	GPa	0.026
Poisson's ratio funnel and auger ^a	ν		0.28
Poisson's ratio catching container ^b	ν		0.21
Poisson's ratio particle	ν		0.3
Particle diameter (after coarse-graining)	d_p	μm	690
Contact law parameters			
Coefficient of restitution particle-particle	e_{pp}		0.2
Coefficient of restitution particle-wall	e_{pw}		0.2
Coefficient of rolling friction particle-wall	$\mu_{r,pw}$		0.2
Coefficients of restitution and friction wall-wall			0.2
Factory settings			
Filling mass		g	6
Number of particles			22705
Simulation time step		s	2.62×10^{-6}
Acceleration due to gravity	g	m/s^2	9.81

^a [139]^b [147]

Table 4.9: The considered DEM input parameters throughout the study.

4.7.2.2 Parameters Varied in the Design of Experiments

On the other hand, one process parameter and three-particle properties were varied (Table 4.10) in the context of the DoE: auger rotational speed (ω), coefficient of static friction ($\mu_{s,pp}$) and coefficient of rolling friction ($\mu_{r,pp}$) between particle-particle and coefficient of static friction ($\mu_{s,pw}$) between particle-wall. Three values were selected for each of the parameters, which include a low, a center, and a high value, see, [59,148] for reference. Based on the DoE, only 20 simulation runs were required.

Parameter	Symbol	Unit	Value
Auger rotational speed	ω	rpm	400, 600, 800
Coefficient of static friction particle-particle	$\mu_{s,pp}$		0.3, 0.55, 0.8
Coefficient of static friction particle-wall	$\mu_{s,pw}$		0.1, 0.25, 0.4
Coefficient of rolling friction particle-particle	$\mu_{r,pp}$		0.1, 0.4, 0.7

Table 4.10 List of the parameters varied in the DoE of auger dosing simulation.

4.7.3 DEM Studied Responses

Four main factor responses were studied in this work, i.e., angle of repose (AoR), bulk density (BD), mass flow rate ($\dot{m}$), and the overall translational kinetic energy (TKE). The BD was determined inside the funnel after the particles are inserted and settled down. After opening the funnel, the $\dot{m}$ was measured and the average of TKE was computed and then normalized (NA-TKE) over the total dosing time (DT) of the particles. The AoR was determined in the catching container after the particles were settled down at the end of the dosing process. Figure 4.32 shows a cross-sectional view of the initial state of the funnel filled with 6 g of powder using ParaView software.

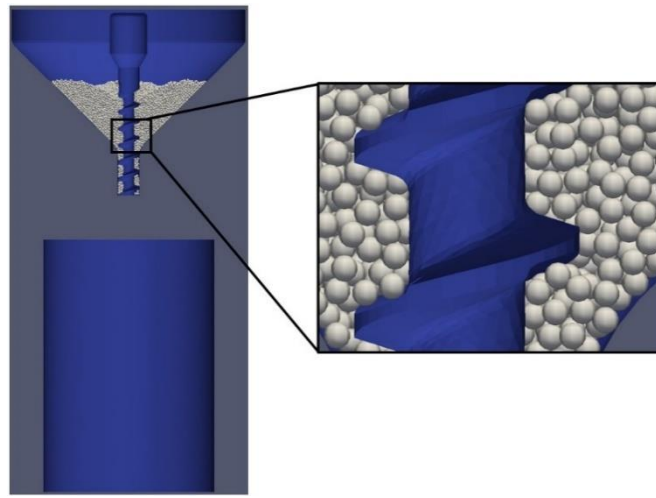


Figure 4.32: The initial state of particles filled inside the funnel in ParaView software.

Table 4.11 presents the DoE matrix followed in performing the parametric and calibration studies. The results cover the dosing of 5.2 g, which is the common linear steady state of the particle flow (dosing) for all the simulations. This is due to the high mass flow fluctuations at the end of the dosing process resulted from the very low amount of powder left inside the funnel.

Run	ω (rpm)	$\mu_{s,pp}$	$\mu_{s,pw}$	$\mu_{r,pp}$	AoR (°)	BD (g/cm ³)	$\dot{m}$ (g/s)	NA-TKE (μ J/s)	DT (s)
1	400	0.3	0.1	0.1	29.3	0.764	0.3995	0.0340	13.0
2	400	0.3	0.1	0.7	35	0.761	0.3712	0.0267	14.0
3	400	0.3	0.4	0.1	29.3	0.767	0.2585	0.0135	20.2
4	400	0.3	0.4	0.7	34.5	0.753	0.2520	0.0107	20.8
5	400	0.55	0.25	0.4	37	0.742	0.2689	0.0112	19.2
6	400	0.8	0.1	0.1	33.5	0.748	0.3780	0.0283	13.8
7	400	0.8	0.1	0.7	40.3	0.730	0.3346	0.0265	15.6
8	400	0.8	0.4	0.1	33.5	0.752	0.2471	0.0139	21.2
9	400	0.8	0.4	0.7	39.5	0.724	0.2301	0.0138	22.8
10	600	0.55	0.4	0.1	32.8	0.759	0.3677	0.0362	14.2

11	600	0.8	0.1	0.4	36.4	0.733	0.4872	0.0594	10.6
12	600	0.8	0.25	0.7	40	0.723	0.3796	0.0352	13.8
13	800	0.3	0.1	0.7	38.4	0.757	0.6674	0.0992	7.8
14	800	0.3	0.25	0.4	36.8	0.759	0.5538	0.0663	9.4
15	800	0.3	0.4	0.1	30.2	0.767	0.4946	0.0778	10.6
16	800	0.55	0.1	0.1	32	0.753	0.6782	0.1102	7.6
17	800	0.55	0.4	0.7	41.5	0.734	0.4523	0.0616	11.6
18	800	0.8	0.1	0.7	40	0.729	0.5974	0.1095	8.8
19	800	0.8	0.25	0.1	33.8	0.754	0.5458	0.0694	9.6
20	800	0.8	0.4	0.4	36.2	0.735	0.4482	0.0689	11.6

Table 4.11: DoE matrix and DEM results for the considered 20 simulations.

4.8 Case Study 2: Results and Discussion

4.8.1 Experimental Results

4.8.1.1 Bulk density

Since the funnel used in the auger dosing experiment is not transparent, the exact height of the powder inside the funnel cannot be determined. Therefore, a transparent glass tube was used instead. After filling 6 g of powder inside the glass tube, the average powder height inside the tube was 17.5 mm (Figure 4.33). As a result, the average calculated BD in the experiment was 0.758 g/cm^3 .

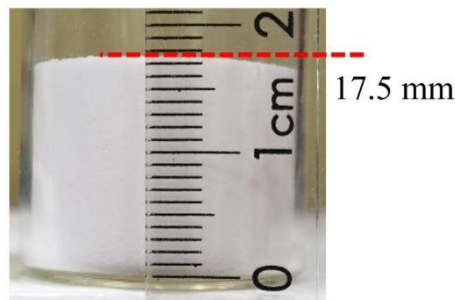


Figure 4.33: The BD experimental result.

4.8.1.2 Mass Flow Rate

The results of the $\dot{m}$ tests of SpheroLac 100 at the three rotational speeds are illustrated in Figure 4.34. A linear regression of the mass curves (with capital M) was done, where the coefficient of determination (R^2) of the linear regression of the mass curves was equal to 1 in all the experiments. The slopes of these linear curves represent $\dot{m}$, namely 0.2568 g/s , 0.3744 g/s and 0.5074 g/s at speeds of 400, 600, and 800 rpm, respectively. The left graph column represents the mass M (g) and the right graph column represents $\dot{m}$ (g/s) where both are plotted over time (s).

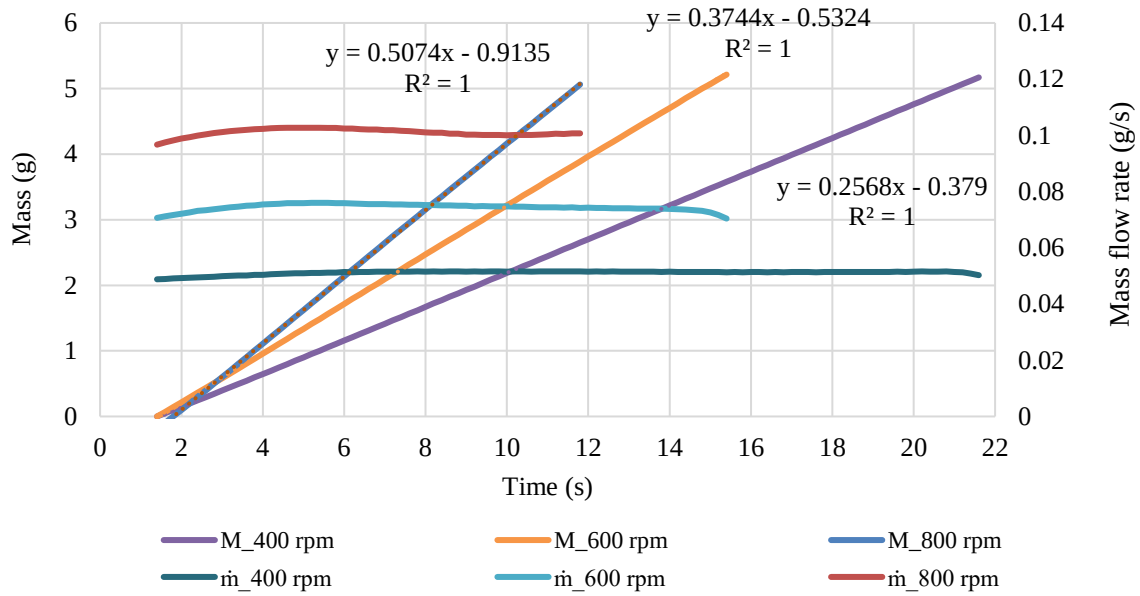


Figure 4.34: The results of the $\dot{m}$ tests at three different auger rotational speeds.

4.8.1.3 Angle of Repose

After the dosing was completed, the powder collected in the catching container formed a heap. For each experiment, the AoR was determined by considering the average value between the left and right angles, i.e., θ_1 and θ_2 , respectively.

Table 4.12 summarizes the results of $\dot{m}$ and the AoR from the auger dosing experiments. The values of $\dot{m}$ at the three different speeds and the averaged AoR, besides the BD, are the macroscopic reference values, which were used in the DEM calibration simulations by validating the prediction models (explained in section 4.9.3). It is also worth mentioning, that it was not possible to measure the TKE experimentally. Therefore, TKE is only discussed in the DEM parametric study and not being part of the calibration process.

ω (rpm)	AoR ($^\circ$)	$\dot{m}$ (g/s)
400	33.2	0.2568
600	33.4	0.3744
800	33.2	0.5074

Table 4.12: Summary of the experimental results of the AoR and $\dot{m}$.

4.8.2 Parametric Study

In this section, the effect of the studied parameters on multiple bulk responses will be discussed in terms of manner and extent. Figure 4.35 illustrates these effects on the studied responses in an adjusted response graph. Each of the presented sections (row-wise) corresponds to a bulk response, and each of

them (column-wise) corresponds to a DEM input parameter. The data points shown in Figure 4.35 represent the points obtained from the simulations and the effect drawn (curve) from them. These data points are adjusted to average out the effects of the other factors included in the model.

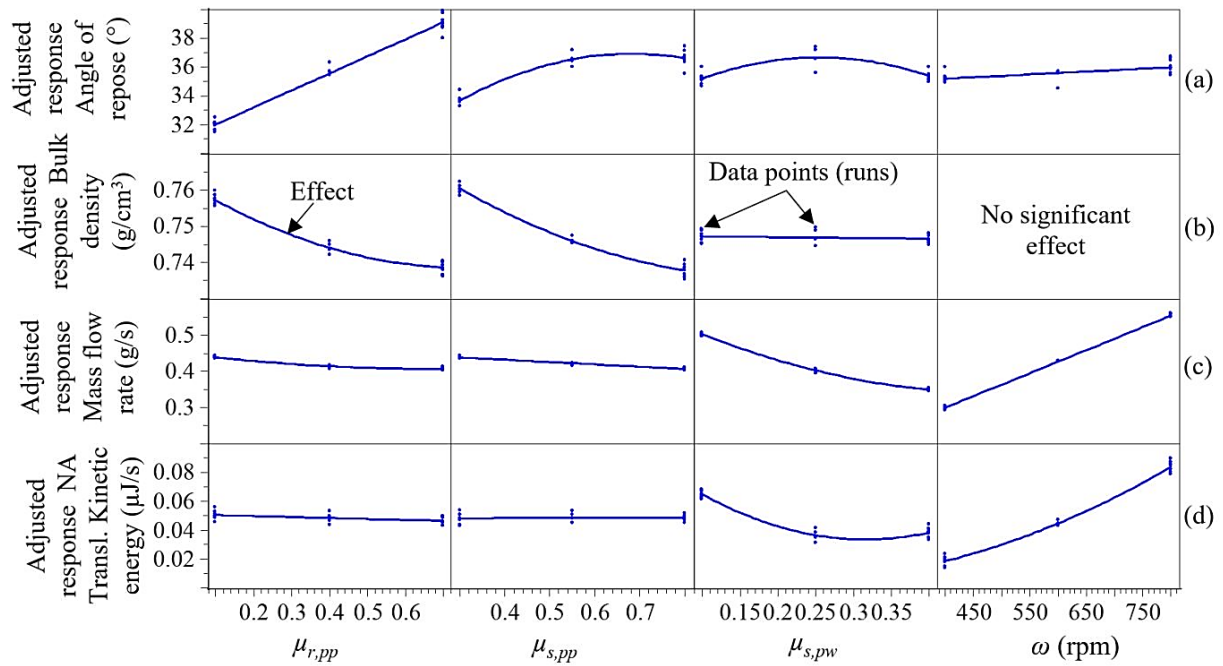


Figure 4.35: Adjusted response graph of the (a) AoR, (b) BD, (c) $\dot{m}$ and (d) NA-TKE as in relation to the four varied DEM input parameters.

The quadratic regression models of the four responses have shown high adjusted R^2 values, which indicates very good fits to the determined data points. Table 4.13 lists the values of the adjusted R^2 of the four studied responses.

	AoR	BD	$\dot{m}$	NA-TKE
Adjusted R^2	0.975	0.983	0.998	0.986

Table 4.13: Summary of the adjusted R^2 values of the four responses' models.

4.8.2.1 Factors affecting angle of repose

As can be seen from Figure 4.35(a), all the studied parameters have a significant effect on the AoR but with varying levels of effectivity. For a more comprehensive illustration of the individual and interactive effects of these parameters, Pareto chart is developed in Figure 4.36. It shows the weight value effect of the input parameters on the AoR. It can be clearly seen that $\mu_{r,pp}$ is the most affecting factor on the AoR in the system, where its increase rises the angles of the heap. This is contradictory to most of the literature, where it is stated that the biggest impact on the AoR is the $\mu_{s,pp}$. This can be explained by that the introduction of the rotating micro-auger in the system caused more rotational motion of the particles around their center. The auger speed has a very small effect on the AoR. Moreover, it is important to

notice the almost negligible effect of $\mu_{s,pw}$, which can be neglected from the list of affecting parameters on the AoR.

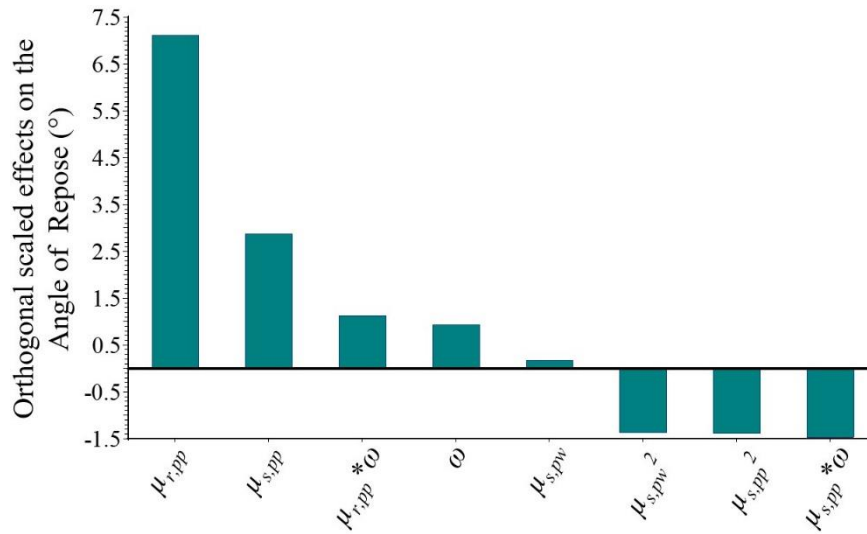


Figure 4.36: Pareto graph of effects of the model factors on the AoR.

4.8.2.2 Factors affecting bulk density

Figure 4.37 illustrates the Pareto graph of effects on the BD. According to the parametric study, $\mu_{s,pp}$ and $\mu_{r,pp}$ can be considered the only effecting parameters on the BD in this study while the $\mu_{s,pw}$ has an insignificant effect. The interactions between $\mu_{r,pp}$ and both of $\mu_{s,pp}$ and $\mu_{s,pw}$ have a negative impact on the BD, where, as they increase, the BD decreases.

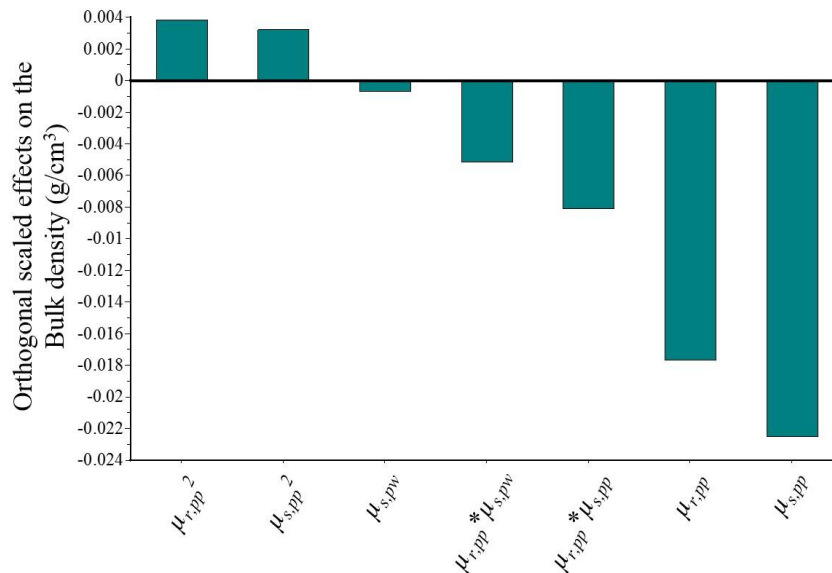


Figure 4.37: Pareto graph of effects of the model factors on the BD.

4.8.2.3 Factors affecting mass flow rate

Similarly, a Pareto graph of effects on the $\dot{m}$ is shown in Figure 4.38. It is evident how the ω has the largest positive significance on $\dot{m}$, whereas $\mu_{s,pw}$ has the largest negative impact on it. As the particle-

wall friction increases, the tendency of particle flow decreases dramatically leading to low mass output. It is important to notice the major effect $\mu_{s,pw}$ is having on $\dot{m}$, where, in many other studies, it has not been considered as a calibration parameter. Thus, after this study, it is important to take it into consideration.

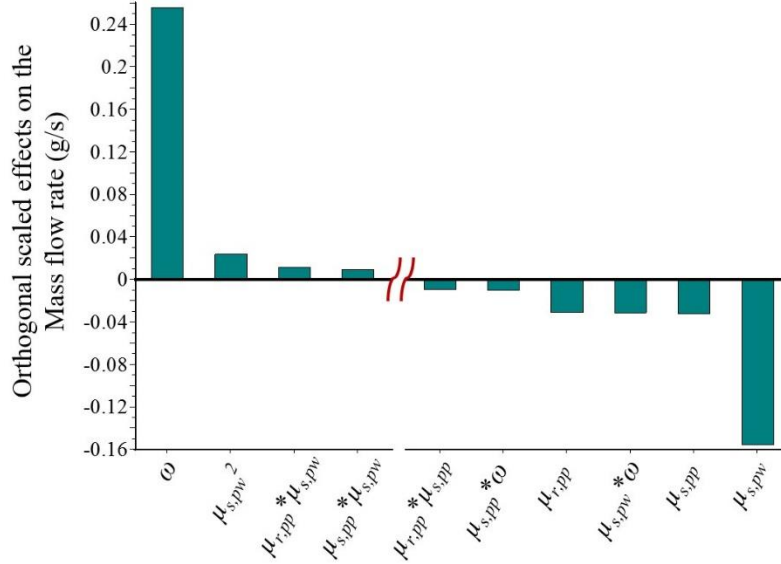


Figure 4.38: Pareto graph of effects of the model factors on the $\dot{m}$.

4.8.2.4 Factors affecting kinetic energy

In Figure 4.39, the influence of the studied DEM input factors on the NA-TKE is demonstrated in a Pareto graph of effects. Once more, ω has the highest positive significant effect followed by $\mu_{s,pw}$, which has the largest negative impact on the NA-TKE. It is noticeable that the $\mu_{s,pp}$ is showing a negligible effect on the NA-TKE. The reason for this is that the particles are basically being exposed to a rotary motion by the auger. Therefore, the translational contact between the particles was almost not existing. Moreover, the rotation of the auger caused the particles to rotate around themselves, rather than rotating around one common axis. This can be seen by the presence of $\mu_{r,pp}$ as having some impact. Although the particles are not showing a translational relative motion with respect to each other, the $\mu_{s,pw}$ is still proven to be a very affecting parameter on the NA-TKE. This means that the particles are sliding on the boundary during discharge. Therefore, it is essential to draw the attention to the importance of calibrating the roughness of the boundaries in DEM. Moreover, as the NA-TKE basically represents the dynamics of the particles, the AoR was measured when the NA-TKE reached the value of 0. This is because the AoR should be measured when the particles get in rest [149].

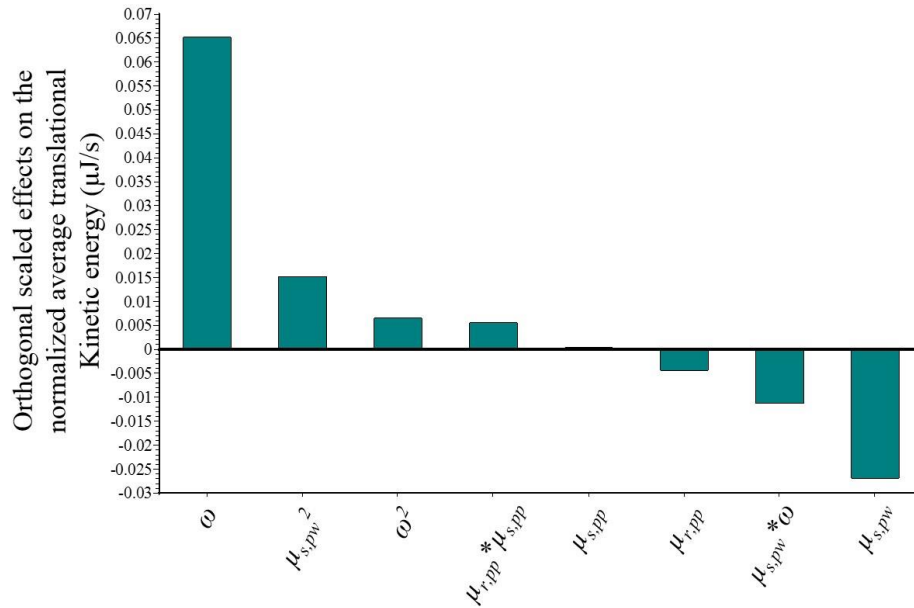
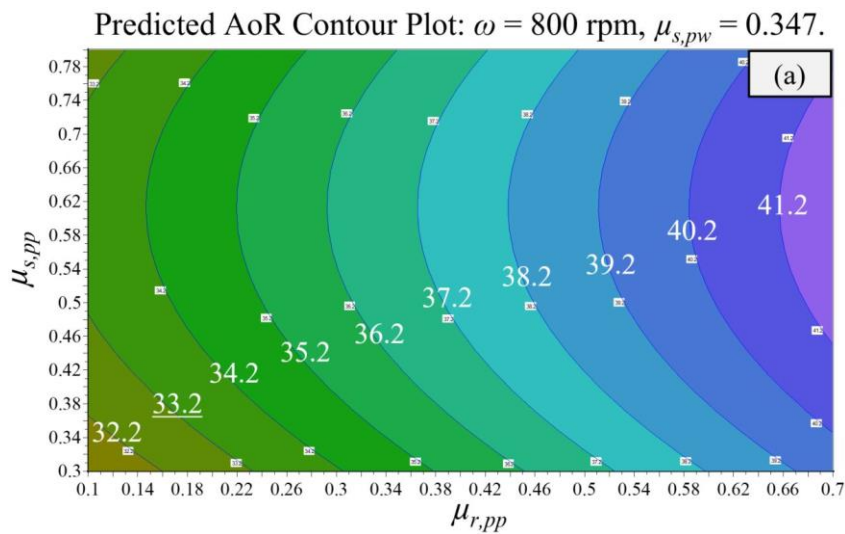


Figure 4.39: Pareto graph of effects of the model factors on the NA-TKE.

4.8.3 Experimental Validation of Predictive Models

A prediction model, based on the MVRA results, is built and used to validate the robustness of the model. As $\mu_{r,pp}$ and $\mu_{s,pp}$ have the highest significant impact on the AoR and the BD, the contour plots were illustrated based on these two parameters (Figure 4.40a and Figure 4.40b). Although, as explained in subsection 4.3.3, ω and $\mu_{s,pw}$ have the highest impact on the $\dot{m}$, but due to the aim of finding the optimal parameter combination between the calibrated parameters by superimposing three contour plots together, $\mu_{r,pp}$ and $\mu_{s,pp}$ were varied on the predicted contour plot of the $\dot{m}$ (Figure 4.40c) as well. As an illustration example to find out the best optimized DEM parameter combination, the predicted contour plots were generated at $\omega = 800$ rpm and $\mu_{s,pw} = 0.347$ (as an optimized value resulting from the optimization algorithms in Cornerstone) as fixed values.



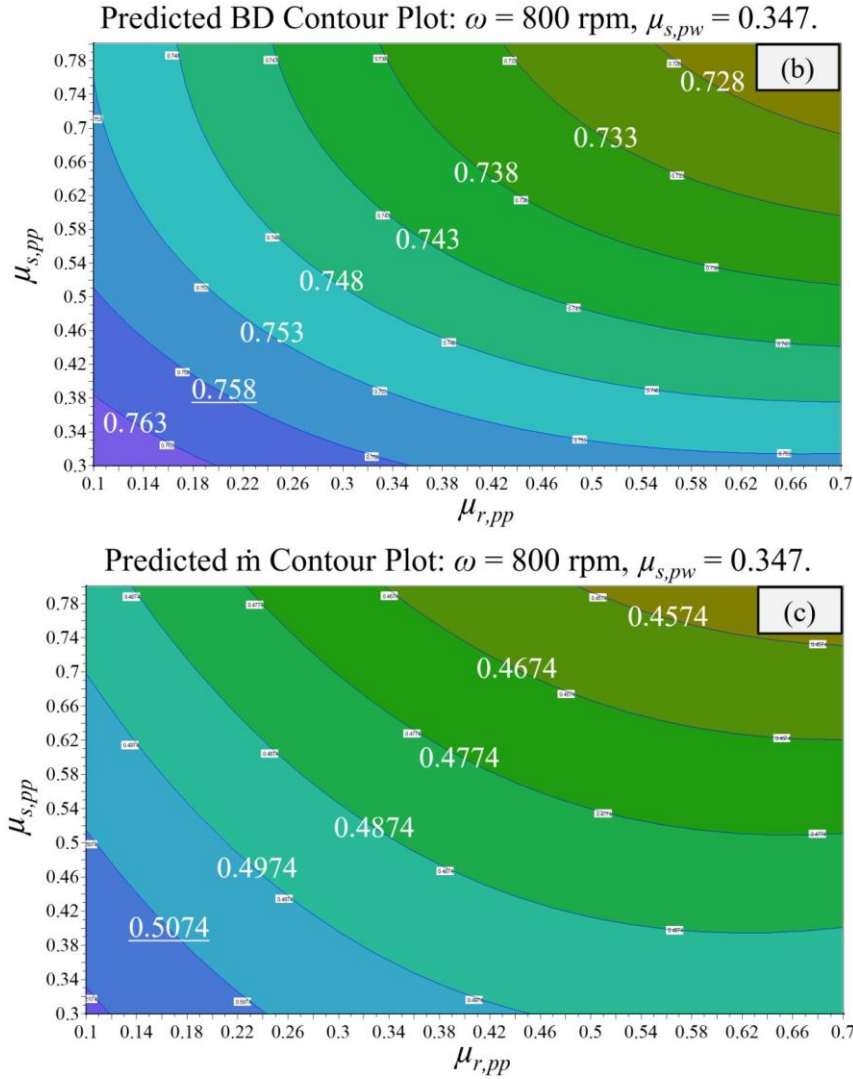


Figure 4.40: Predicted contour plots of the (a) AoR, (b) BD and the (c) $\dot{m}$ by varying $\mu_{s,pp}$ and $\mu_{r,pp}$ and by having ω and $\mu_{s,pw}$ fixed.

As can be seen from the above figures, wide ranges of combinations between $\mu_{r,pp}$ and $\mu_{s,pp}$ in each response lead to delivering the experimental reference values which are the underlined values in the graphs. Therefore, considering a single experimental bulk behavior highlights the problem of ambiguous parameter combinations in the calibration process. Consequently, the three predicted contour plots in Figure 4.40 were superimposed. By this, the number of admissible combinations is dramatically minimized, where a very small intersecting region between the reference target values was found (see Figure 4.41). Taking into account the minimized area of permissible combinations, the valid range for $\mu_{r,pp}$ is between 0.1 and 0.2 combined with $\mu_{s,pp}$ which is between 0.34 and 0.5. For more clarity, the intersecting contour lines outside the region of the reference target values are blurred in the graph. The contour lines of the BD and the $\dot{m}$ evidently intersect, whereas the contour line of the AoR (33.2°) is shifted around 0.3° ahead, which is accepted and within the range of the prediction model. In the context of the superimposing approach, one of the problems that may occur is the non-intersection between the contour lines

[150]. Roessler et. al [48] added that the measurement error needs to be taken into consideration to enable a significant overlap of the test results.

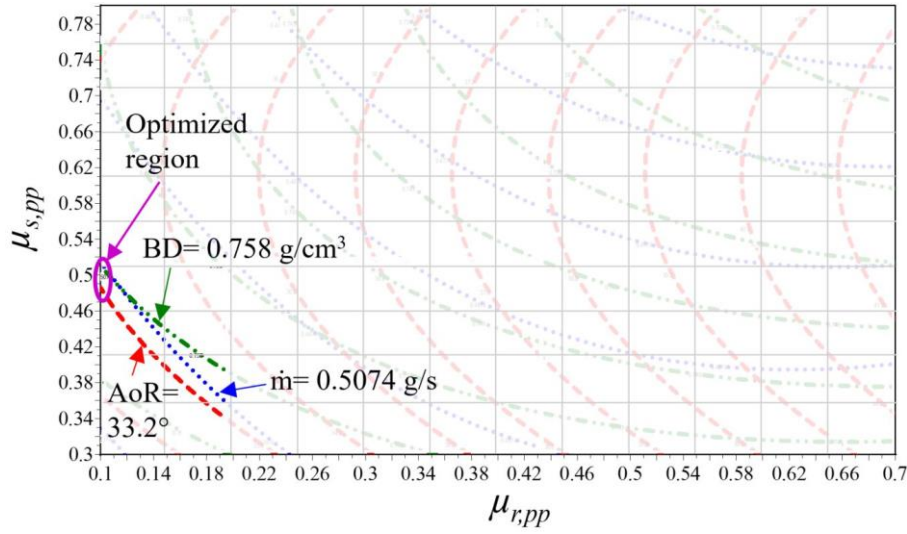


Figure 4.41: Superimposed contour plot showing the permissible combinations of the AoR, BD, and $\dot{m}$ that fulfill the experimental reference values.

Cornerstone software offers an integrated optimization algorithm to select an optimal unique set of parameters combination that predicts the response values based on the experimental reference target values. Based on the optimization feature algorithm, that aims to match the target values with minimum deviation errors, a simulation at 800 rpm was executed to validate the prediction model (confirmation run). In addition, two more simulations, at speeds 400 rpm and 600 rpm, took place similarly based on their prediction models as well, which are not explained here. Table 4.14 lists the different considered combinations and related bulk responses.

Run	ω (rpm)	$\mu_{s,pp}$	$\mu_{s,pw}$	$\mu_{r,pp}$	AoR (°)	BD (g/cm³)	$\dot{m}$ (g/s)	NA-TKE (μJ/s)	DT (s)
21	400	0.523	0.375	0.1	33.1	0.757	0.2542	0.0119	20.6
22	600	0.313	0.36	0.322	33.9	0.758	0.384	0.0313	13.6
23	800	0.51	0.347	0.1	33.5	0.757	0.5053	0.0692	10.4

Table 4.14: DEM parameter combinations of the prediction model and the bulk responses.

Figure 4.42 compares the experimental AoR and the simulation ones. The results show a very close matching between the simulated heaps and the reference ones.

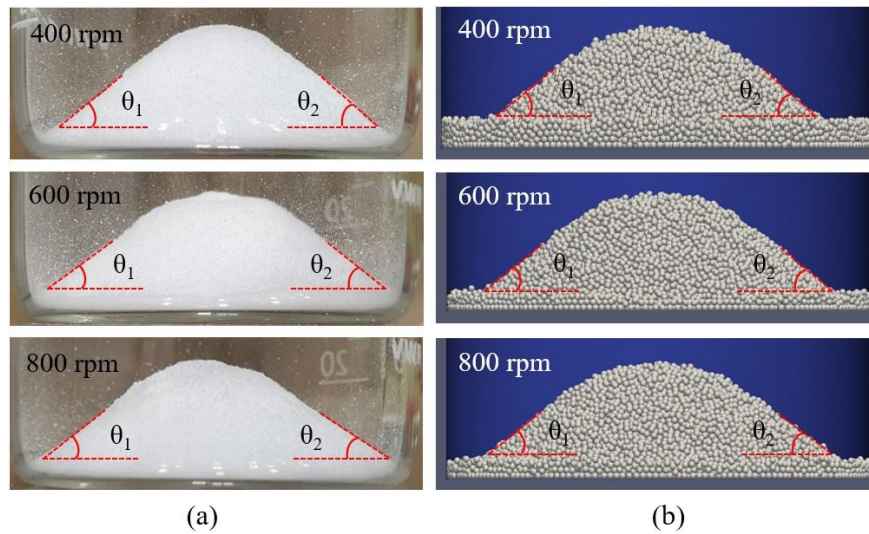


Figure 4.42: Comparing the AoR of the (a) experiments and the (b) DEM simulations.

Using the selected optimized DEM input parameters, a DEM simulation is performed to calibrate the BD with the experiment. Figure 4.43 represents the powder height of the BD test in the experiment compared to that of the DEM simulation, where they both showed the same height (17.5 mm) and thus the determined BD was matching (0.758 g/cm^3).

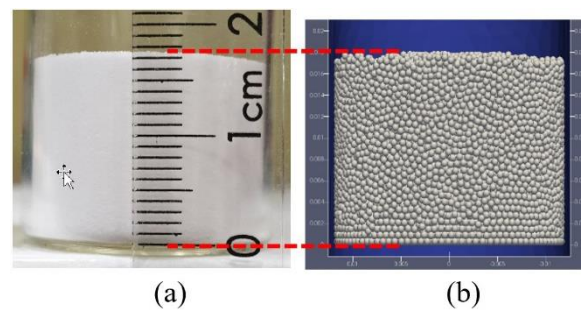


Figure 4.43: The BD results in experiment (a) and in the DEM simulation (b).

Figure 4.44 represents the $\dot{m}$ graphs of the three prediction simulations, where the results of the DEM clearly match with the experimental findings. This adds up to validate the robustness of the followed systematic MVRA using the auger dosing technique and thus consider it as a standard calibration device. It is shown in the $\dot{m}$ graphs, especially with speeds 600 rpm and 800 rpm that, at the beginning of the powder flow, there is a small deviation between the experiment and the simulation. In fact, this is due to the acceleration of the motor until the auger filling volume was full with powder.

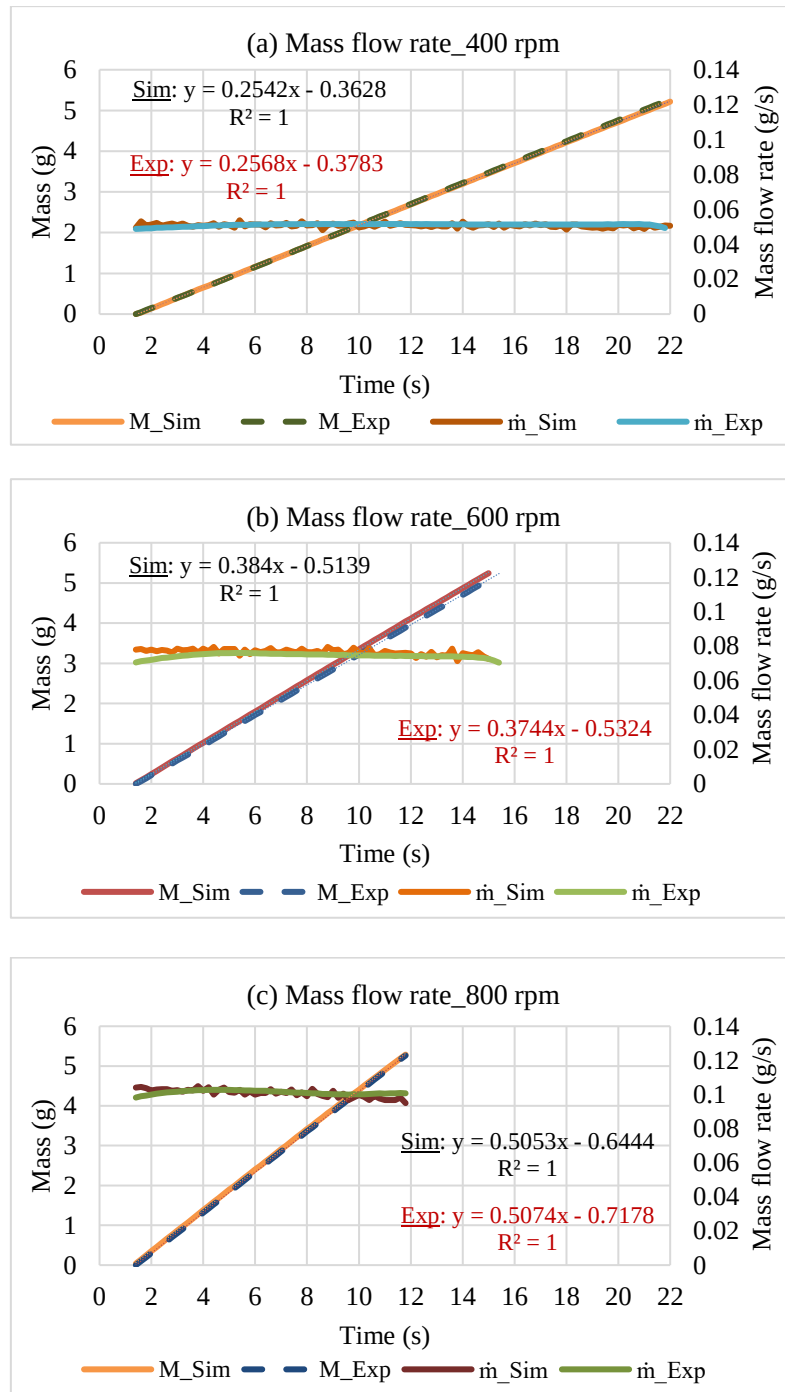


Figure 4.44: Validating the $\dot{m}$ of the DEM with the experiments at speeds (a) 400 rpm, (b) 600 rpm, and (c) 800 rpm.

As it has been pointed out, the quadratic regression prediction models qualitatively and quantitatively predicted the results of the three experimental calibration tests. Table 4.15 summarizes the results from the validation process and compare them with the experimental reference values. The maximum deviation is 2.86 % (in the case of the DT), where the overall mean deviation of the whole responses (12 results listed in Table 4.15) is 0.87 %.

ω (rpm)	Bulk behavior	Experiment	Simulation	Deviation (%)
400	AoR ($^{\circ}$)	33.2	33.1	-0.3
	BD (g/cm^3)	0.758	0.757	-0.13
	$\dot{m}$ (g/s)	0.2568	0.2542	-1.01
	DT (s)	20.4	20.6	0.98
600	AoR ($^{\circ}$)	33.4	33.8	1.2
	BD (g/cm^3)	0.758	0.758	0
	$\dot{m}$ (g/s)	0.3744	0.384	2.56
	DT (s)	14	13.6	-2.86
800	AoR ($^{\circ}$)	33.2	33.5	0.9
	BD (g/cm^3)	0.758	0.757	-0.13
	$\dot{m}$ (g/s)	0.5074	0.5053	-0.41
	DT (s)	10.4	10.4	0

Table 4.15: Evaluation of the calibrated parameter combinations of the simulation results compared to that of the experiments.

Compared to Roessler et. al [48], the followed approach in this study proved to be more efficient and more reliable, where only 20 simulations (instead of 384 in their case) were executed to reach the same accuracy in the results. Thus, our systematic approach using DoE and MVRA is an optimized procedure that saves time and costs.

4.8.4 Summary

In case study 2, a standard calibration method for calibrating a non-cohesive powder, by tremendously reducing the ambiguity of the DEM parameters selection, is established. It is based on an auger dosing principle, which calibrates three bulk material properties at the same time, namely AoR, $\dot{m}$ and BD in relation to the auger speed, particle-particle beside particle-wall static friction coefficients and particle-particle rolling friction coefficient. The study analysis was done through a highly systematic statistical approach using MVRA. With only 20 simulations, a parametric study of DEM input parameters on the bulk behaviors was performed followed by generating very robust prediction models. This parametric study explained how the aforementioned bulk behaviors depend highly on the particle level interactions and properties. The simulation results used to build a prediction model showed how a high number of different parameter combinations can lead to the same bulk responses. This study proved that using the superimposing principle by combining the prediction contour plots of the bulk responses evidently led to a very narrow region that matches greatly with the experimental reference values. One optimized solution combination of DEM parameters through the prediction model was selected, where their results agreed significantly with the experimental data. A number of conclusions extracted from this study can be summarized as follows:

- Individual calibration tests are not sufficient to calibrate a material in DEM where the ambiguity for the admissible parameter combinations is high.
- The auger dosing machine can be used to extract three reference bulk responses, i.e., AoR, BD, $\dot{m}$.
- Using the D-optimal design model, the superimposing principle of three predicted contour plots from the MVRA and the optimization algorithm in Cornerstone proved to be a robust approach to eliminate ambiguity.
- $\mu_{r,pp}$ was shown to have an overwhelmed effect over $\mu_{s,pp}$.
- Using the presented auger dosing machine, a unique optimized parameter combination was identified and experimentally validated.

For the future work, process optimization should be carried out by studying and validating more material properties and bulk behaviors. In addition, the study should be extended to cover cohesive powders.

5 Design of the Vertical Loss in Weight Feeder

In this study, a systematic design approach for realizing a vertical LiW feeder in the pharmaceutical industry is developed. This LiW feeder is used to dispense intermediate quantities (low rate feeding) of powder in the range from mg up to several hundred grams. The first crucial steps towards designing and building a new vertical LiW feeder were defining the requirements and identifying the problems to be solved. As part of a systematic design approach, feasible design solutions for different challenges were selected in a similar way as for morphological analysis. The functionality of the feeder is experimentally validated through sensitivity analysis and powder dosing tests for the aim of achieving high dosing accuracies. An external catching load cell is used to accurately measure the amount of dosed mass and dosing stability (i.e., mass flow rate consistency).

5.1 Methodology

5.1.1 Problem Statement

Dosing with the horizontal orientation has a number of drawbacks. For instance, the powder residuals inside the dosing container or hopper or around the auger cause a longer cleaning time, and thus higher costs [81]. Cleaning can be done either manually or wash-in-place and rinse-in-place, which means that the feeder will not be operating/dosing at that time (lower production output). In other words, when powder A is exchanged with powder B, the feeder will not be able to run continuously. Some manufacturers run a back-up (second) feeder, which means double costs in order to ensure a continuous dosing process. Others interchange the feeder with another one and perform the cleaning externally [151]. In many cases, the manufactures use one feeder for each powder, which reveals less flexibility. Besides, one of the big negative impacts of using the horizontal dosing orientation is the huge loss due to the remaining residuals that cannot be dosed, where mainly some active pharmaceutical ingredients (APIs) are very expensive and they are only available in a limited amount. The automatic refilling process is still a challenge in the horizontal LiW feeders, where a lot of disturbances in the dosing accuracy take place during the continuous dosing process [87]. In addition, the dosing of cohesive powders, especially the fluffy ones, using the horizontal LiW feeders is still facing different problems, where the dosing accuracy is poor [8]. In the current dosing machines, one motor is used to control the speeds of the auger and the stirrer using gears transmission [8,81]. This means that the speed of the stirrer is set to a fixed ratio to that of the auger (cannot be controlled independently), and thus has less flexibility. Accordingly, mechanical parts exchange is required to modify the transmission ratio for any future speed change

depending on the material attributes of the new powder. Moreover, 100 % full dosing (emptying) of the hopper cannot be reached and the overall weight of the feeders is too heavy. Therefore, the goal of this study was to develop a prototype dosing machine to overcome a number of these stated drawbacks.

5.1.2 Design Workflow

A systematic design approach was followed to build a LiW feeder prototype. After identifying the challenge objectives and the needs for building a vertical LiW feeder, functional design specification used to set the guidelines was identified to develop and select possible solutions. After defining the solutions of different aspects of the dosing machine, a risk analysis was performed to ensure the safe components' functionality of the machine. Subsequently, 3-dimensional (3D) design was followed by purchasing and manufacturing the parts, and then mounting the entire machine. The 3D CAD models were constructed in SolidWorks version Premium 2016 from Dassault Systèmes (Vélizy-Villacoublay, France). Evaluation of the machine is a significant phase of the total design process to fulfill the proof of concept and thus the success of the design. Five powders of diverse flowing behaviors were utilized in evaluating the dosing accuracy of the vertical LiW feeder. Testing and evaluating the functionality of the machine and suggesting possible future improvements of the machine design were identified. Figure 5.1 demonstrates the flowchart of the engineering design process followed in building the vertical LiW feeder.

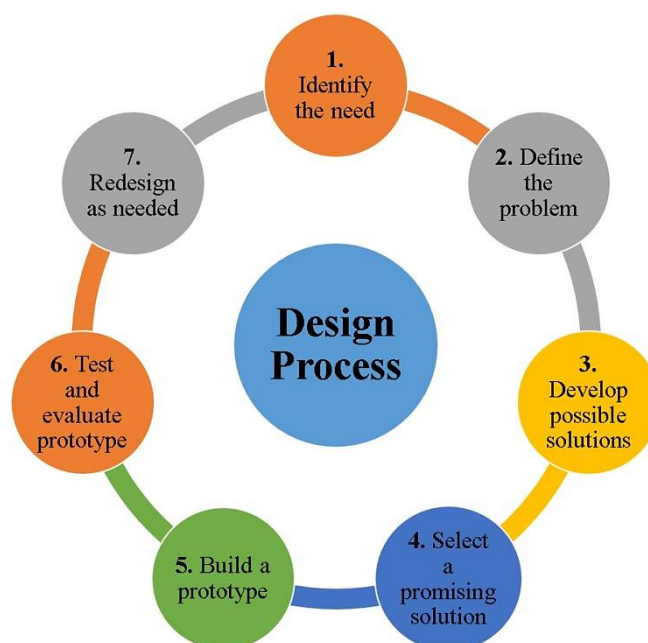


Figure 5.1: The engineering design process flowchart.

5.1.3 Functional Design Specification

The functional design specification (FDS) is created to define the requirements and specifications of the new dosing machine as well as the constraints of the design. In the development process, the FDS represents the interaction of mechanics, pneumatics, electrical hardware, and software. The new dosing module should include the interaction between mechanics, electrical hardware, and software. Six main

criteria parameters were designated for the evaluation and selection of different possible design solutions and then weighed according to their importance based on the demanded FDS. These main parameters are listed according to their weighing ranking order as following: accuracy, functional performance, lightweight and fewer space requirements, automation, manufacturing and procurement costs, and production simplicity.

The accuracy of the powder dosing results is of high importance since it has big impacts on the entire production process of the drug manufacturing and thus it was one of the crucial parameters to be achieved in this study. The functional performance of the motors and their couplings with the powder container is an essential design parameter. Moreover, a lightweight system is highly prominent, where it will be placed on a weighing module that has a limited acceptable weight capacity. Besides, the occupied space of the machine when integrated into the potential continuous manufacturing line is of great interest. The fourth evaluation criterion was the automation, where the LiW machine design should be easily automated with control systems to regulate its functionality and thus achieve more optimized results. In addition, its operation should be user-friendly. Besides the other required parameters, in order to ensure market acceptance, it is important to offer a low-cost manufacturing solution. Finally, yet importantly, ease of manufacture of the components and the entire machine was considered in the evaluation criteria.

Furthermore, some additional requirements are needed to be taken into consideration. Two detached motors are required to operate the auger and the stirrer independently. The motors should be easily coupled and decoupled from the powder container, where the powder container should be removed outside the system to be cleaned and filled up again with powder. It is also essential to eliminate the noises and vibrations coming from the surrounding into the dosing module and have special care for the motors' cables to be positioned in a way that they have very less influence on the accuracy of the load cell. In addition, an electric switch cabinet and software are required to obtain an automated controlled system.

5.2 Design Concept

5.2.1 System Description and CAD Design

The developed system, which is used to accurately dose pharmaceutical powders, especially APIs, is based on LiW feeder principle. After thoroughly evaluating and fulfilling the main selection criteria parameters discussed in the FDS section, a final powder dosing machine design prototype is realized. The main components included in the system are a high precision bench weighing platform (PBK989-AB15) from Mettler Toledo, three small synchronous servomotors from Beckhoff, a linear module, a powder container enclosing the auger and the stirrer, and a programmed automation software. The entire system is placed on a weighing platform that has a maximum capacity of 15 kg, a resolution of 20 mg, and a linearity deviation at a half load of 80 mg. The weighing platform is accurate, reliable, robust, versatile, and well applicable for industrial applications. It is provided with user-friendly software,

where the weighing settings can be easily controlled and adapted to different environmental and measuring conditions. It can be directly connected to control systems such as programmable logic controller (PLC) systems. The three servomotors have a standstill torque of 0.5 Nm, a rated speed of 3000 1/min at rated supply voltage of 48 V, and standstill current of 4 A. The motors are characterized by their one cable technology that combines power and feedback signals in one standard motor cable. This is a considerable advantage for the designed machine here by decreasing the number of external cables connected to it. The linear module has a maximum stroke of 75 mm with a spindle pitch of 1 mm. Moreover, an electric switch cabinet was installed for the aim of controlling the motion of the three motors that were programmed using TwinCAT 3 automation software. CX2030 embedded computer from Beckhoff, as a powerful PLC and motion control system, was installed. To add up, since the total capacity of the weighing platform is 15 kg, an optimization of the weight of the components was performed. Different materials in the design were utilized, i.e., aluminum, stainless steel, glass, and plastic to ensure having a total weight assembly within the limits and at the same time achieving good functional performance and stability. The total weight of the assembly that was placed on the weighing platform was around 14 kg.

5.2.1.1 Powder container design

A new unique powder container design was developed (Figure 5.2). It has an inner diameter of 75 mm and a total length of 267 mm. As mentioned before, despite most current horizontal LiW feeders which use only one motor to control the speed of the auger and the stirrer, in this design, the auger and the stirrer rotate on separated axes, where their axes are connected to motor 2 and motor 3, respectively. Two ball bearings for each axis, to ensure rigid mounting [84], are used to obtain frictionless rotations. Two deep groove ball bearings for combined linear and rotary motion from Rexroth (Germany) were purchased to ensure axial and radial motions of the auger shaft. The components of the powder container were designed and manufactured with close tolerances and were then axially aligned to ensure their centricity with each other. The stirrer and auger shafts were accurately positioned between the ball bearings and the housing frame components. The stirrer and the auger run independently, where their speeds are easily controlled through the installed software. Speed controlling is essential to adapt and deal with different kinds of powders. Besides, the flexible design allows easy interchange of different designs of stirrers and augers. The dosing tube can be exchanged as well depending on the outer diameter of the installed auger. The powder container is designed for utilizing small auger design geometries of outer diameters range between 3 and 10 mm. These small to medium sizes are well applicable to dose powders accurately in the range of mg and several grams. One of the most important aspects of designing the auger geometry is having a conical form which proved, as discussed in chapter 3 and [114], to have its positive influence on the dosing accuracy over the standard straight form. It served, by light compression, in transferring the powder into the dosing tube and thus increasing the density around the auger. The conical geometry aided in forwarding the powder downwards into the dosing tube and thus helped

in consistently having a high powder mass concentration around the auger [114]. Besides, the powder container consists of a transparent glass that allows the monitoring of the powder behavior inside the container during discharge. The lower part of the container is a conical funnel made of stainless steel 316L electro polished surface finish to ensure a smooth flow of powder. Due to its design and its surface finish, and due to the gravity force, the powder flows well downwards and thus a better dosing accuracy can be achieved. The powder container was assembled using two long screws. Moreover, powder dispensing involves the displacement of product and air, where the hopper supplying the product must admit air [83]. Thus, two air filters were added on the top of the container to ensure venting and consequently eliminate negative pressure inside the container during the dosing process. In the case of a closed container, negative pressure during powder discharge increases especially when dosing very fine powders, where the distance between the particles is very small, and thus the permeability of air is high (less air penetration). This would prevent air to escape inside the container and thus causing a vacuum that affects negatively the continuous supply of powder [114]. A spring is integrated inside the powder container to work as a closing mechanism so no powder can flow outside the container when the operator transports it outside the machine. This ensures no powder leakage during transport and thus more cleanliness and less contamination could occur. Two main radial shaft seals made of PTFE were mounted inside the container to prevent any leakage of powder through the internal parts of the powder container, i.e., spring and ball bearings area. A cross-sectional view of the powder container design is depicted in Figure 5.2. It shows furthermore the stirrer geometry, where, as it rotates clockwise, it serves in conveying the bulk material downwards into the feed auger and the dosing tube. Blue arrows represent rotation and the green arrow represents upward and downward movements, where the auger is at its initial closed position. The auger axis rotates completely independently from the stirrer assembly.

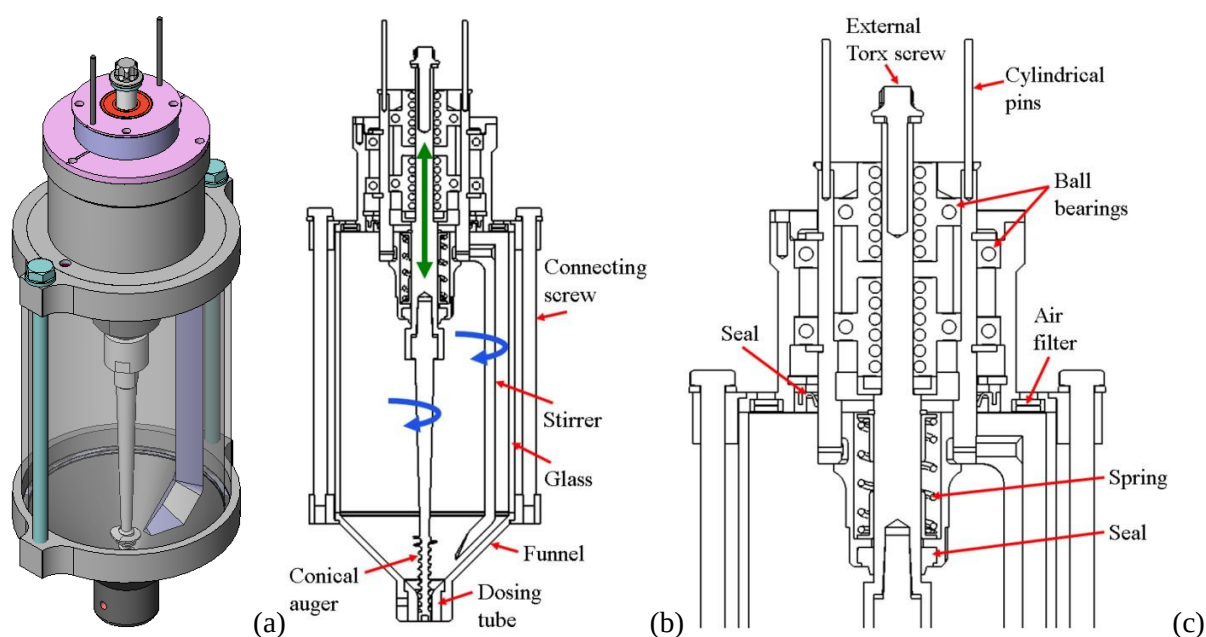


Figure 5.2: (a) 3D isometric view of the powder container and the cross-sectional view representing (b) the entire powder container and (c) the upper part of it.

5.2.1.2 Overall machine design

The entire dosing system was placed on a weighing platform as depicted in Figure 5.3. The weighing platform has a size of 280 mm x 350 mm with assembly total height of 650 mm. The whole assembly was mounted with the weighing platform through an L-shape aluminum plate and was fixed with four screws. Motor 1, with brake integration, is coupled with the linear module which allows the upward and downward movement of the other two motors coupled with the powder container. Two-timing or toothed pulleys of ratio 2:1 (30 and 60 teeth aluminum alloy pulleys) and a timing belt pulley used for power-transmission are coupled with motor 2 to rotate the stirrer. This configuration results in no slippage, and therefore the driving and driven shafts remain synchronized. The motor is coupled to the powder container through two hardened cylindrical pins fixed on the stirrer shaft. Motor 3 is accounted for rotating the auger through a coupling between an internal Torx screwdriver fixed on the motor shaft and an external Torx screw fixed directly on the auger shaft. The speed of each motor is controlled separately using a programmed user interface on a computer monitor. Moreover, Figure 5.3 shows the position of motors 2 and 3 before coupling them with the powder container.

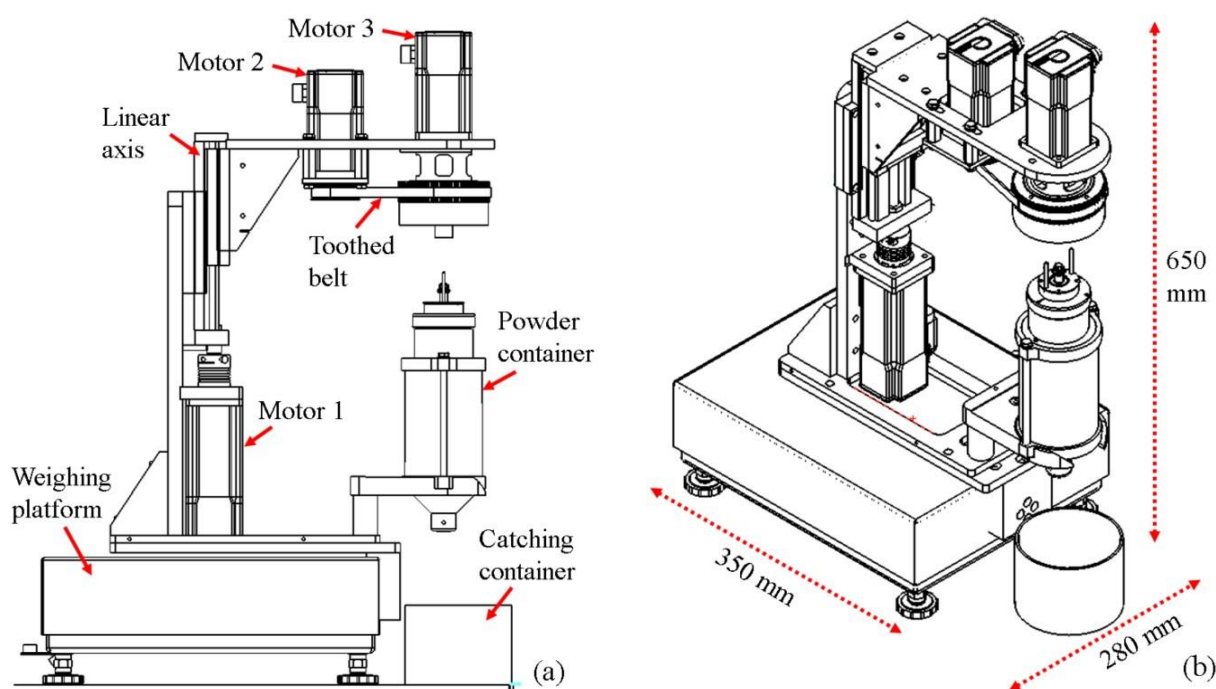


Figure 5.3: LiW feeder machine design before coupling, (a) side view, and (b) isometric view.

Figure 5.4 shows the coupling of motors 2 and 3 with the powder container. Motor 2 is coupled with the powder container through two hardened cylindrical pins, whereas motor 3 is coupled with the powder container through an external Torx screw.

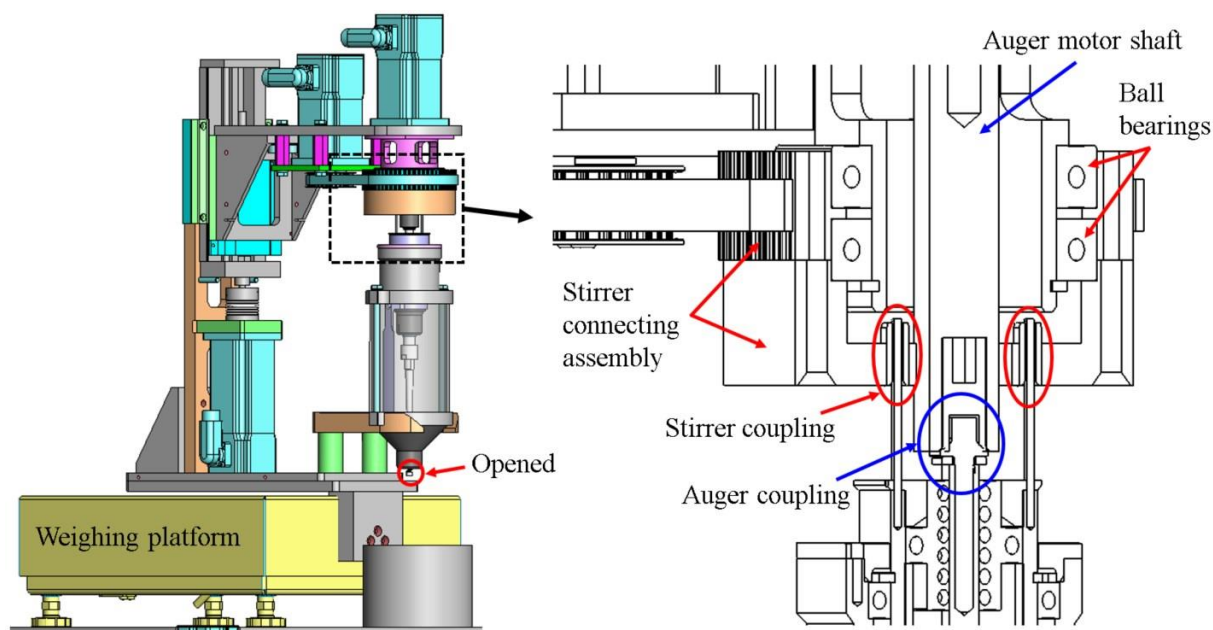


Figure 5.4: 3D view of the entire LiW feeder machine after coupling.

5.2.2 Mode of Operation

The mode of operation starts with cleaning the powder container and then placing it without powder in the machine. The powder container is then coupled with motors 2 and 3, where motor 1 moves them downwards with a stroke of around 55 mm through the linear module till the dosing tube is opened by having around 7 mm of the auger looking outside. This is called a dispensing position. During the downward movement, motor 3 rotates slowly at 5 rpm to adequately couple with the external Torx screw, which is fixed on the auger shaft. At this state, a stabilization time of few minutes should take place until stabilized weight readings in the weighing platform are reached. Thereafter, the taring of the weighing platform takes place. Then, the container is removed and filled with powder and placed again in the machine after repeating the previous steps and waiting to reach stabilization. Subsequently, dosing starts by setting defined speeds to the auger and the stirrer. At the end of the dosing, motors 2 and 3 stop and motor 1 transports both motors up to decouple them from the container. With the help of the integrated spring inside the powder container, the dosing tube is closed again so no powder can fall when the operator transports the container.

The initial weight read by the weighing platform is just the filled powder inside the container. During the dosing process, the weight was monitored and recorded, where the weight started decreasing gradually until the desired weight to be dispensed was reached. The falling weight can be measured every few milliseconds. APW-Link software from Mettler Toledo was used on a computer to record the values of mass flow over time through a serial interface connection. Dispensed amount, dosing time, and other data for every single test were stored into Excel file. The mass flow rate or the volumetric feed rate ($\dot{m}$) was calculated using the difference in mass (Δm) obtained from the weighing platform divided by the difference in time between consecutive weighing balance measurements (Δt).

For the proof of concept validation, a laboratory weighing load cell XPE 204, used as a catch scale, was placed under the outlet of the powder container to record the powder mass flow every 0.5 s, where the dosed powder was collected in a transparent acrylic catching container. Engisch et al. [152] proposed a similar methodology approach to characterize LiW feeders by implementing GiW load cells. The feed rate of the XPE 204 was considered as GiW since the mass is increasing on it during the dosing process, whereas the dosing machine is losing mass and thus it is considered as LiW feeder. The aim was to confirm the feed rate and dosed mass from the LiW feeder with that of the GiW. The XPE 204 has a capacity up to 220 g, readability down to 0.1 mg, and a linearity deviation of ± 0.2 mg. HTerm software was used on a computer to record the values of weight (in g) over time. Furthermore, Static Line LC ionization device was utilized to neutralize electrostatic charges.

5.2.3 Key Features of the Design

The key features of the new invented vertical LiW feeder machine are custom engineered for easier handling, cleaning, and product changeover. Highlights of these features include:

- First-in-first-out is ensured, which means very fewer powder residuals (high yields) are remaining inside the container and thus saving money, where many APIs are very expensive.
- Modular quick-change design concept allows the easy exchange of the auger, the stirrer, and the whole powder container design.
- Parallel operations can be executed, e.g., placing and dosing powder container 1 while cleaning and preparing other containers to have a quick change over of the material. Moreover, one container for each type of product, eliminates contaminations, infections, and impurities.
- Replacing the empty powder container with another container holding the same or different material is done within few seconds, which is an alternative option for the refilling process found in the horizontal LiW feeders. This process saves the refilling and cleaning time processes, where no need for in-place cleaning. Therefore, the production time (continuous manufacturing process) is preserved and thus money is saved.
- High potent APIs can be easily handled in this machine, where the filling of the container can be done in a safe workbench, and with the help of the auger-closing mechanism, the container transfer is safe.
- Ideal sealing of the powder container for ultimate product containment.
- The LiW feeder is a lightweight design, where the total assembly without the weighing platform is around 14 kg.
- The machine is user-friendly, where it is automated with the help of a software.
- Relatively cheap machine and easy maintenance.
- This machine can be used as a standalone machine as well as being integrated into many processes, such as continuous manufacturing line for mixing, wet and dry granulation, extrusion,

and filling capsules and tablets. It can be used in different industries other than the pharmaceutical, e.g., food, chemicals, painting, cosmetics, etc.

5.2.4 Implementation of the Prototype

Many parts were manufactured in house at the manufacturing shop floor of Bosch in Waiblingen, Germany and the rest of the components were purchased from external suppliers, e.g., weighing platform, servomotors, linear module, ball bearings, toothed pulley and belt, etc. As an initial step, the weighing platform was calibrated and then the entire machine parts were mounted on a leveled and stable table support. Four leveling element foot for the table (BNVS), from Bilz Vibration Technology AG (Germany), were used for vibration and structure-borne noise insulation. A red silicon rubber of 3 mm thickness was placed under the bottom plate between the assembly and the weighing platform to help in absorbing vibrations. The electric switch cabinet was mounted and connected to the three motors. In addition, the motion of the motors was programmed through TwinCAT 3 software from Beckhoff. The cables of the three motors were installed horizontally with sufficient length on two aluminum profiles so that they hang without influencing the weighing platform. A user interface (monitor display) was placed, where it contained the installed software to control the motion of the motors. As mentioned before, a laboratory weighing balance XPE 204, used as a catch scale, was placed under the outlet of the powder container. A protective draft shield (carton) was placed to block or divert the stray air currents coming from the surrounding into the feeder to ensure maximum accuracy of the weighing platform in unstable environments. The assembled LiW feeder is shown in Figure 5.5(a), while Figure 5.5(b) represents the backside of the whole machine and showing the electric switch cabinet.

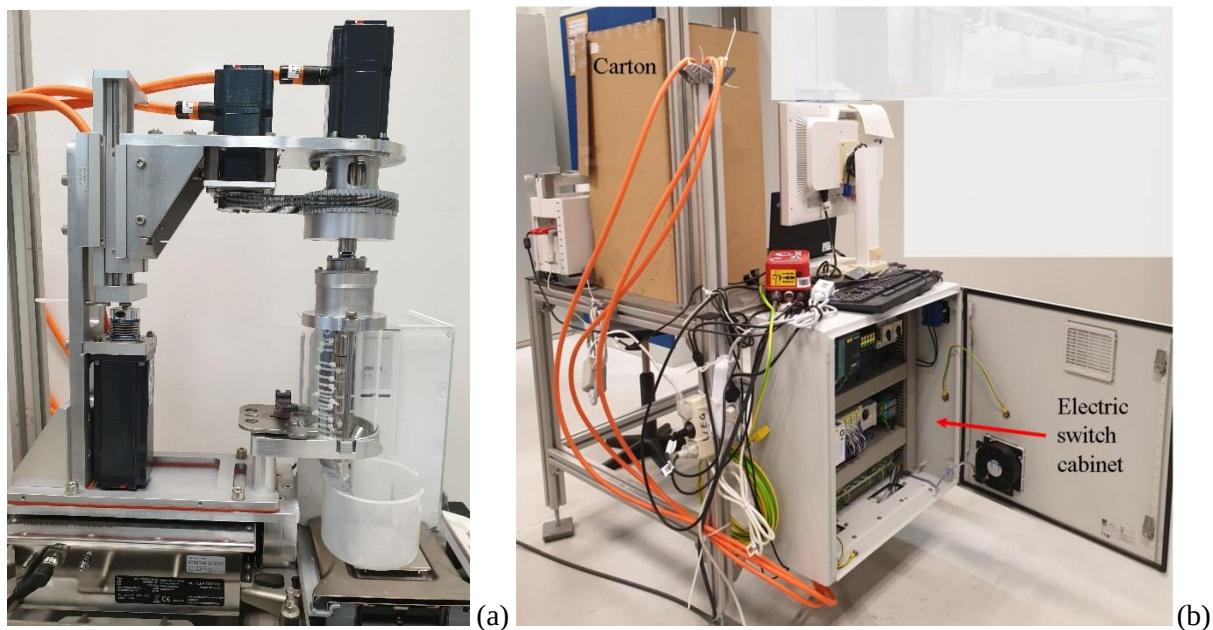


Figure 5.5: Setup of the entire LiW feeder machine showing (a) a side view of the feeder and (b) an isometric backside of the whole assembled table.

5.3 Experimental Validation of Proof of Concept

5.3.1 Sensitivity Analysis

In the process of the proof of concept of the vertical LiW feeder prototype, a sensitivity analysis was performed to study the functionality of the machine and the accuracy of the weighing platform, in addition to figuring out the best applicable weighing parameter settings for the application. The experimental setup of the dosing machine during the sensitivity analysis and the powder dosing tests is shown in Figure 5.6.

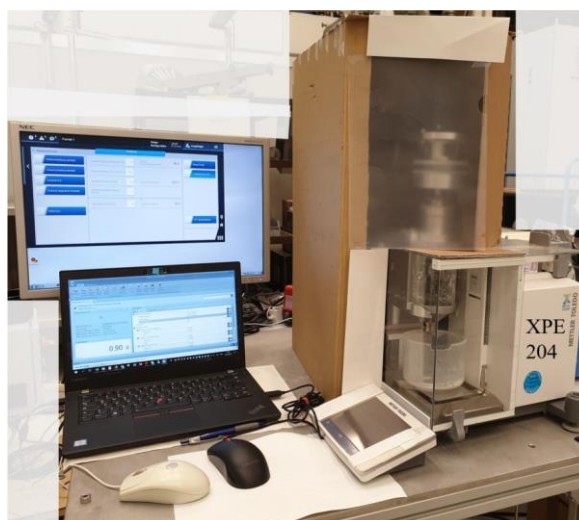


Figure 5.6: Experimental setups of the LiW feeder.

Since three servomotors were placed on the weighing platform, we had to verify that the rotation motors and their connected electric and signal cables did not cause any disturbances. Motors 2 and 3 were mainly our focus since no dosing took place during the rotation of motor 1, which is only responsible for coupling and decoupling the powder container before and after dosing, respectively. Principally, five test conditions were executed to investigate the functionality of the prototype. Table 5.1 lists the conditions of the tests used in performing the sensitivity analysis and proof of concept experimental validation.

Testing condition	Material availability	Varied parameters
No motor operating	without powder	Filters
Auger running		Speeds
Stirrer running		
Auger and stirrer running	with powder	5 powders, 3 augers and stirrers' geometries, speeds
Auger and stirrer running		

Table 5.1: The test conditions of the sensitivity analysis and the proof of concept experimental validation.

The tests were repeated three times to ensure the repeatability of the results. The first test was performed to determine the variability of the weight values over time when no motors were running. Without using the filters integrated into the weighing platform, i.e., filter with a cut-off frequency of 20 Hz, the results showed weight fluctuations over time, whereas in case of using the maximum possible filter with 0.001 Hz, the results were then very stable as represented in Figure 5.7 which reveals reality. Therefore, to increase the accuracy of the feeder and avoid incorrect weighing results, the signal coming from the weighing platform was filtered so that disturbances that do not correspond to a real weight change were removed. Such disturbances or mechanical noises can be, for instance, a vibration caused by someone walking close to the feeder or touching its table, transporters of goods moving on the shop floor or a draft of air. Therefore, sensor weighing mode intended for continuous dosing process along with filtration with a cut-off frequency of 0.001 Hz were utilized in conducting all the test cases.

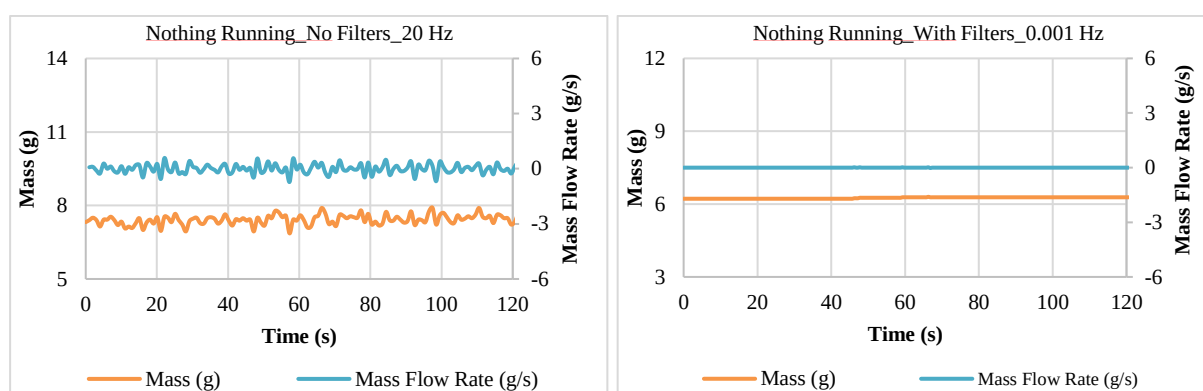


Figure 5.7: Mass flow rate sensitivity tests when no motors were running without filters (left) and with filters (right).

The second test was done when only the auger was rotating at different speeds between 100 rpm and 600 rpm. The weight change over time was recorded before turning on the motor, during rotation and after turning off the motor. The results revealed that just after turning on the motor (i.e., motor 3), there was a small negative shift deviation (dropdown) of around 1-2 g in the weight values, where the weight readings remained almost constant overtime during the continuous rotation until the motor was turned off. Thereafter, the weight value shifted back to its initial value (i.e., before turning on the motor).

Similarly, in a third sensitivity test, the impact of the rotation of the stirrer through motor 2 on the weight variability over time was investigated with different speeds between 10 rpm and 200 rpm. The results showed that just after turning on the motor, there was a positive shift deviation in the weight values (around 2-3 g) that remained constant at that high value over time until the motor was turned off, where then the weight value converged almost back to its initial state. This phenomenon can be explained by the sudden shock loading and mechanical resonance that took place at the moment of turning on the motors. A torque perpendicular to the weighing platform due to the rotation motion may have caused the whole system to lift up or down and deviate from its center of gravity and thus influenced the weight.

Moreover, considering the continuous constant speed rotation, the shifted weight values remained constant until the motor was turned off.

The shifts in weight values caused by motors 2 and 3 are represented in the subsequent illustrations in Figure 5.8. The illustrations show two examples of sensitivity analysis tests, where the effects of motors 2 and 3 were investigated in the same test as part of the fourth sensitivity analysis test conditions. Initially, the recording of the mass over time started when nothing was running for 10 sec before turning on the auger motor (motor 3), where a small negative shift in the weight value took place. After around 12 sec, the stirrer motor (motor 2) was turned on while motor 3 was still running. As a result, a high positive shift occurred in the weight value. After around 65 sec, both motors were turned off which caused a negative shift of the weight value, where then within a few seconds the weight was stabilized back to its initial state value. It is obvious in Figure 5.8 the occurrence of repeatable patterns of weight value variations (high and low peaks), where it is in fact due to the continuous constant speed rotation of the two motors. The weight deviation error between the final and initial state was around 1-2 %. Hence, the rotation of the stirrer has a larger impact on the weighing platform readings than the rotation of the auger. Future optimization work will tackle the deviation problem.

In the first example, the rotation speeds of the auger and the stirrers were 200 rpm and 120 rpm, respectively, while in the second example the speeds were 400 rpm and 200 rpm, respectively. It was obvious to see more weight fluctuations when both motor speeds were relatively low and more stable values with higher speeds which means that the shock loading peaks were faster converged due to a higher number of rotations per second. Nevertheless, regardless of the peaks, the weight values were stabilized almost back to their initial values in both tests after turning off the motors. The mass flow values over time for the two mentioned test cases, i.e., with low and high rotation speeds, are represented in Figure 5.8.

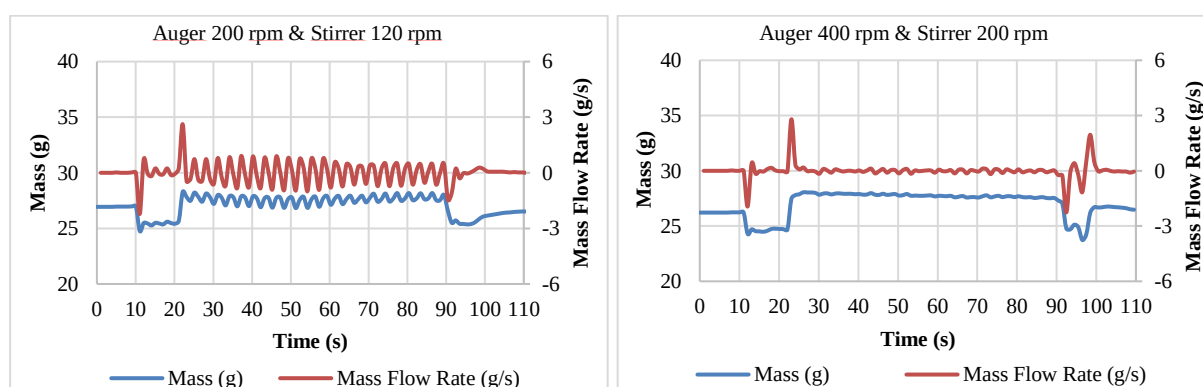


Figure 5.8: Mass flow rate sensitivity tests when running the auger and the stirrer with different rotation speeds.

An additional test was performed by rotating motors 2 and 3 without coupling them with the powder container. The results indicated no significant impact or disturbances on the weighing accuracy.

5.3.2 Powder Dosing Tests

For conducting the confirmation powder dosing tests, the same weighing conditions of the weighing platform were applied in the laboratory XPE 204 load cell, where the results of XPE 204 (as GiW) were considered as reference values. The results from the LiW feeder were compared to that of the GiW load cell. Five different powders were studied in this work to investigate how successfully and accurately the LiW feeder prototype can dispense them. Five powders ranging between free-flowing and cohesive flowability characteristics were selected to evaluate the functionality of the LiW feeder. The powders are SpheroLac 100, Vivapur 12, Maize starch extra white powder, Magnesium stearate (NutriMag ST-v), and micronized Paracetamol (Acetaminophen). Micronized Paracetamol is used as an API, where it was purchased from Sigma-Aldrich (Germany). Table 5.2 summarizes several raw material attributes of the five considered powders.

	d10 (μm)	d50 (μm)	d90 (μm)	MC (%)	BD (kg/m^3)	TD (kg/m^3)	CI (%)	AoR ($^\circ$)
SpheroLac 100	53.78	123.8	197.57	0.08	714	850	16	31.8
Vivapur 12	38.96	161.74	323.81	2.68	365	472	22.67	34.7
Maize Starch extra white	7.86	14.26	21.05	10.55	543	714	24	50
Magnesium Stearate	0.69	2.46	12.08	4.56	265	378	30	47
Micronized Paracetamol	2.3	11.2	40.8	0.12	242	360	32.78	52

Table 5.2: Raw material attributes of the five studied powders.

Three micro-conical augers with different outer and inner diameters and pitches were utilized in conducting the LiW feeder tests. Table 5.3 summarizes the geometry dimensions of these three conical augers used in the experiments to validate the functionality of the dosing machine prototype.

Auger number	Outer diameter (mm)	Inner diameter (mm)	Pitch (mm)
25	6	2.9	3.5
34	8	3.3	4.5
43	10	5	4.5

Table 5.3: Augers used in auger dosing tests.

Figure 5.9 represents the geometry of the three conical augers, namely augers numbers 25, 34, and 43 from left to right, respectively. The conical augers have a total length of 125 mm.

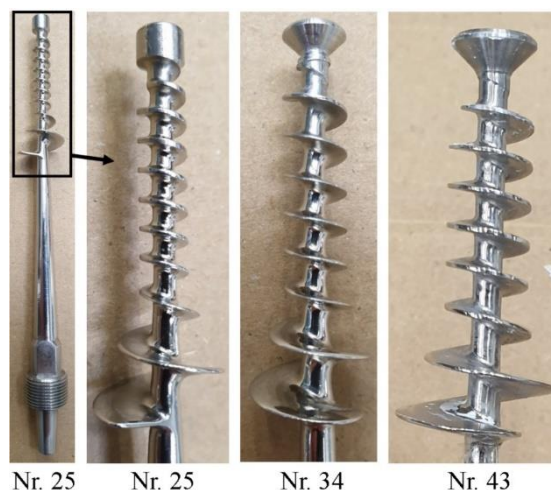


Figure 5.9: The three conical augers used in running the powder tests.

5.3.2.1 Free-flowing powders

5.3.2.1.1 SpheroLac 100

SpheroLac 100 was tested with auger number 25 running at three different rotation speeds (200, 400 and 600 rpm) along with three different stirrer rotation speeds (30, 120 and 200 rpm). The dosing of SpheroLac was performed very successfully, where the feed rate results from the LiW and GiW were almost in good agreement with each other and no blockages were occurred. As a first test example, SpheroLac 100 was dosed with auger speed 200 rpm and stirrer speed 120 rpm, where within 805 s, 80.1 g were dosed in the LiW while 79.66 g were recorded by the GiW which represents 0.55 % deviation error. The powder bed was decreasing continuously without sticking on the glass wall as shown in Figure 5.10, where at the end of the dosing process the powder container was almost fully dosed (emptied).



Figure 5.10: During an auger dosing test of SpheroLac 100.

In Figure 5.11, the left graph represents the LiW mass flow rate results while the right graph represents the GiW mass flow rate results. A linear regression of the mass curves was executed, where the coefficient of determination (R^2) of the linear regression was almost equal to 1 in both graphs. The R^2 represents the goodness of fit of the model. The slopes of these linear curves represent the $\dot{m}$, i.e., -0.0998 g/s (LiW) and 0.1022 g/s (GiW). The left graph column represents the mass (g) and the right graph column represents the $\dot{m}$ (g/s), where both are plotted over time (s). As discussed previously, in the LiW graph when the motors are turned on and off, sudden shifts in the mass values occurred and then the values were converged. Although the weight readings in the LiW were fluctuating due to vibrations caused by the motor rotation, the confirmation results with the GiW at the end of the dosing process were accepted.

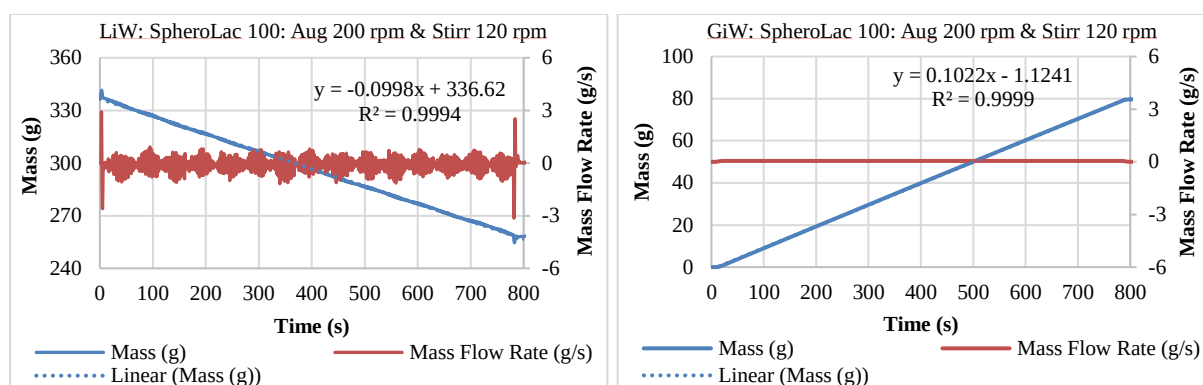


Figure 5.11: SpheroLac 100 dosing test results with auger 25 at speed 200 rpm and stirrer at speed 120 rpm.

In a second test example, SpheroLac 100 was dosed with auger speed 400 rpm and stirrer speed 200 rpm, where within 454 s, 91.42 g were dosed in the LiW while 90.44 g were read by the GiW, which then reveals 1.08 % deviation error. The slopes ($\dot{m}$) of the linear regression mass curves, i.e., -0.214 g/s (LiW) and 0.2103 g/s (GiW) are represented in Figure 5.12. In the LiW results, a positive shift in weight at the moment of turning on the motors was recorded and then a negative shift took place when the motors were turned off. This up and down shifting is averaged through the whole dosing process leading to similar results with the reference GiW result values. The GiW results revealed a smooth linear regression with a high coefficient of determination.

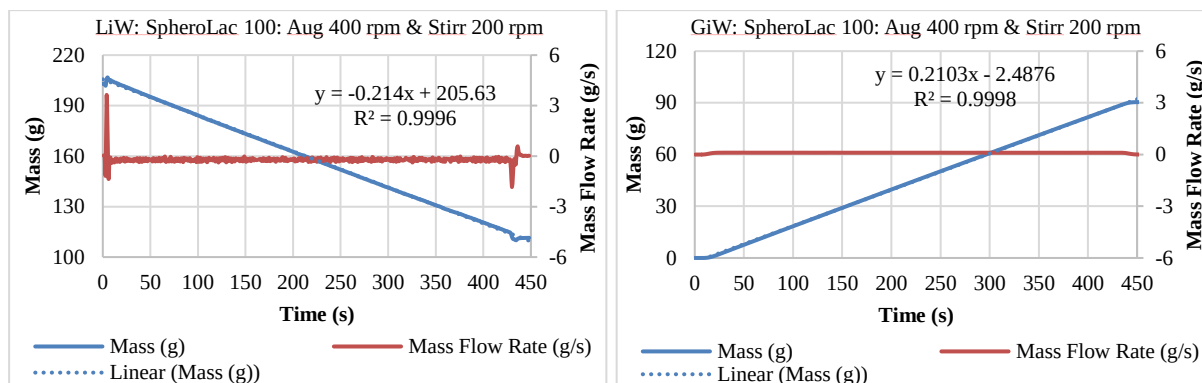


Figure 5.12: SpheroLac 100 dosing test results with auger 25 at speed 400 rpm and stirrer at speed 200 rpm.

In a third test example, SpheroLac 100 was dosed at auger speed 600 rpm and stirrer speed 200 rpm (Figure 5.13). In 291 s, 85.8 g were dispensed in the LiW while 86.25 g were recorded by the GiW, which represents a very small deviation error of -0.52 %. It is obvious to see the less fluctuations in the weight readings when running with higher rotation speeds, especially the stirrer rotation, where in other tests with same auger speed but with higher stirrer speeds, the fluctuations were vastly eliminated. The reason for that would be due to the mechanical connection of the stirrer to the motor using the pulley and belt, which has a negative impact on the accuracy of the weighing platform. The continuous repeated coupling and decoupling of the pulley and belt tooth would be the main cause of these arising fluctuations.

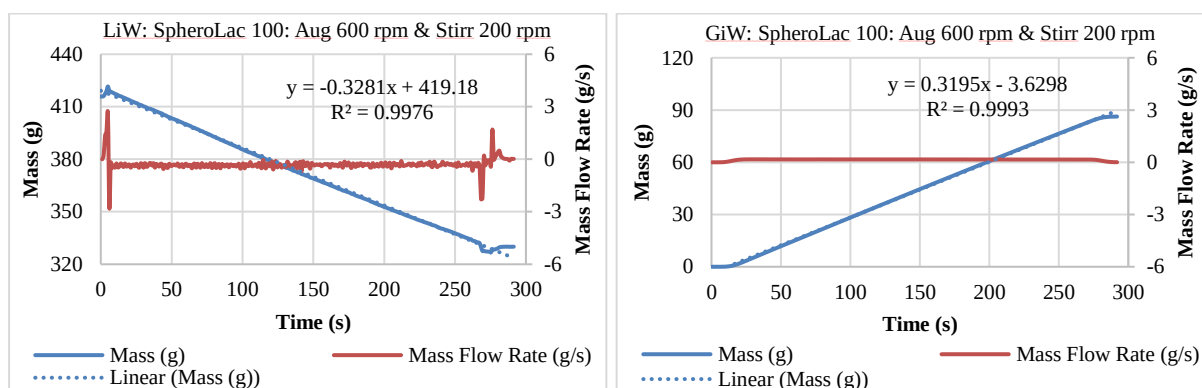


Figure 5.13: SpheroLac 100 dosing test results with auger 25 at speed 600 rpm and stirrer at speed 200 rpm.

5.3.2.1.2 Vivapur 12

Vivapur 12 was tested with auger number 25 running at speed 600 rpm and stirrer running at speeds 30, 120, and 200 rpm. Vivapur 12 running with auger speed 600 rpm and stirrer speed 30 rpm indicated weight reading fluctuations in the LiW, where within 304 s, 48.34 g were dosed in the LiW while 46 g were recorded by the GiW, which represents a deviation error of 5.1 %. The slopes of the linear regression of the mass curves were -0.1703 g/s (LiW) and 0.1666 g/s (GiW) as depicted in Figure 5.14. Small amount of powder remained in the container at the end of the dosing process.

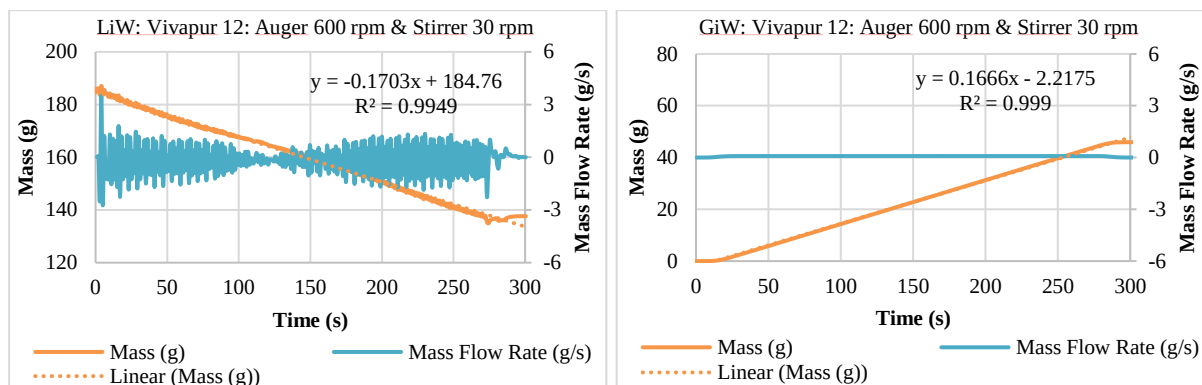


Figure 5.14: Vivapur 12 dosing test results with auger 25 at speed 600 rpm and stirrer at speed 30 rpm.

In the second case study, with higher stirrer speed at 200 rpm, the dosing results of Vivapur 12 were relatively better than that with lower stirrer speed (30 rpm). In the LiW, 50.9 g was dispensed while in the GiW 49.1 g were recorded at the same time (322 s). The deviation error is 3.66 %. As represented in Figure 5.15, the slopes of the linear regression of the mass curves were -0.1704 g/s (LiW) and 0.1648 g/s (GiW) which are similar to the previous test because the auger ran in both tests at the same speed (600 rpm). The higher stirrer speed showed a positive effect in dosing Vivapur 12 by having a lower deviation error. Therefore, compared to the previous test example and to that of SpheroLac 100 powder, the stirrer speed and/or its mechanical connection to the motor alter the LiW weight readings, where the high rotation speed proved to cause higher convergence of the weight values and thus eliminated the large peaks in the reading values.

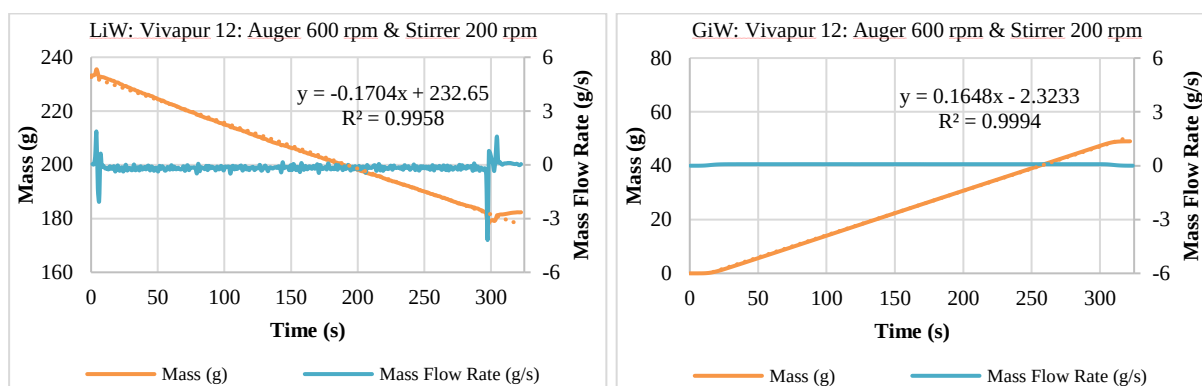


Figure 5.15: Vivapur 12 dosing test results with auger 25 at speed 600 rpm and stirrer at speed 200 rpm.

In general, dosing the two free-flowing powders, i.e., SpheroLac 100 and Vivapur 12, showed good results. In that, the results proved the functional validity and accuracy of the vertical LiW feeder in dosing such kind of powders. More process investigations and analyses along with the potential mechanical improvements of the stirrer connection with the motor would improve more the results.

5.3.2.2 Cohesive powders

On the other side, the other three tested powders, i.e., Maize Starch extra white, Magnesium Stearate, and Paracetamol are considered cohesive.

5.3.2.2.1 Maize Starch extra white

Maize starch extra white was dosed with auger number 34 at speeds 100, 200, and 600 rpm. Using the original stirrer geometry as previously depicted in Figure 5.2, the dosing of Maize Starch extra white was very challenging where the powder was sticking on the inner walls of the powder container, namely the glass, and the stainless steel funnel, and thus the flow discharge was poor. Therefore, as an iteration improvement process, white plastic cable ties were fixed on the stirrer and on the auger as well to break down the accumulations of powder and lightly wipe the inner walls. Consequently, this aided in facilitating the powder flowability. Figure 5.16 shows how the white cable ties were positioned on the stirrer (named as stirrer version 2) and on the auger and how they then assisted in wiping the glass wall. In addition, Figure 5.16(c) displays how the powder container was almost empty at the end of the dosing test, where the geometry modifications on the stirrer and the auger proved their positive effect on the process results.

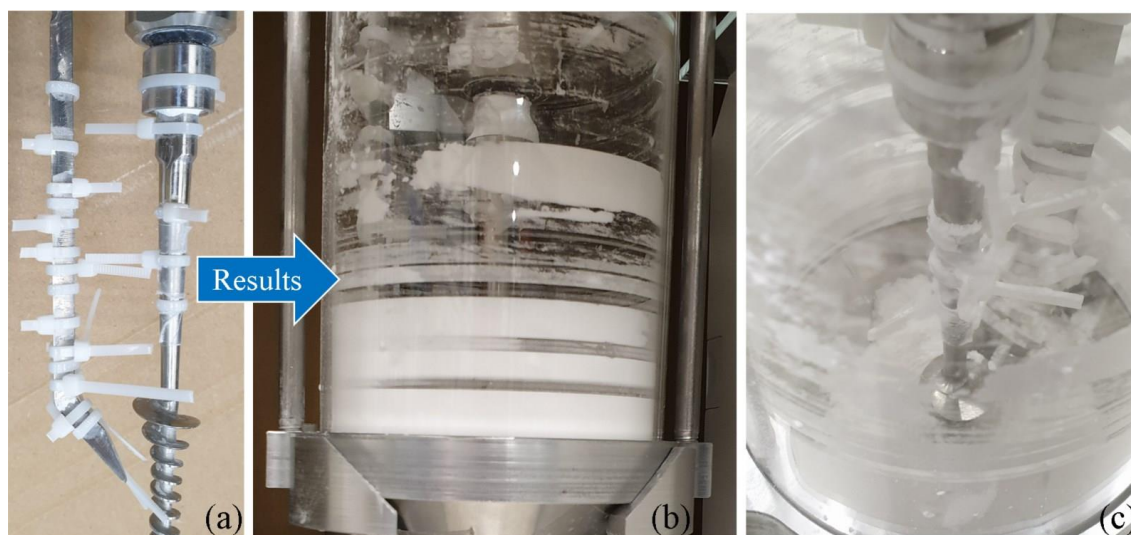


Figure 5.16: Maize Starch extra white dosing using (a) stirrer version 2 and (b) its effect on the glass wall and (c) almost full emptying the container.

Maize Starch extra white was studied under low motors speeds, where pre-tests with high motors speeds caused discontinuous discharge of this powder. With auger rotation at 100 rpm and stirrer rotation at 10 rpm, 59.86 g were dispensed in the LiW, whereas 59 g were measured by the GiW and that was within 477 s as depicted in Figure 5.17. The deviation error is 1.46 %, which reveals a very good result for this cohesive powder that contains high moisture content. The $\dot{m}$ of both LiW and GiW results were relatively close to each other.

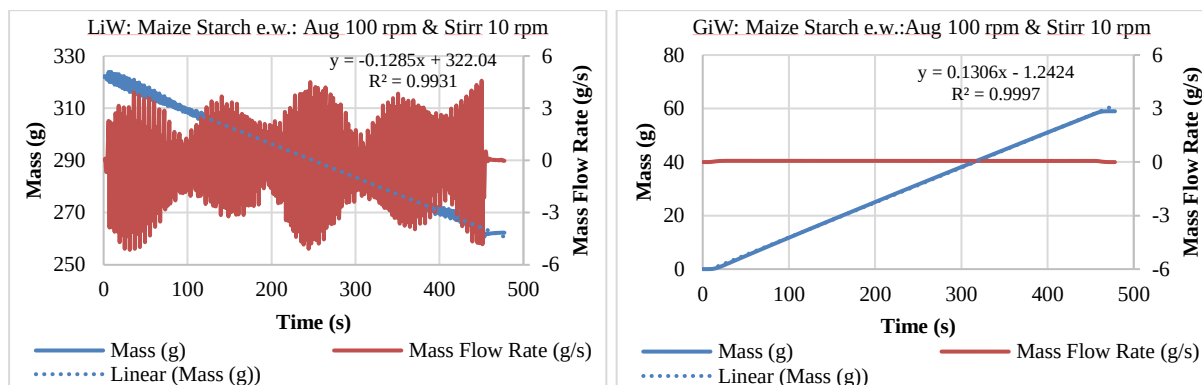


Figure 5.17: Maize Starch extra white dosing test results with auger 34 at speed 100 rpm and stirrer at speed 10 rpm.

In different test conditions with higher motors speeds, i.e., auger rotation at 200 rpm and stirrer rotation at 20 rpm, good dosing results were achieved as well (Figure 5.18). In 244 s, 47.8 g were dispensed in the LiW while 48.57 g were recorded by the GiW which indicates -1.58 % deviation error.

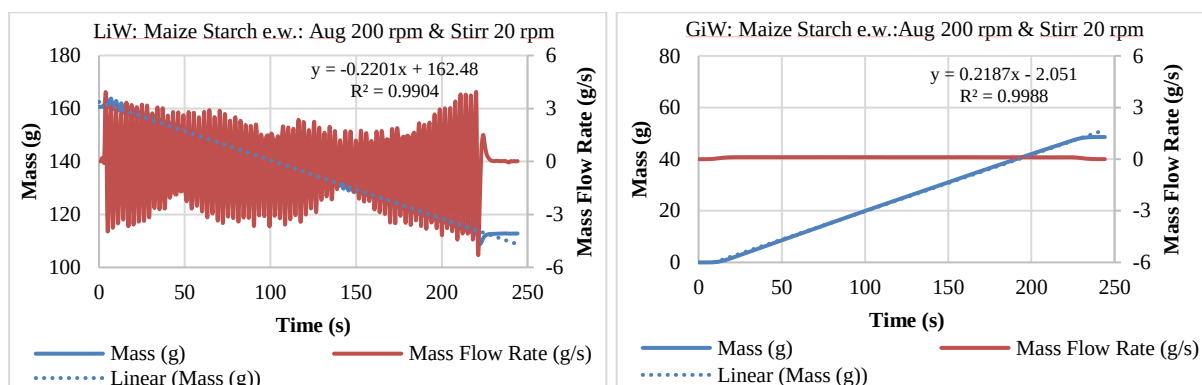


Figure 5.18: Maize Starch extra white dosing test results with auger 34 at speed 200 rpm and stirrer at speed 20 rpm.

5.3.2.2.2 Magnesium Stearate

Augers 34 and 43 were handled to dose Magnesium Stearate. Similarly, as with Maize Starch extra white, using the original stirrer geometry, very small quantities of Magnesium Stearate were dispensed at the beginning of the dosing process, where after less than one minute, no more powder was discharged. After waiting a few minutes, the test was stopped and the powder container was then opened. Figure 5.19 represents how the original stirrer geometry caused Magnesium Stearate to completely adhere to the inner walls of the container.



Figure 5.19: Magnesium Stearate blockage when using the original stirrer geometry.

Equivalently, the use of stirrer version 2 procured to ease the discharge of Magnesium Stearate but with limited manner, where the powder was still sticking on the inner walls. Consequently, an additional modification on stirrer version 2 took place by adding a long plastic cable tie to wipe more the inner surface wall of the glass. The new modified stirrer, stirrer version 3, is represented in Figure 5.20(a), where it led to very good results by dispensing most of the Magnesium Stearate outside the container. At the end of the test, the glass was almost transparent as depicted in Figure 5.20(b) and fewer residuals were left over on the conical funnel as shown in Figure 5.20(c).

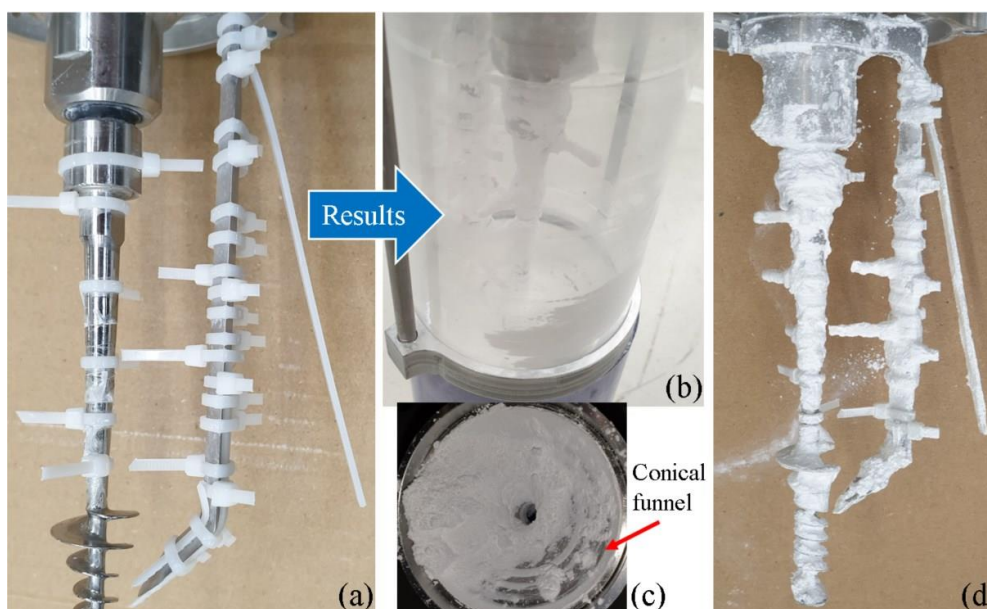


Figure 5.20: Results of Magnesium Stearate when using (a) stirrer geometry version 3, (b) and (c) left overs inside the container, and (d) sticking powder on the auger and stirrer.

Using auger number 34, Magnesium Stearate was tested at auger speed 400 rpm and stirrer speed 30 rpm. The illustration in Figure 5.21 represents the results when the powder bed level was high at the beginning of the test. In 244 s, 38 g were dispensed in the LiW while 37.3 g were measured by the GiW.

The deviation error is 1.88 %, which is a good result. In addition, the values of the $\dot{m}$ in both LiW and GiW were very close to each other.

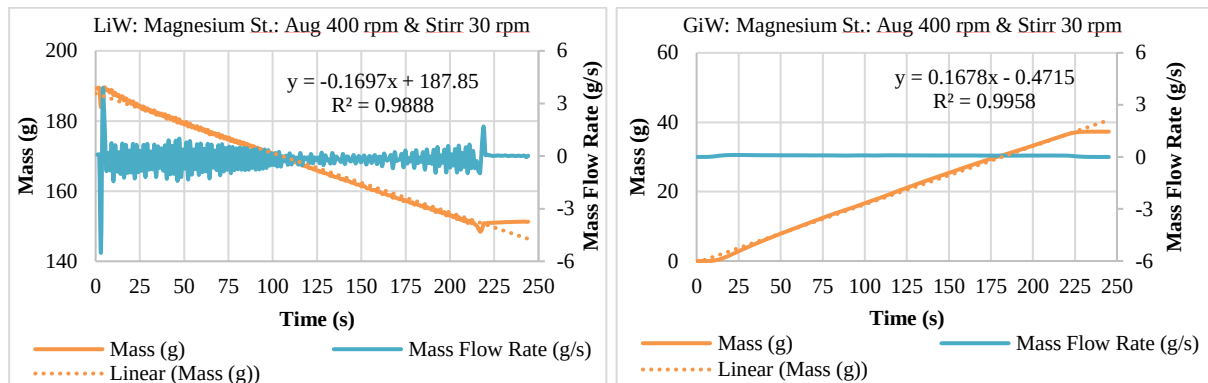


Figure 5.21: Magnesium Stearate dosing test results with auger 34 at speed 400 rpm and stirrer at speed 30 rpm.

On the other side, with low powder bed level, Magnesium Stearate was tested with the same motors speeds as the previous test, i.e., at auger speed 400 rpm and stirrer speed 30 rpm, though by using auger 43 instead of auger 34. The results showed that in the LiW, 26.26 g were dispensed while the GiW recorded 25.18 g in 189 s, which reveals 4.3 % deviation error. Figure 5.22 represents the mass flow rate results of this test case. Although auger 43 has a bigger filling capacity than that of auger 34, the comparison between the results of both tests, i.e., with low and high powder bed levels, showed that the $\dot{m}$ was lower when using auger 43 in both LiW and GiW. Therefore, this adds to the fact that the powder bed level plays a significant role in affecting the discharge rate ($\dot{m}$).

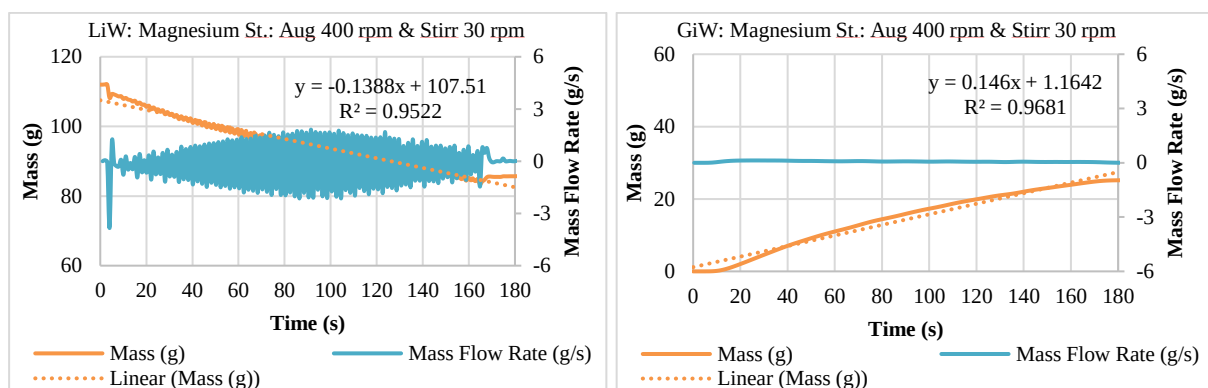


Figure 5.22: Magnesium Stearate dosing test results with auger 43 at speed 400 rpm and stirrer at speed 30 rpm.

Using auger 43 with auger speed 400 rpm and stirrer speed 120 rpm along with high powder bed level, good dosing results were obtained by having a relatively high increase in the dispensing rate of Magnesium Stearate as shown in Figure 5.23. In 100 s, 32.52 g were discharged in the LiW while 33.18 g were recorded by the GiW. The results indicate -1.93 % deviation error.

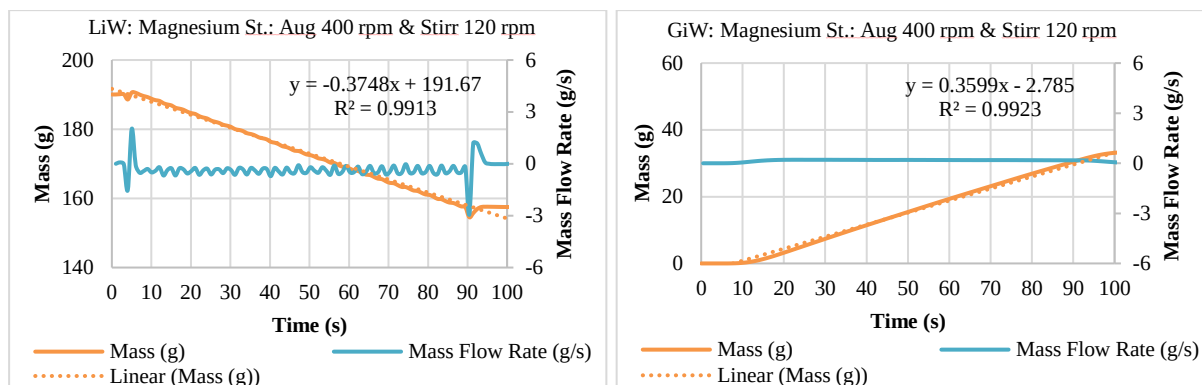


Figure 5.23: Magnesium Stearate dosing test results with auger 43 at speed 400 rpm and stirrer at speed 120 rpm.

5.3.2.2.3 Micronized Paracetamol

Dosing micronized Paracetamol with the powder container of the vertical LiW feeder was unfortunately unsuccessful. The original and varied versions of the stirrer geometry did not work well in agitating micronized Paracetamol inside the container. The stirrer motor stopped automatically after a few seconds, where micronized Paracetamol was accumulated and thus blocked the rotational movement of the stirrer. The maximum standstill torque of the servomotors (i.e., 0.5 Nm) was not enough to apply sufficient torque against such kind of cohesive powder. Therefore, as an alternative solution, an old small mixer powder container from the lab, as shown in Figure 5.24(a), was used in running the dosing test of micronized Paracetamol. This mixer powder container has a stirrer mixer system linked with gears to the motor drive shaft. The stirrer mixer system consists of three whisks and one squeegee as depicted in Figure 5.24(c). The motor drives the screw connected to the auger which in turn is linked to a large gear of 55 teeth (Figure 5.24(b)). This gear then drives the three little whisks through their small gears, where each gear contains 15 teeth (Figure 5.24(d)). Accordingly, the gear ratio between the large and small gears is 3:11. The whisk rotates around itself and simultaneously makes a complete rotation inside the container with the same rotation speed of the auger. As the squeegee is directly connected with the auger shaft, it rotates with the same speed as the auger. The role of the squeegee is to scrub the inner wall of the container and thus to avoid the powder from blocking and sticking. Figure 5.24(e) shows how the mixer powder container was integrated inside the vertical LiW feeder. A welded long screw was mounted to couple the mixer container with the motor shaft since the linear module reached its maximum stroke limit.

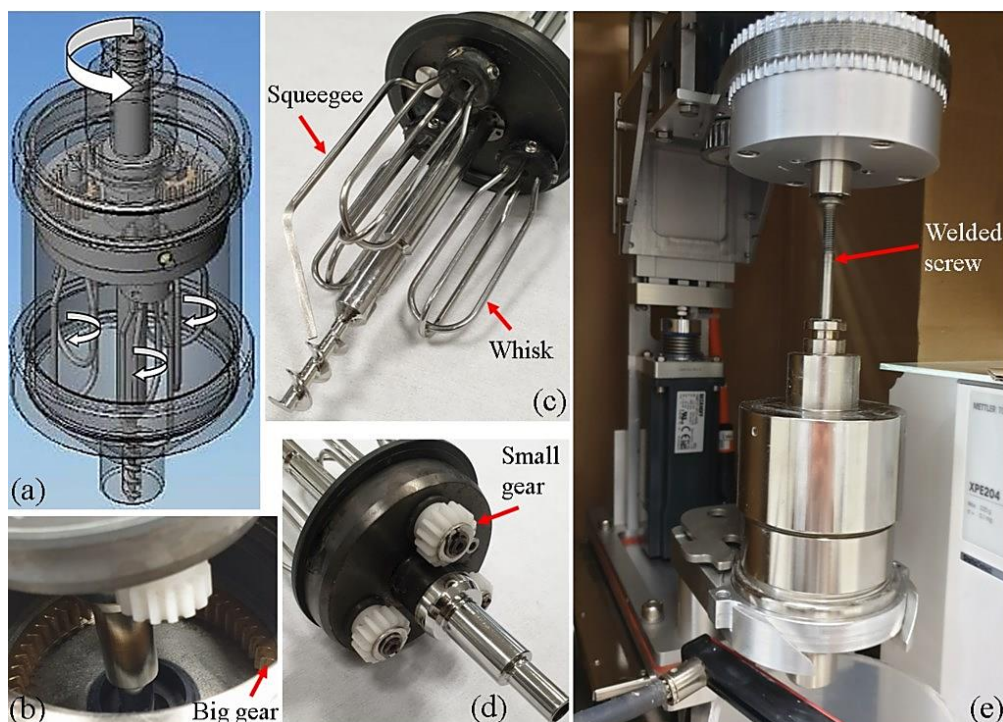


Figure 5.24: (a) Geometry of the mixer powder container, (b) and (d) the internal gears connected to the stirrer mixer, (c) the mixer system and (e) how the mixer container was placed in the LiW feeder.

As the gear ratio is 3:11, with auger motor shaft running at speed 200 rpm, the speed of the stirrer mixer was around 733 rpm. As revealed in Figure 5.25, the micronized Paracetamol was successfully dosed, where the container was almost completely drained. Although some powder is left over on the inner wall of the container, the results indicate that the mixing system was efficient.



Figure 5.25: Mixer powder container after dosing showing Paracetamol leftovers on the (a) inner surface and (b) on the conical funnel.

Although the powder flow during the dosing of micronized Paracetamol was almost linear and continuous as represented in Figure 5.26, the confirmation results between the LiW and GiW showed high

deviation. At the beginning, after turning on the motor, the discharge of powder took some time until the powder was well mixed inside the container. In 282 s, 11.68 g was dosed in the LiW feeder while only 10.45 g were measured by the GiW, which indicates a high deviation error of 11.8 %. As it was a good achievement to dispense such cohesive powder, a design improvement should take place to decrease the high vibrations caused by the stirrer mixer system that affected negatively the dosing results. The stirrer mixer geometry proved its efficiency but it would be rather good to control its speed with a separate motor than that of the auger.

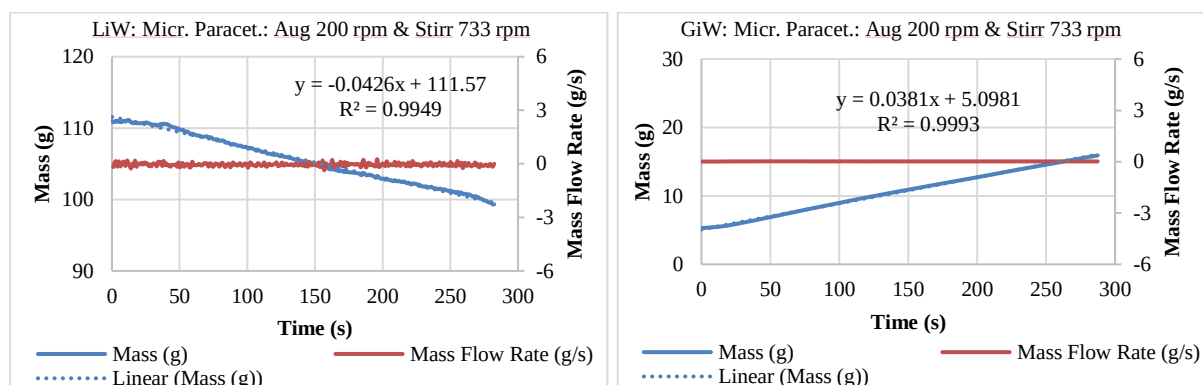


Figure 5.26: Micronized Paracetamol dosing test results using mixer powder container with auger running at speed 200 rpm and stirrer mixer running at speed 733 rpm.

In brief, good results were achieved by dosing the previous five powders through the vertical LiW feeder that validated the proof of concept prototype. Rotating the motors with higher speeds caused the weight readings fluctuations to converge, and thus more accurate dosing results were obtained. It is important to mention that the weighing accuracy of the laboratory load cell (GiW) is much better than that of the weighing platform. Moreover, very few vibrations were transmitted to the laboratory load cell since no motors ran on it. In that, all the GiW mass flow rate graphs showed almost linear mass flow without fluctuations. It was evident in the results that with high powder bed level, the $\dot{m}$ was high. High powder bed level causes the powder to be compressed under the influence of the high powder mass inside and around the dosing tube [25]. During the dosing process, the powder mass started decreasing gradually over time, where the powder then became less compressed and thus led to a reduction of the density [25]. Consequently, this aided to a decrease in the $\dot{m}$.

5.3.3 Future Work

Several additional improvements should take place to increase the dosing accuracy and functionality of the prototype. It is necessary to work on the root causes of the current problems and consequences. In concern of improving the dosing accuracy and having a very low deviation error between the LiW and GiW, the height of the total assembly should be shortened because the height affects negatively the weighing accuracy. Having a shortened assembly would make it easier to match the center of gravity of the weighing platform and thus improve the weight reading results. This can be done, for instance, by

placing the motors horizontally and coupling them with the powder container through bevel gears. Besides, the current L-shape mounting plate that holds motors 2 and 3 should be more stable and the stirrer connection should be optimized to decrease the fluctuations and eliminate the repeatable weight peaks variability. Moreover, new stainless steel stirrers similar to stirrer version 3 should be manufactured to better handle cohesive powders. The stirrer mixer system proved its efficiency in dispensing micronized paracetamol, where integrating this mixing system in the developed powder container is very essential to acquire better results in combination with the other proposed solutions in this chapter. In addition, the stirrer mixer and the auger should rotate on two separate axes to control them independently.

One of the suggested solutions to optimize the coupling connection between motor 3 and the auger shaft is to place an elastomer polymer coupling instead of the rigid Torx screw to better ensure the centricity and eliminate any transmitted vibrations. The elastomer polymer coupling damps vibrations and achieves a backlash-free torque transmission. Furthermore, a more stable environment and a more rigid table assembly should be established to insulate the dosing machine from any external perturbations and noises as possible. To add up, regular calibration of the weighing platform should take place to prevent its malfunction. It is important to prevent any force transmit to the system that would falsify the weight reading. One advanced back-up option to reach higher reading accuracy is to implement a new advanced weighing platform with better industrial filtration technology.

Regarding the automation, the weighing platform PBK989-AB15 should be connected to the PLC through a ConBlock and a Fieldbus module. A control unit system must be integrated so that the speeds of the motors regulate automatically depending on the density variability and rate of dispensing to ensure a more stable mass flow rate. A level sensor for detecting a minimum powder bed level should regulate the motors' speeds to ensure complete dispensing of the powder container if required. As part of a full future automation solution, a powder dispensing robotic tool can be combined with other tools to accommodate multiple workflows. The robotic tool can, for instance, pick and exchange the empty powder containers with full ones to acquire a machine that can run continuously, e.g., overnight, without the interference of any operator, and thus increase the productivity and eliminate any human error or process delay. In addition, future work could address the functionality optimization of the auger dosing process via, e.g., including machine learning techniques. These machine learning approaches find nowadays application in different fields of science, such as in computational mechanics [153], machine optimization, solid-state materials science and regression tasks of large sets [154–156]. Machine learning could be introduced to predict and identify the vibrations and damages of parts besides implementing digital transformation (industry 4.0) and connecting the dosing machine to a cloud to facilitate its operation by the user/customer and thus easily to collect any required data.

5.4 Summary

In this study, the development of a novel machine, the semi-automated vertical LiW feeder for pharmaceutical applications was presented. A design process approach to define needs and identify problem statements and then select possible solutions based on functional design specifications was discussed. The proof of concept of the developed dosing machine was verified by standalone and powder dosing experiments. A sensitivity study of dosing five powders having a wide range of material attributes was performed. The weight variability of a weighing platform was investigated and experimentally validated by a GiW load cell by successfully confirming the mass flow rate and the amount of dosed mass of the LiW feeder. Therefore, the results proved the functionality of the machine.

The developed LiW feeder exhibits several substantial improvements compared to horizontally oriented feeders, such as

- first-in-first-out powder discharge,
- high dosing accuracy,
- flexible and modular design,
- independent speed control of auger and stirrer units,
- simple changeability of the powder container,
- and lightweight design.

Further development and improvements of the design should take place to reach a more robust performance and build a final version of a market-ready product with optimized features and functions. As a further improvement, the assembly should be designed in a more stable manner in order to reduce vibrations transmitted to the weighing platform. Potentially, a more sophisticated weighing platform including enhanced noise filtering software could be employed for achieving more precise mass flow data. For an industrial application, a dedicated closed-loop control system is required for adjusting the motors speeds.

6 Conclusions

6.1 Summary

The underlying work aimed to develop a statistical methodology that identified the correlations between critical material attributes, critical process parameters and critical quality attributes of a volumetric vertical micro-auger dosing process and to end up with building a loss in weight (LiW) feeder. In this, shear cell analysis and other various empirical tests were used to characterize 30 material attributes of ten different powders ranging between free-flowing and cohesive ones that are widely used in the drug industry. The analysis was done using principal component analysis, which revealed the (dis)similarities between the powders and identified overarching attributes. Thereafter, regression analysis was applied to find possible correlations between the studied factors. Moreover, as a numerical approach, Discrete Element Method (DEM) simulations were performed to model the granular material flow in an auger dosing process. Two main studies were carried out to calibrate and validate the studied powders with the experimental reference values. The successful application of the multivariate regression analysis resulted in predicting the bulk behavior of the powder by clustering different predicted behaviors of various responses together which allowed to dramatically reduce the admissible parameter combinations. Consequently, an optimized set of calibrated DEM parameters was chosen, where the simulation results accurately matched with the experimental data. The essential goal of this contribution was to present the alternative vertical LiW feeder dosing concept. In a systematic design process, based on functional design specifications, the semi-automated vertical LiW feeder for dosing a wide range of powders, especially cohesive ones, was developed. Experimental sensitivity analysis and powder dosing tests to study the functionality of the machine and to investigate the weight variability of a weighing platform was successfully accomplished. As a result, the proof of concept prototype was validated.

6.2 Conclusion and Discussion

In this thesis, analyzing the flowability of ten granular powder materials using a multivariate data analysis is thoroughly discussed. The characterization techniques of 30 material attributes are explained. Identifying possible correlations between critical material attributes, critical process parameters and critical quality attributes are essential in analyzing and optimizing the powder solid handling auger dosing process performance in the pharmaceutical industry. A systematic methodology is followed to gain a deep process understanding by identifying robust correlations for the aim of process optimization. This methodology is based on optimized Design of Experiments, principal component analysis and multivariate regression analysis in the framework of Quality by Design. The results demonstrated the feasibility

of developing robust correlations between the measurable powder attributes and the auger dosing performance. In that, the results showed the significant influence of a number of material attributes and the auger geometry parameters. It is therefore essential to adjust the design of a given system to the material being handled. To conclude, the demonstrated approach in the underlying research work allowed efficiently to identify the influence of different factors on an auger feeding process. This approach is apparently important to be further used for a Quality by Design dosing approach, building predictive models and optimizing processes for further industrial applications.

Discrete Element Method (DEM) numerical simulation of the auger dosing process is implemented. The development of a standard calibration method for calibrating powders by incredibly reducing the ambiguity of the DEM parameters selection is introduced. Two main case studies, i.e., angle of repose and auger dosing, are validated with experimental data results. In these two studies, using multivariate regression analysis, good agreements of the numerical and experimental results are observed. By applying the coarse graining method, the calculation time is decreased from a couple of days to several hours. This simple dynamic numerical calibration tool proves to strongly verify and predict the flowability of free-flowing bulk materials. This tool can be used to reduce the number of experimental investigations and aid in less time and costs. It can be used, for instance, to research the impact of different auger and stirrer geometries, and other process parameters.

The new designed vertical loss in weight feeder prototype is an alternative solution for some existing powder dosing problems. It eliminates some of the weaknesses of the current horizontal loss in weight feeders. It is modularly designed, where this modularity offers the possibility to respond to different challenges of various powders especially the cohesive ones by altering different process parameters. As a proof of concept, the feeder is experimentally validated through sensitivity analysis by examining the functionality of the machine and performing powder dosing tests for the aim of achieving high dosing accuracies.

In conclusion, the present work delivers the modular designed vertical loss in weight feeder, optimized design of experiments along with multivariate statistical data analysis and the numerical treatment approach for further research and development of powder dosing systems.

6.3 Future Work

To better understand the implications of these results, future studies could address more powders to be added in the developed regression database to be able to predict the behavior of new powders. This would dramatically optimize and speed up the drug development process by building robust prediction models and thus ensure overall process stability. In addition, machine learning techniques can be introduced in regression analysis to build more robust prediction models regarding the flowability behavior of powders and optimize the dosing process parameters. Moreover, machine learning can identify and predict possible malfunctioning, e.g., vibrations and damage of parts, of the developed vertical loss in

weight feeder. A number of automation solutions could be added to the dosing machine, such as installing a control unit to self-regulate the speeds of the motors depending on the discharge rate of the powder and digitally transforming the machine by connecting it to a cloud (industry 4.0). Further research should focus also on additional numerical and experimental studies under different conditions to address very cohesive powders and thus optimize the design of the dosing machine. Research works should consider introducing more cohesive contact models in the numerical simulations to facilitate dealing with low accuracy dosing results of some challenging powders in the industry.

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A Powder Behavior During Dosing

The experimental study of the ten different investigated powders showed how these powders behave differently and that their flow is dissimilar. In some powders, e.g., Emcompress and SpheroLac 100, their discharge was freely and smoothly, whereas in other powders, e.g., Ascorbic acid and Magnesium Stearate, the discharge was more agglomerated and discontinuous. Figure A.1 illustrates some flow behavior of the ten studied powders during auger dosing experiments.



Figure A.1: Illustrational examples of powder flow behavior of the ten powders.

B Box-Cox Plot and Model Transformation

A transformation was applied to the measured values of the percentage of dosed mass, the dosing time, and the feed rate. The transformation was selected based on the Box-Cox plot. The Box-Cox plot in Figure B.1 compares the residual sums of squares (the error term) of different transformations. Transformations with a lower error term indicate a better fit. The significance line illustrates the border between transformations having a worse fit and the best fit (based on a likelihood-ratio test). Mainly, if the untransformed response is above the line, one of the transformed responses below the line will give better fit. Lower values indicate better fit, points above the significance line result in a significantly worse fit than the best fit in the comparison.

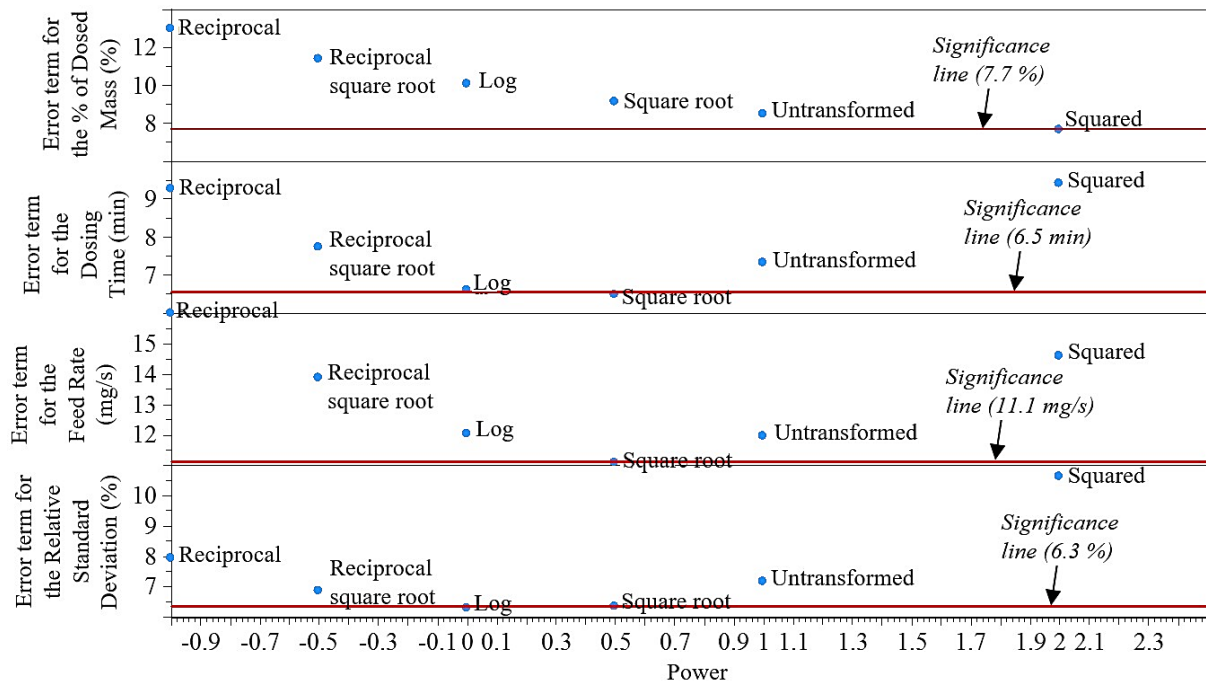


Figure B.1: Box-Cox plot for the percentage of dosed mass (PoDM), dosing time (DT), feed rate (m) and the relative standard deviation (RSD).

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